Technology Experts Company

Radiation Protection Center

Radiation Protection Training Course

Prepared by:

Eng. Ashraf Mohammed Enany

Master of Science in Nuclear Engineering

Head of Radiation Protection Center

Mobile : +966 568360241

Email : ashraf@ t-experts.com

Website : www.t-experts.com

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4. External & Internal Dose Calculations and Shielding
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Introduction to Radiation Physics

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What is the "Radiation Source"



Simply,

It is a source of " Energy "

But, ... What kind of Energy !!







Why it is called "Nuclear" ??

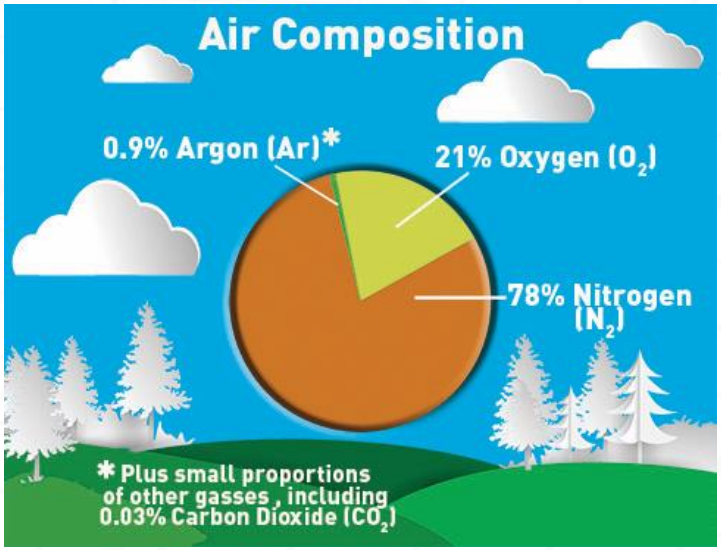
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MATTER

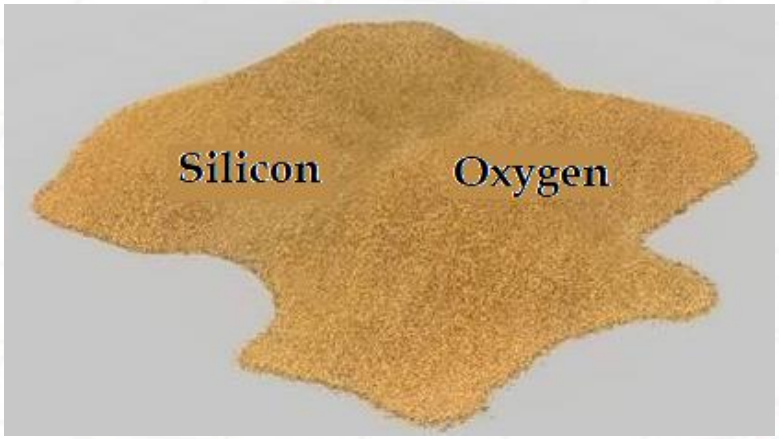
- All **matters** in the **natural** are made up of **Elements**.





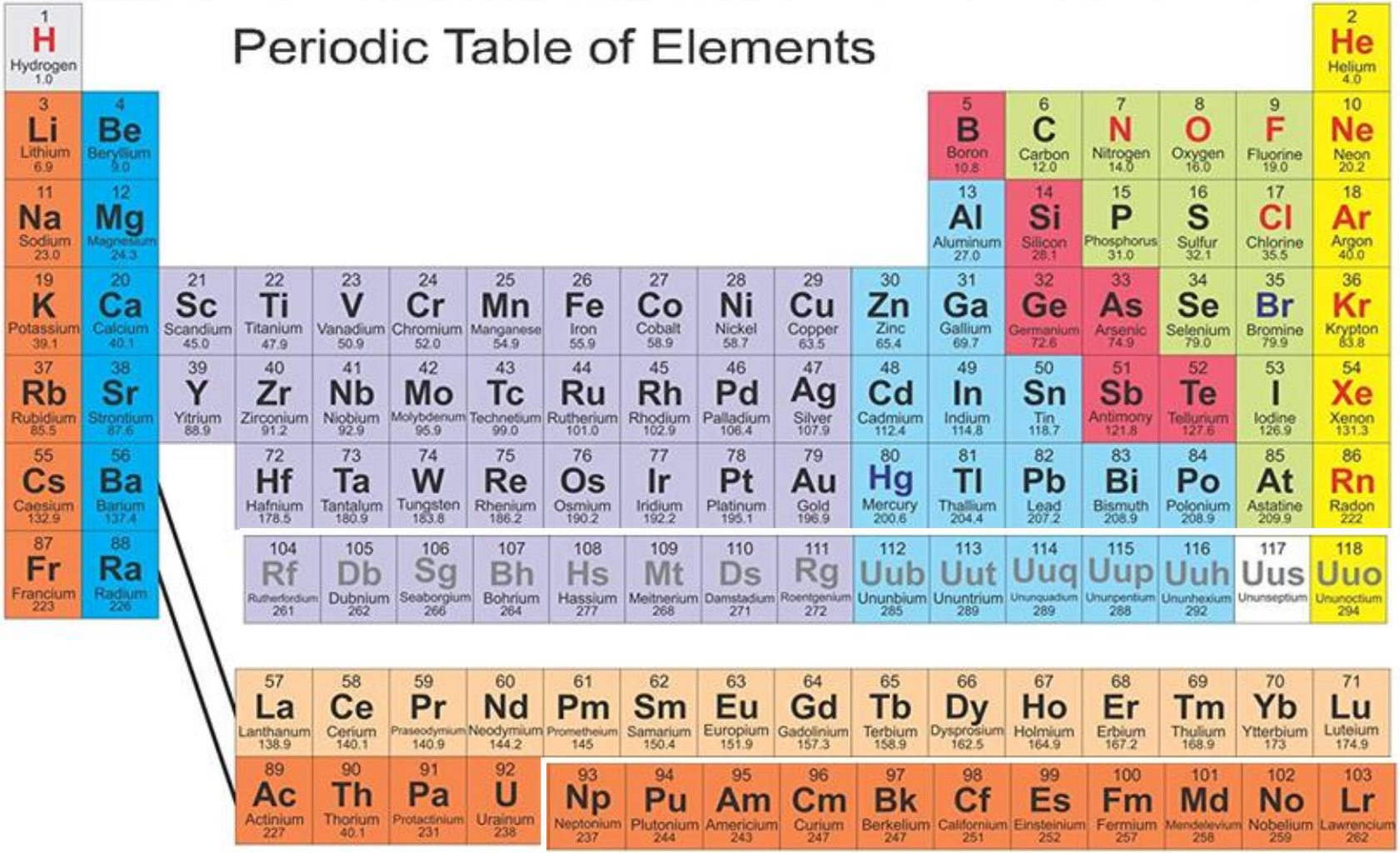


- There are **92 chemical elements** in **nature**, which form the basis of all the materials around us.



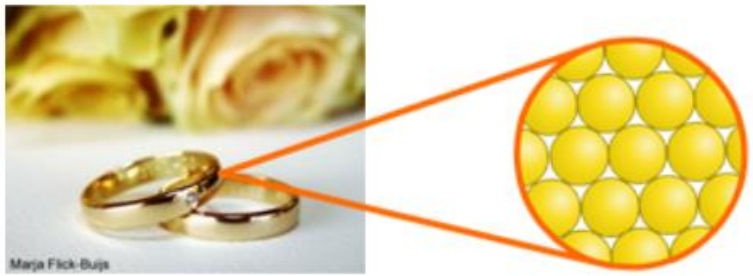
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MATTER



MATTER

- Any **element** consists of tiny basic building units called **ATOMS**.



Gold is an element made up of only gold atoms.



Carbon is an element made up of only carbon atoms.

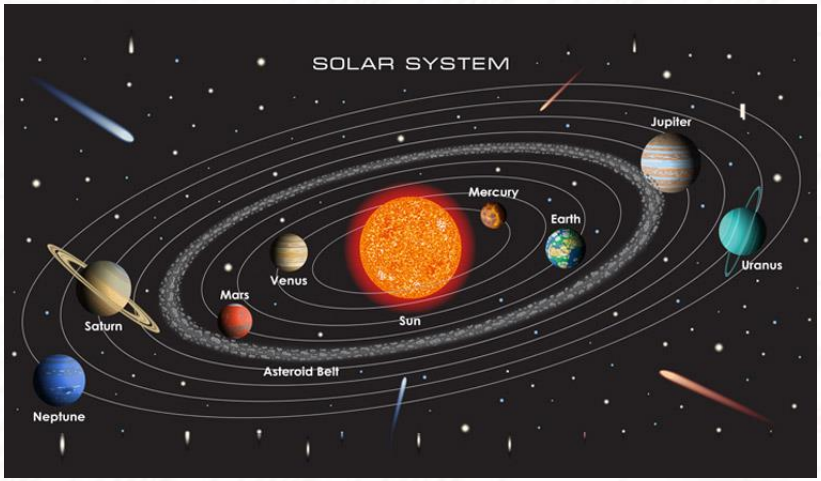
- **Atom:** is the smallest building unit of an element with its characteristic properties.

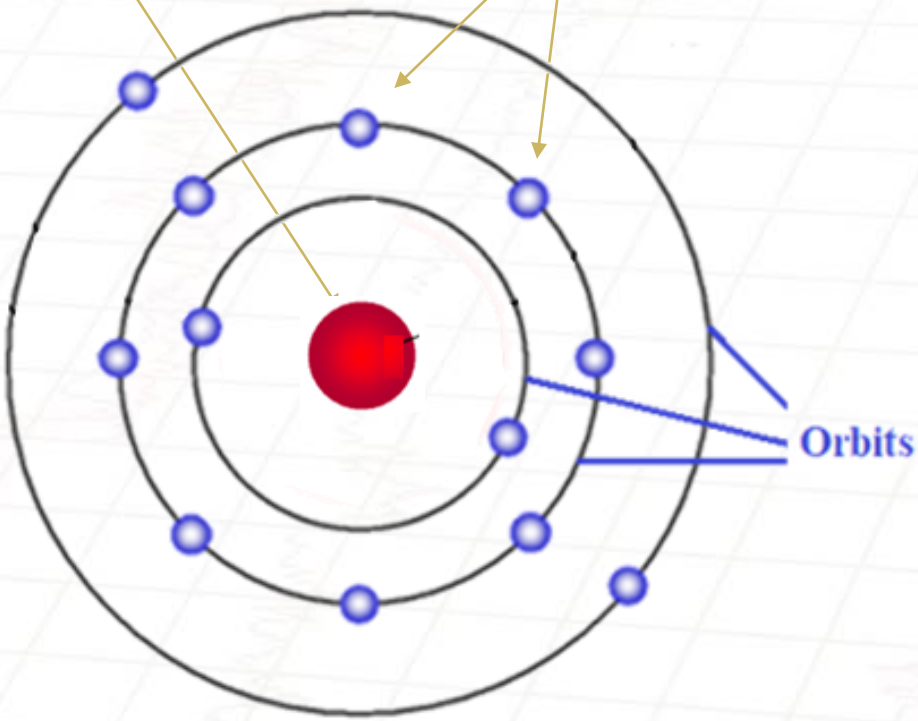
1 gram of an element has more than 10^{21} atoms

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MODEL OF ATOM

- The atom consists of: **1. Nucleus** and **2. Electrons**





Simple Model of Atom

Why dose it call "Nuclear Energy" ??

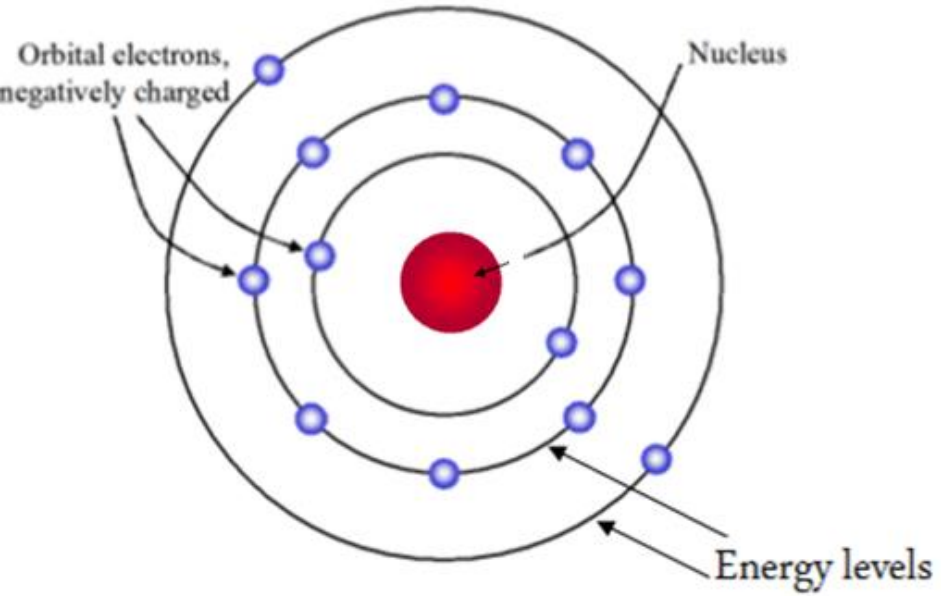
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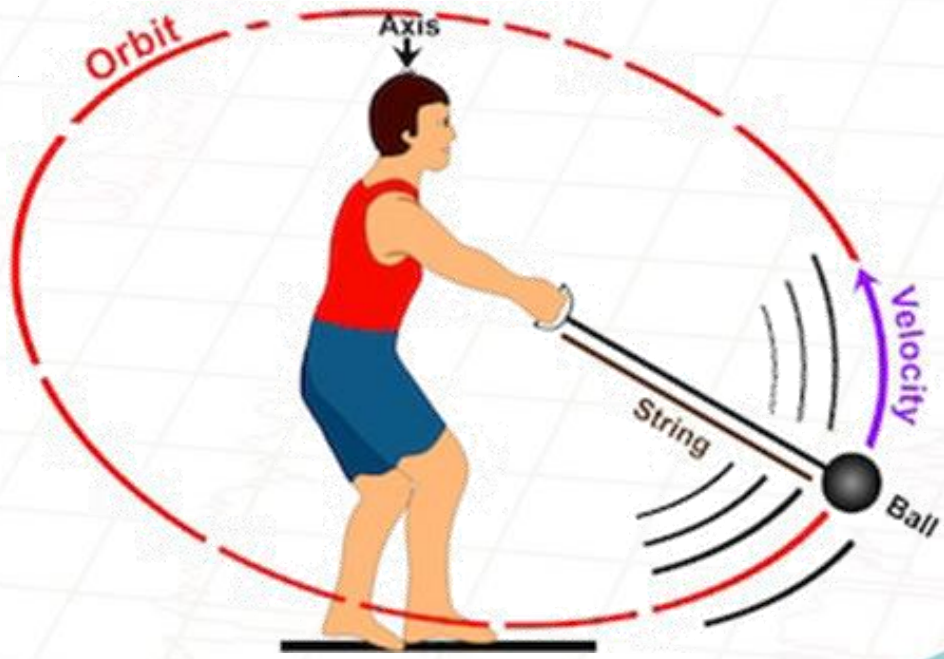
Because the Nucleus is the origin of the emitted radiation (energy).

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ORBITAL ELECTRONS

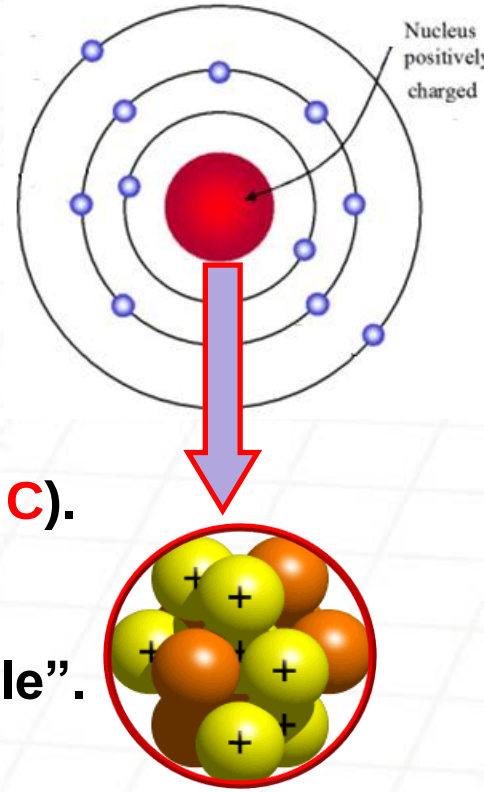
- Electrons are negatively charged particles ($1e^- = -1.6x10^{-19}$ Coulomb).
- Electrons are moving so fast around the nucleus in certain **Orbits** (**Energy Levels !!!**).





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NUCLEUS



- Nucleus is the central part of the atom, and it composes of two different types of particles: **Protons** and **Neutrons.**

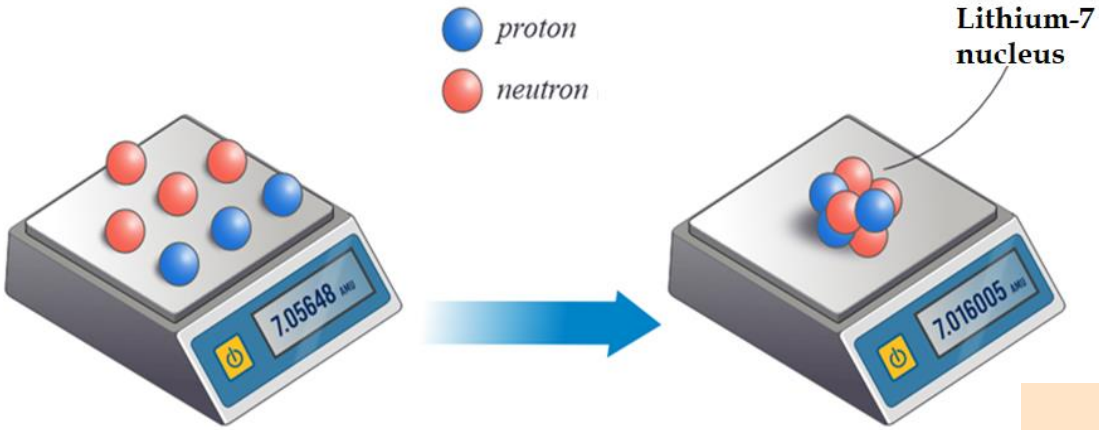
Proton is positively charged particle ($+1.6 \times 10^{-19}$ C).

Neutron has NO electrical charge "neutral particle".

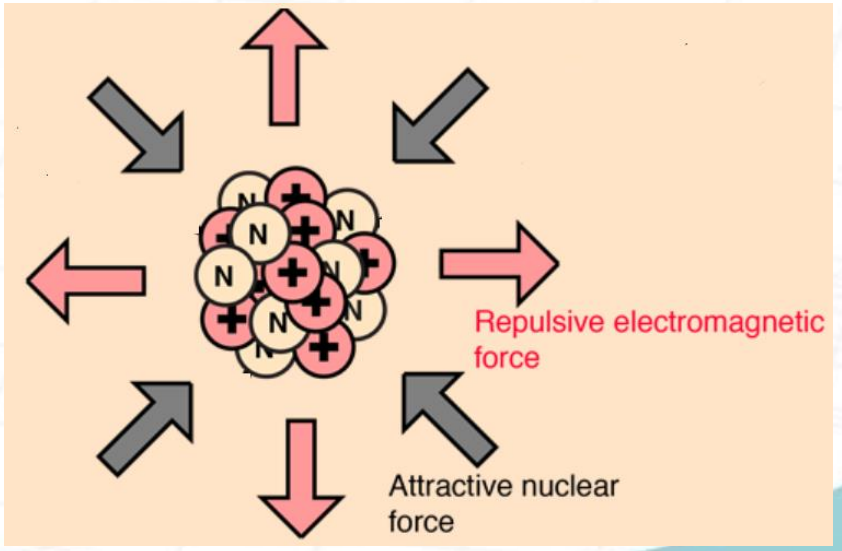
ATOM is neutral (net electrical charges = 0)

NUCLEUS

Mass Defect: The difference between the mass of the nucleus and the sum of the masses of its parts.



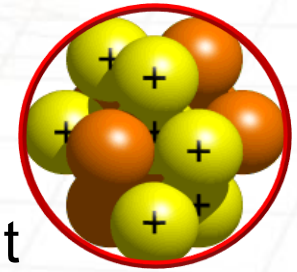
The mass defect is converted into nuclear binding energy (glue)



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ATOMIC NUMBER (Z)

- **Atomic Number (Z): Number of Protons in the nucleus.**
- **Atoms of same element always contain the same Z.**



The atomic number **(Z)** is the "ID Card" of the element

Example:

Z = 1 is **Hydrogen**.

Z = 6 is **Carbon**

Z = 55 is **Cesium**

Z = 77 is **Iridium**

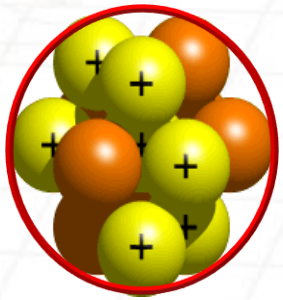
Periodic Table of Elements

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	*	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	**	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
		*	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
		**	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

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MASS NUMBER

- **Mass Number (A):** Total number of Protons and Neutrons in the nucleus.



Atoms of the same element may not contain the same number of neutrons (different mass number).

Example:

^{6}Li 7.5%	^{7}Li 92.5%
3p 3 n	3p 4 n

iron
Fe
26

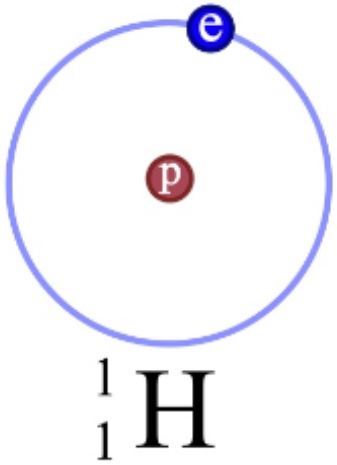
26p 28n Fe-54 STABLE	26p 30n Fe-56 STABLE	26p 31n Fe-57 STABLE	26p 32n Fe-58 STABLE

Ir-191 NA: 37.3%	^{193}Ir NA: 62.7%
77 P	77 P
114 n	116 n

ISOTOPES

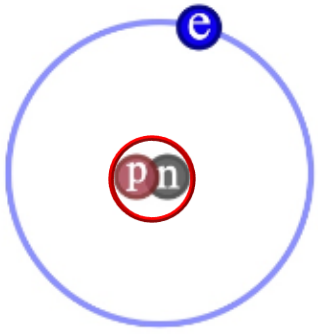
Isotopes: Atoms of the same element that have different number of neutrons in the nucleus (different Mass number "A").

Example: Hydrogen element



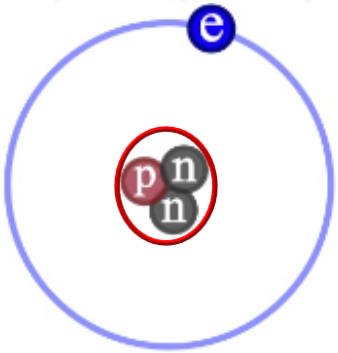
$^{1}_{1}H$

Hydrogen -1



$^{2}_{1}H$

Hydrogen -2 (Deuterium)



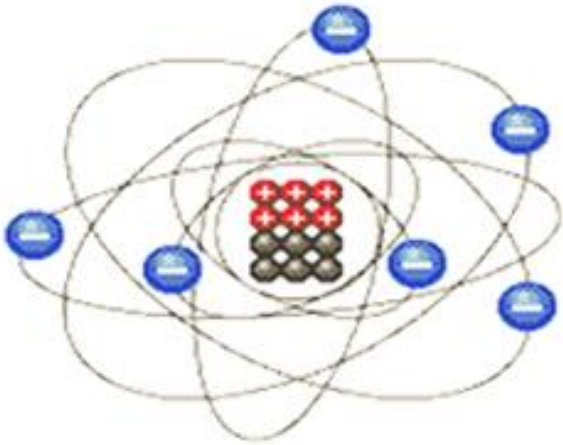
$^{3}_{1}H$

Hydrogen -3 (Tritium)

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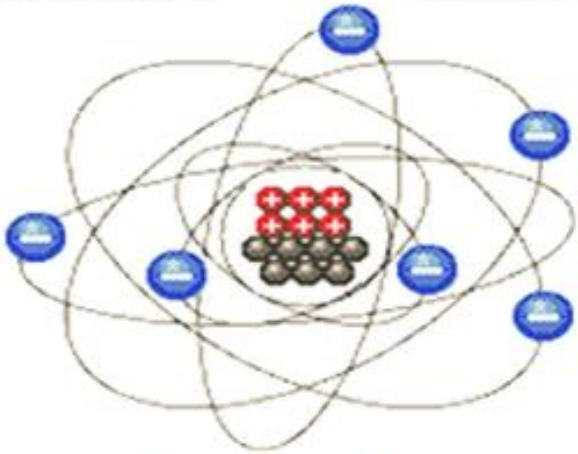
ISOTOPES

Example: Carbon element



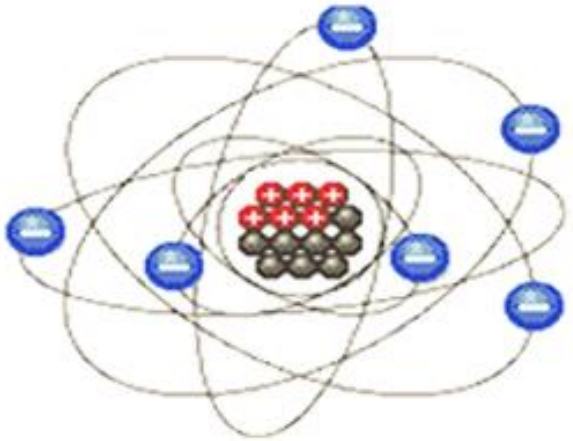
Carbon-12 (C-12)

- 6 Electrons
- 6 Protons
- 6 Neutrons



Carbon-13 (C-13)

- 6 Electrons
- 6 Protons
- 7 Neutrons



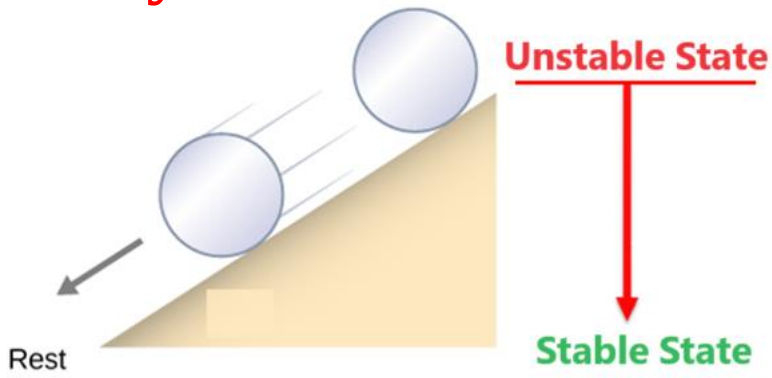
Carbon-14 (C-14)

- 6 Electrons
- 6 Protons
- 8 Neutrons

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Stability and Unstability

Some isotopes are Stable, and others are unstable (Radioactive Isotopes)



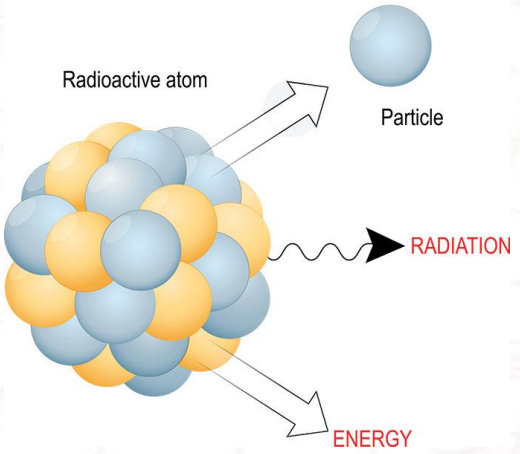
Stable Isotope:

The **numbers** of protons and neutrons in the nucleus remain the same forever.



Unstable Isotope:

The **numbers** of protons and neutrons in the nucleus are changing with time



All elements with Z > 83 is unstable

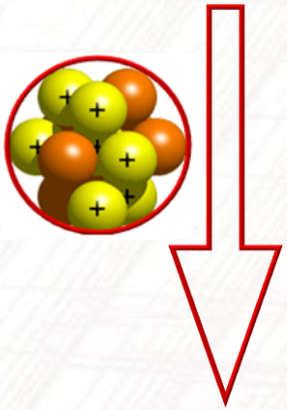
Naturally Occurring Radioactive Materials "NORM" like U, Th, Ra, …

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Unstability

Why some isotopes are unstable (Radioactive Isotopes)





There is **NO EQUILLIBRIUM** between the **NUMBERS** of neutrons and protons inside the nucleus.

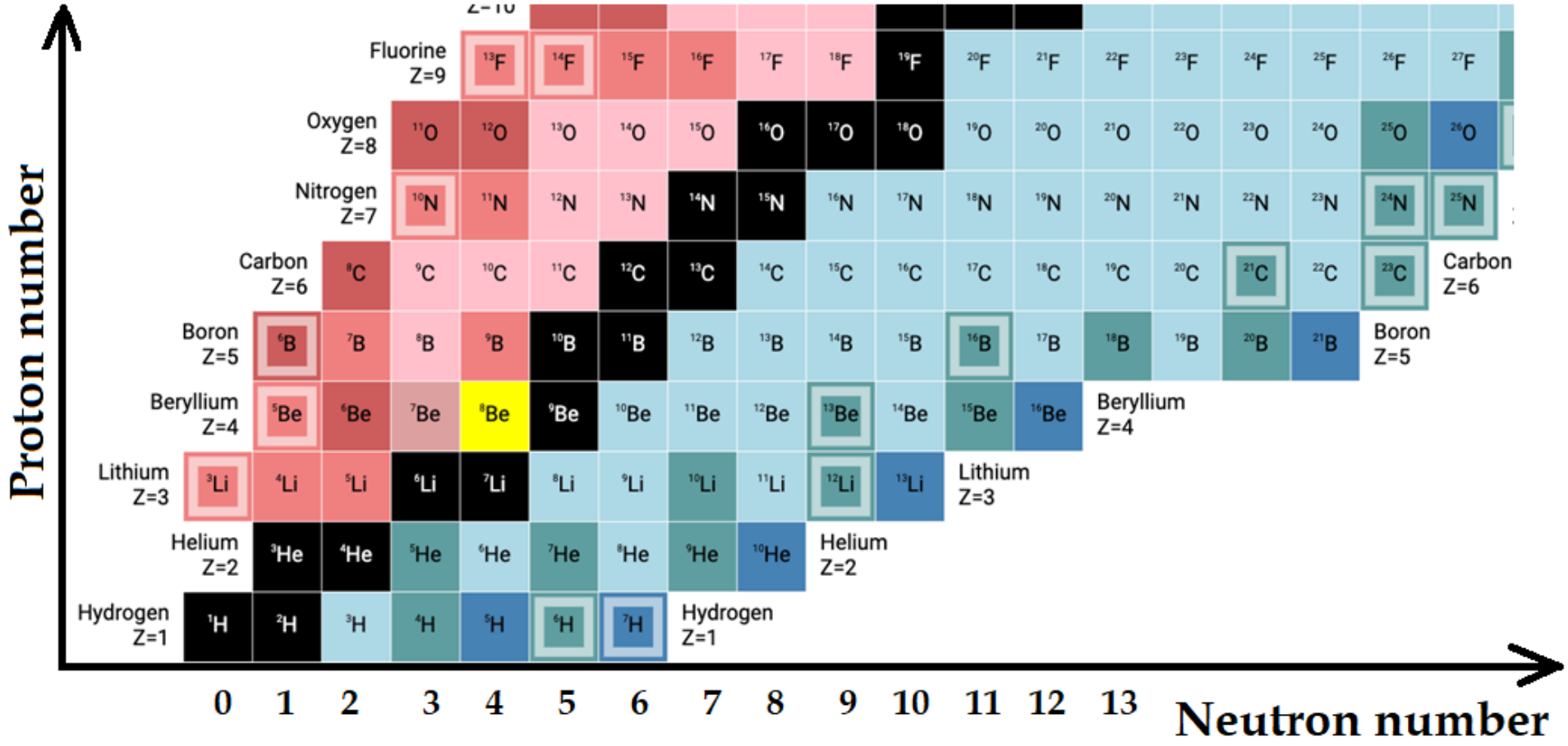


Nucleus has

EXCESS ENERGY inside it

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Stable and unstable Nuclides



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NUCLEAR STABILITY

Example:

Isotopes of Hydrogen



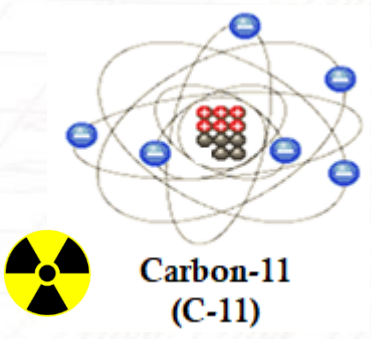
# of Protons	# of Neutrons	Status	Reason
1	0	Stable	OK
1	1	Stable	OK
1	2	Unstable	More neutron than nucleus needs

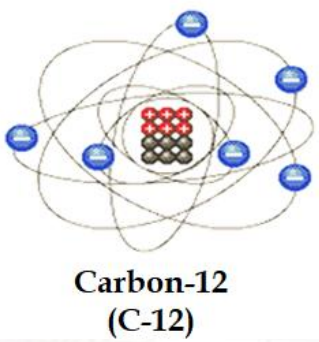
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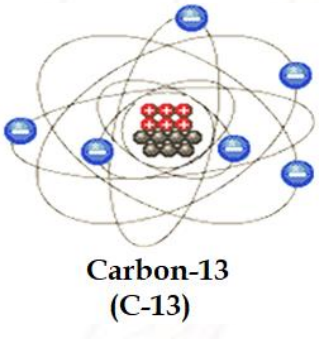
NUCLEAR STABILITY

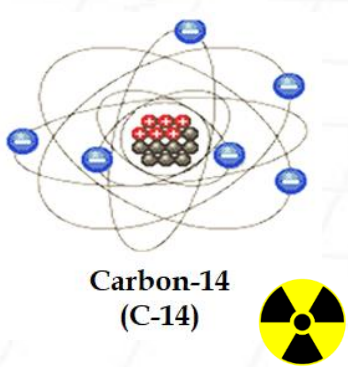
Example:

Isotopes of Carbon









# of Protons	# of Neutrons	Status	Reason
6	5	Unstable	Less neutron than nucleus needs for stability
6	6	Stable	OK
6	7	Stable	OK
6	8	Unstable	More neutron than nucleus needs for stability

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NUCLEAR STABILITY

Does the unstable nucleus stay in this state forever?

Certainly, NO

What will happen?

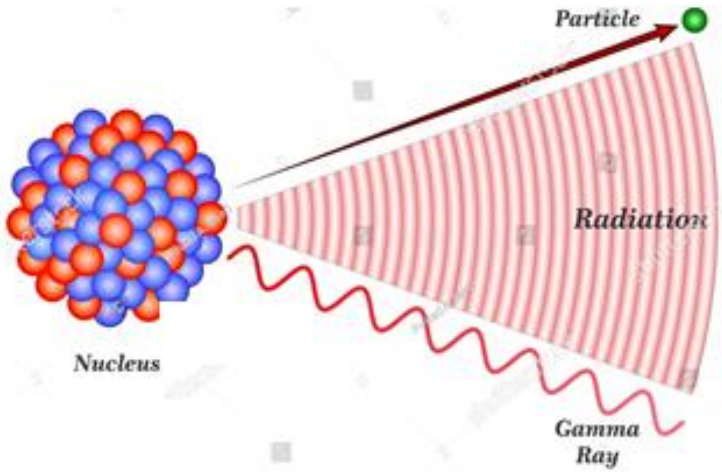
The unstable nucleus will decay (breakdown) to another isotope/element to adjust the ratio between the number of Neutrons and Protons to reach the stability and also emits its excess energy



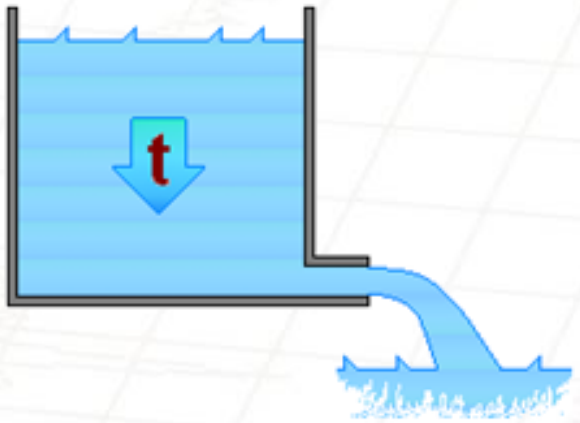
RADIOACTIVITY

Radioactivity:

Spontaneous emission of radiation (energy) from the unstable nucleus of the atom.



Radioactivity is **NOT affected** by any external conditions such as Temperature, Pressure, Physical State, etc.

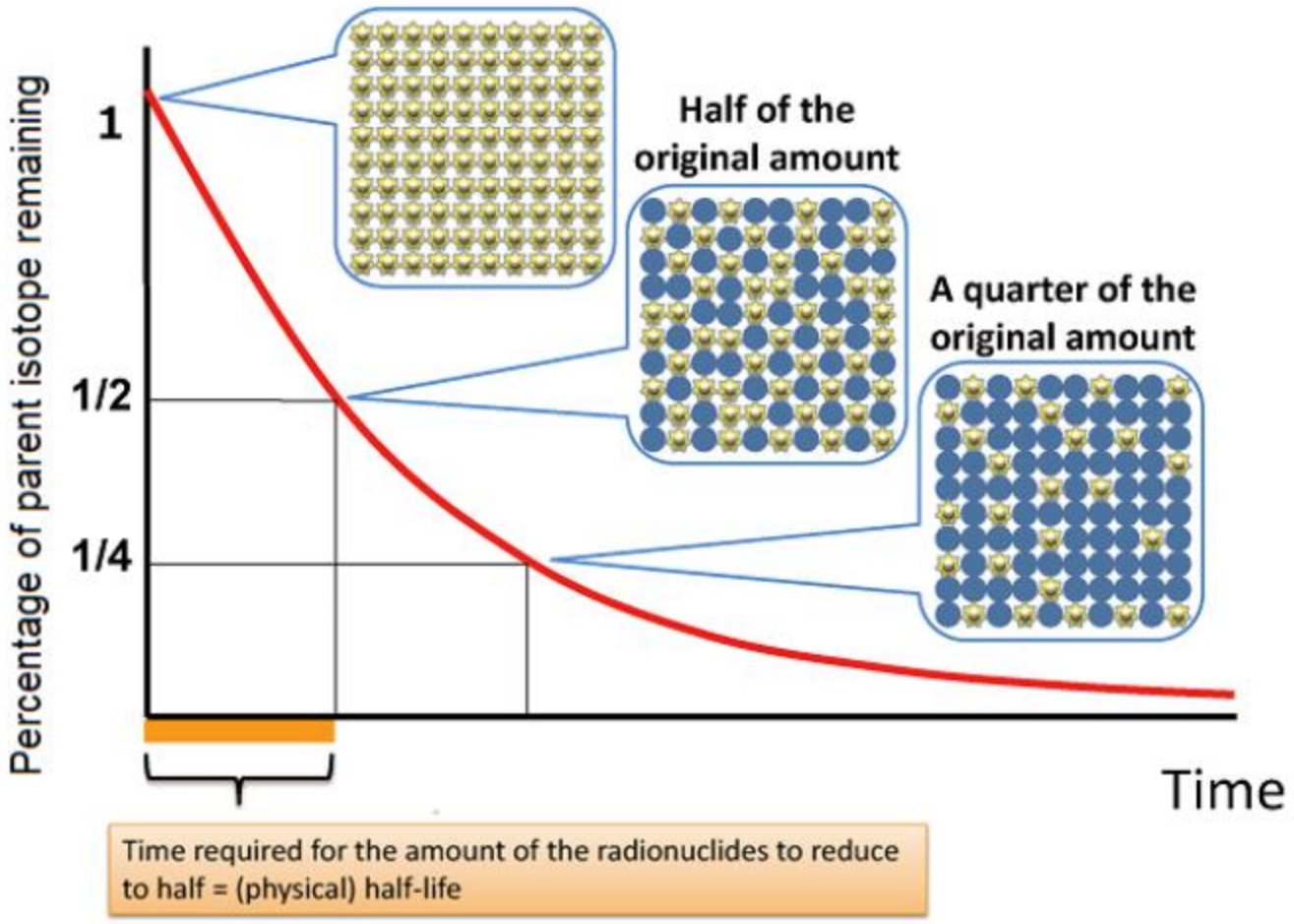


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DECAY & TIME

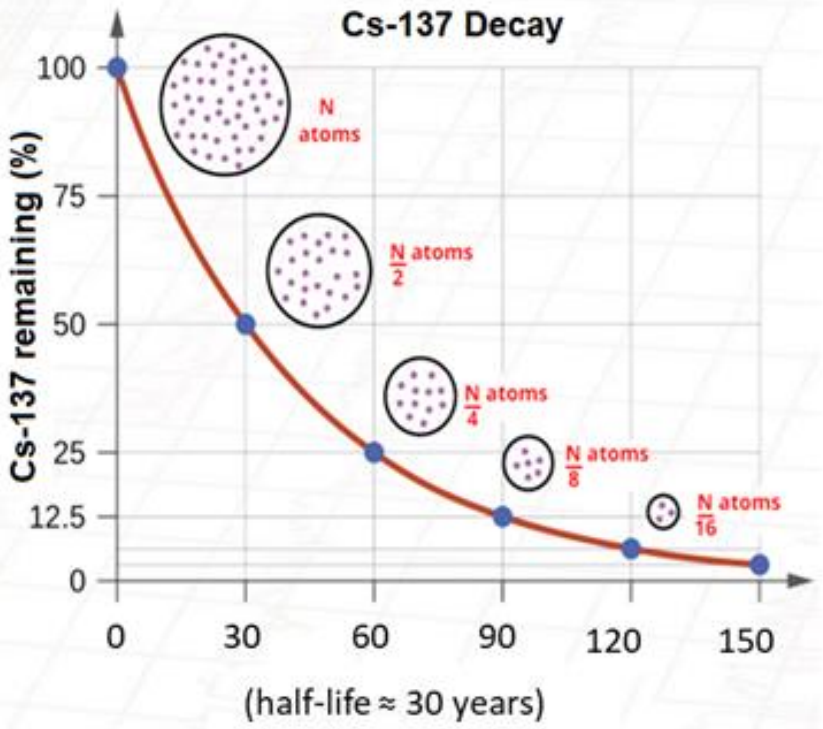
$T_{1/2}$ (half-life):

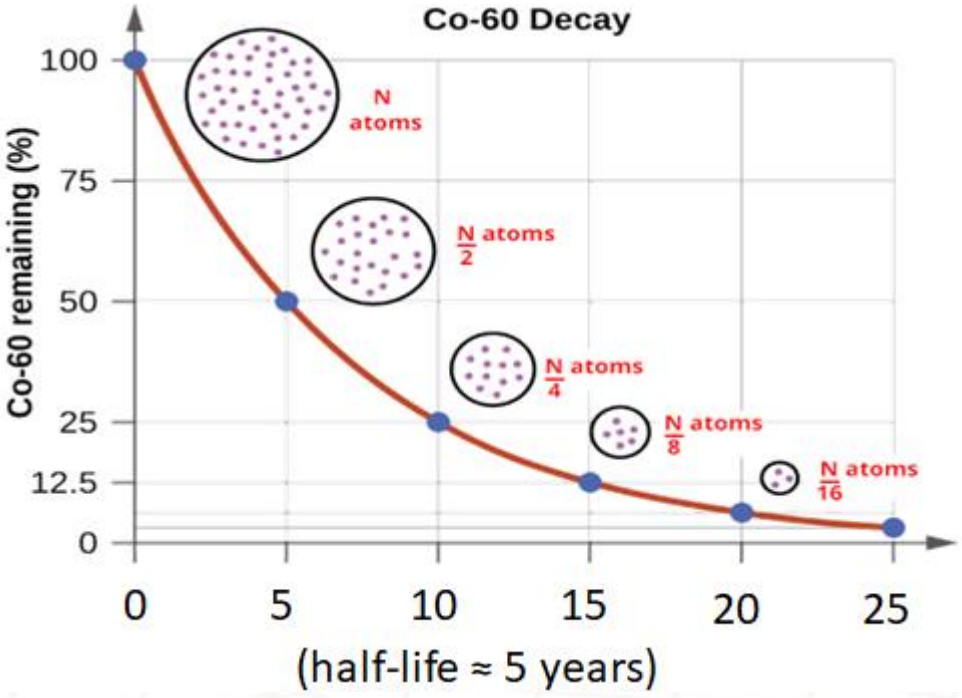
It is the time required for one half of the amount of radioactive atoms to decay into a more stable atom.



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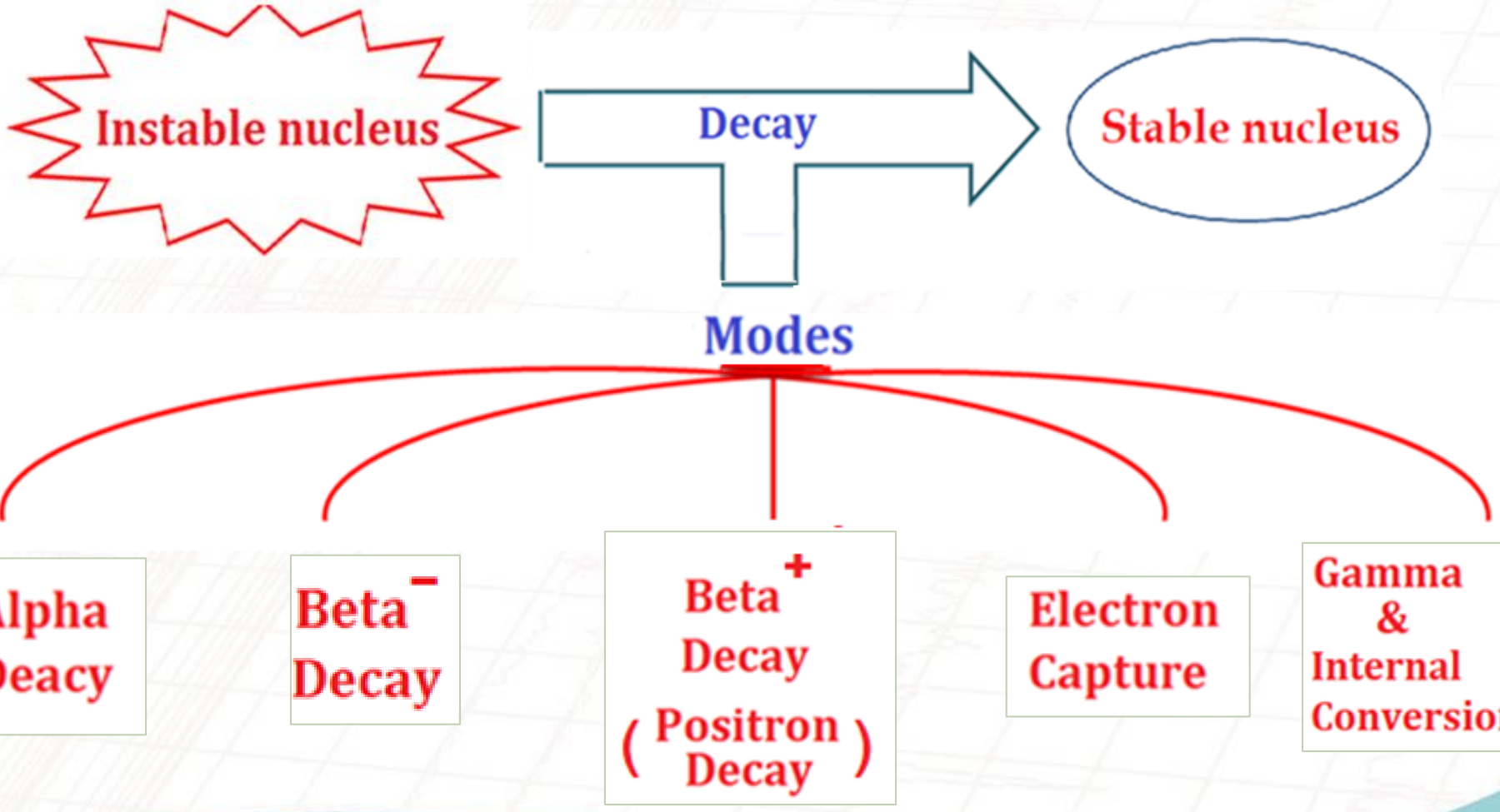
DECAY & TIME





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MODES OF DECAY



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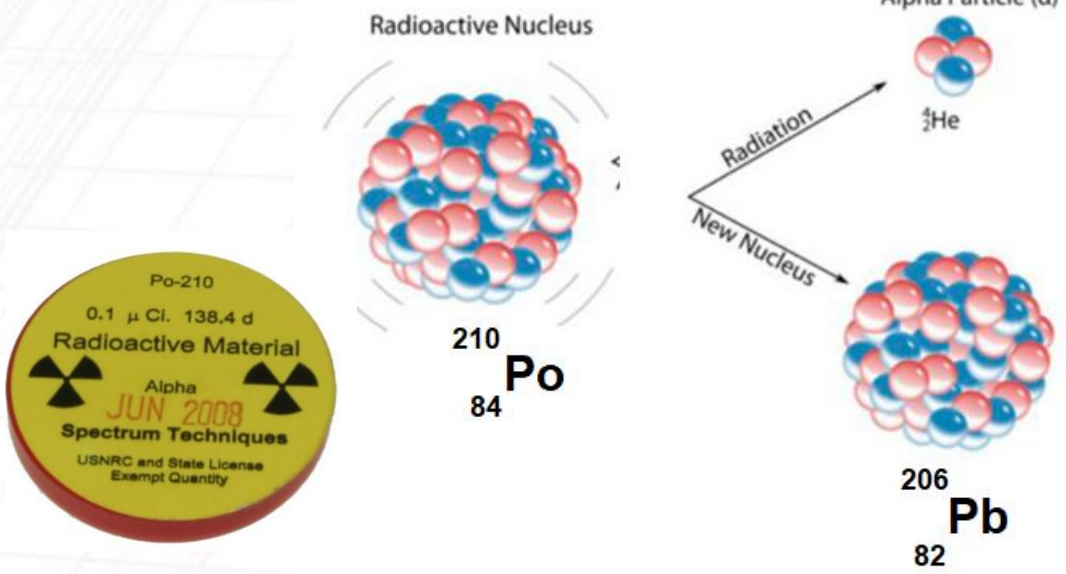
ALPHA DECAY (α)

- **Alpha decay, mainly, occurs with isotopes of HEAVY nucleus (Generally Z > 82).**

OR

Proton to neutron ratio is higher than stable ratio (Proton rich, too many protons)

- **In α decay, the atomic number decreases by (2) and the mass number decreases by (4).**

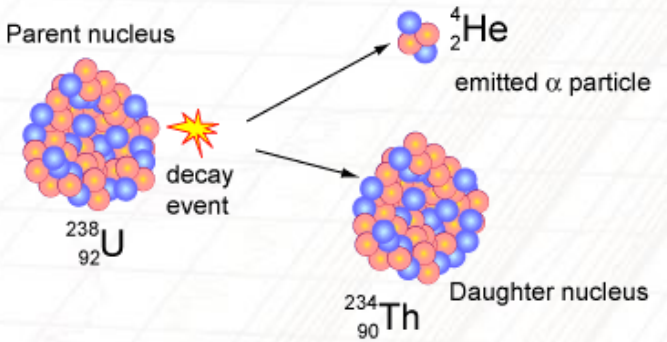


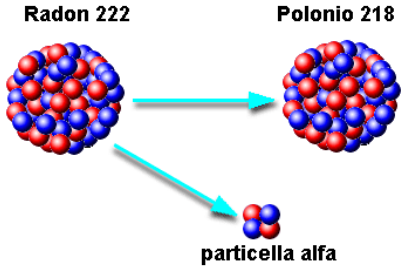
$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + {}^{4}_{2}\alpha + \text{Energy}$$

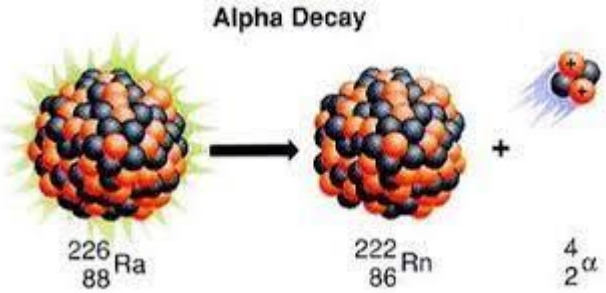
Parent — Daughter — nucleus of helium atom

$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + {}^{4}_{2}He$$

Alpha Decay





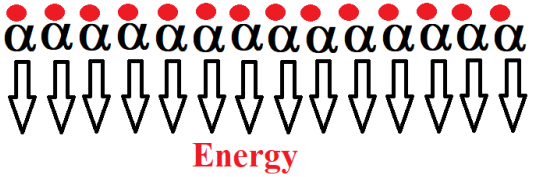


All emitted **α** from a certain source have the same value(s) of energy "*mono-energetic particles*" (*has defined values of energies*) (*has discreet spectrum*).

Alpha energy is a Finger Print of the radioactive source

Decay process	Half-life
$^{226}Ra \rightarrow ^{222}Rn + \alpha(4784.34\,keV)$	1600 year
$^{222}Rn \rightarrow ^{218}Po + \alpha(5489.48\,keV)$	3.8232 day
$^{218}Po \rightarrow ^{214}Pb + \alpha(6002.35\,keV)$	3.071 min
$^{214}Po \rightarrow ^{210}Pb + \alpha(7686.82\,keV)$	162.3 μs
$^{210}Po \rightarrow ^{206}Pb + \alpha(5304.33\,keV)$	138.3763 day

Energy of principal α particles

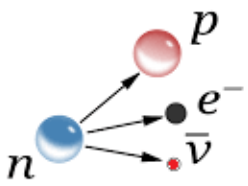


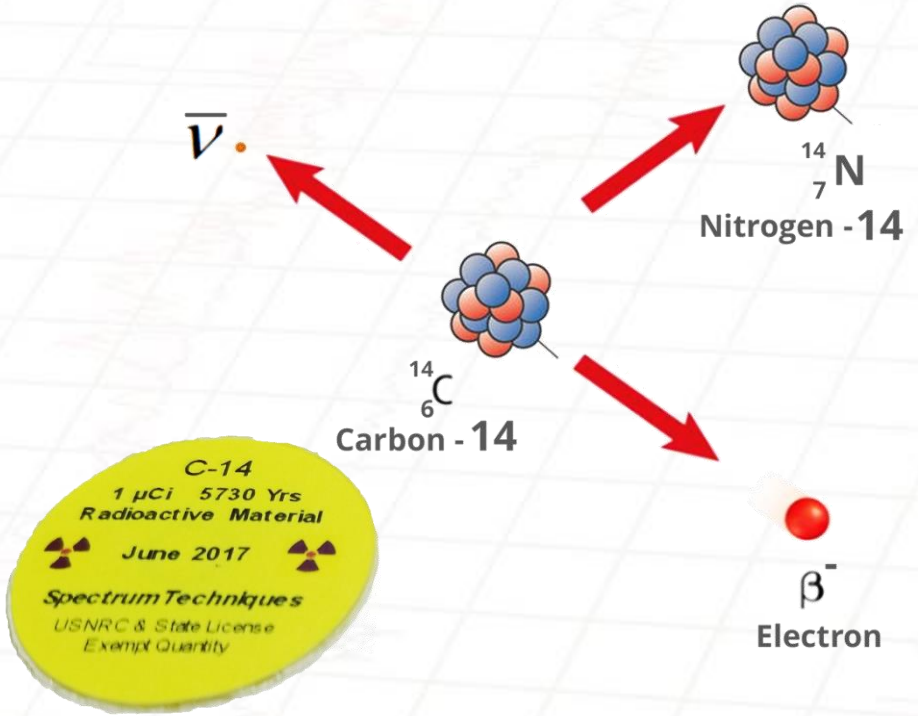
BETA DECAY (β^-)

(High-energy electrons)

- A nucleus will undergo Beta minus decay if it has excess numbers of neutrons than it needs for stability (Higher "N/P" ratio) (Neutron-rich) , too many neutrons.

$$ {}^{1}_{0}n \longrightarrow {}^{1}_{+1}p + {}^{0}_{-1}\beta + \bar{\upsilon} $$





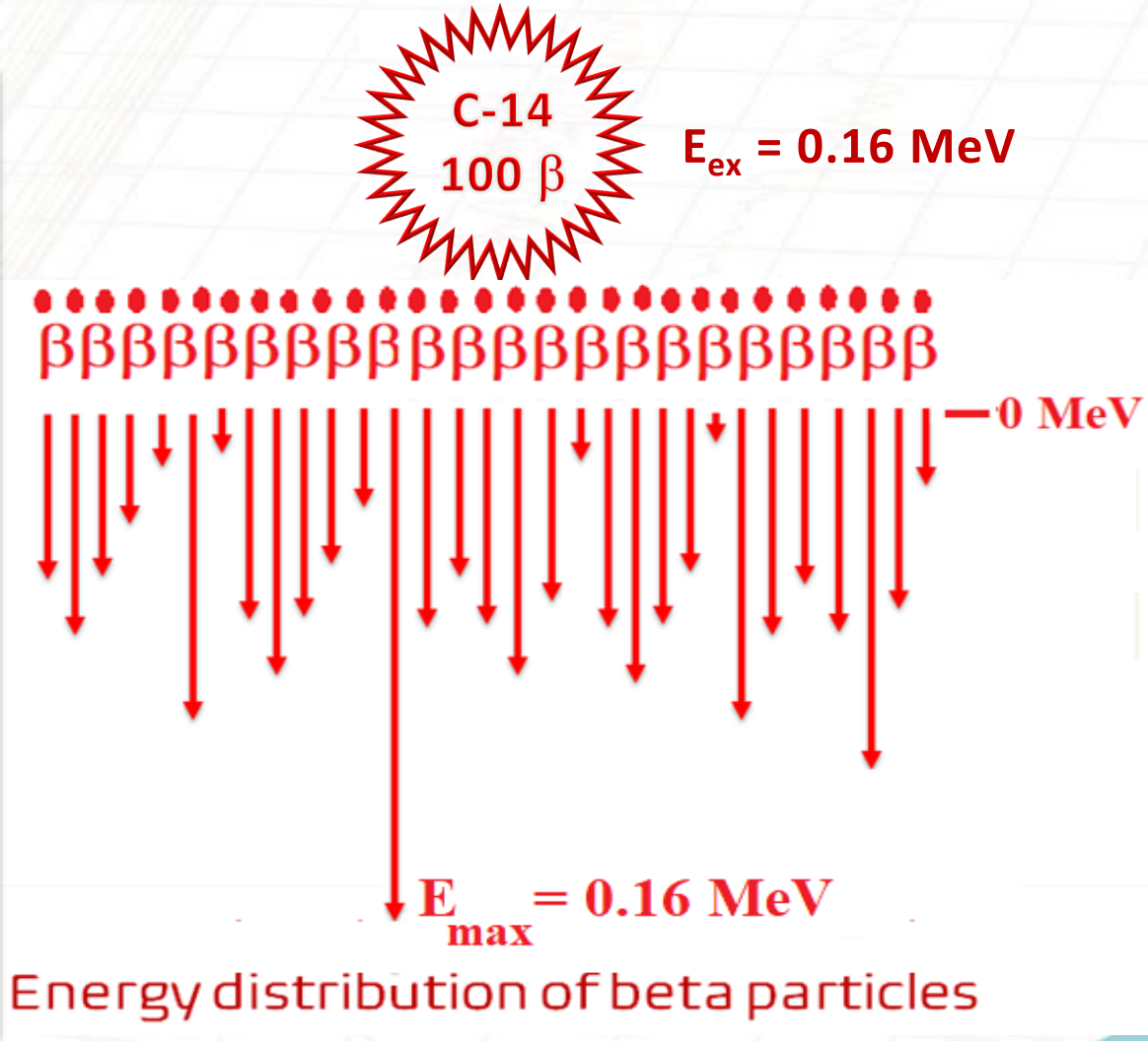
$$ {}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}\beta + \bar{\nu} $$

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ENERGY OF BETA

- Because the total energy of the decay process is divided randomly between the β particle and antineutrino therefore, Beta particles are emitted with a range of energies from zero to a maximum value.

Thus, Beta energy spectrum is NOT a Finger Print of the radioactive source

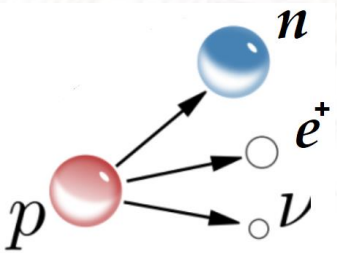


POSITRON DECAY (β^+)

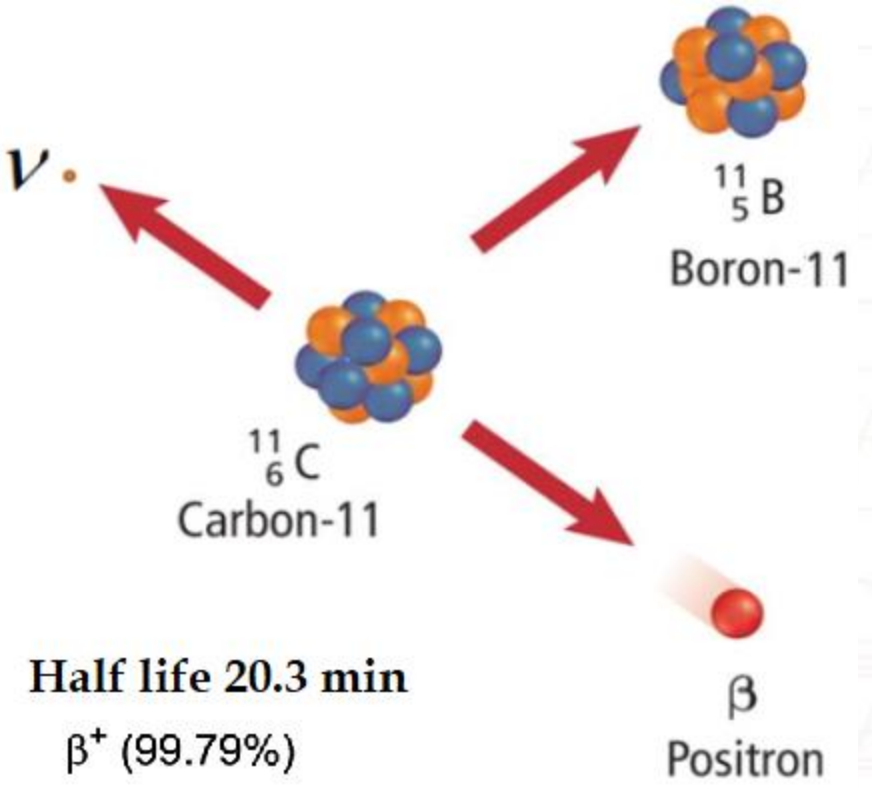
- **A nucleus will undergo Positron decay if it has:**

1. lower number of neutrons than it needs for stability "Lower (N/P) ratio" "Proton-rich.

2. Excess energy ≥ 1.022 MeV.



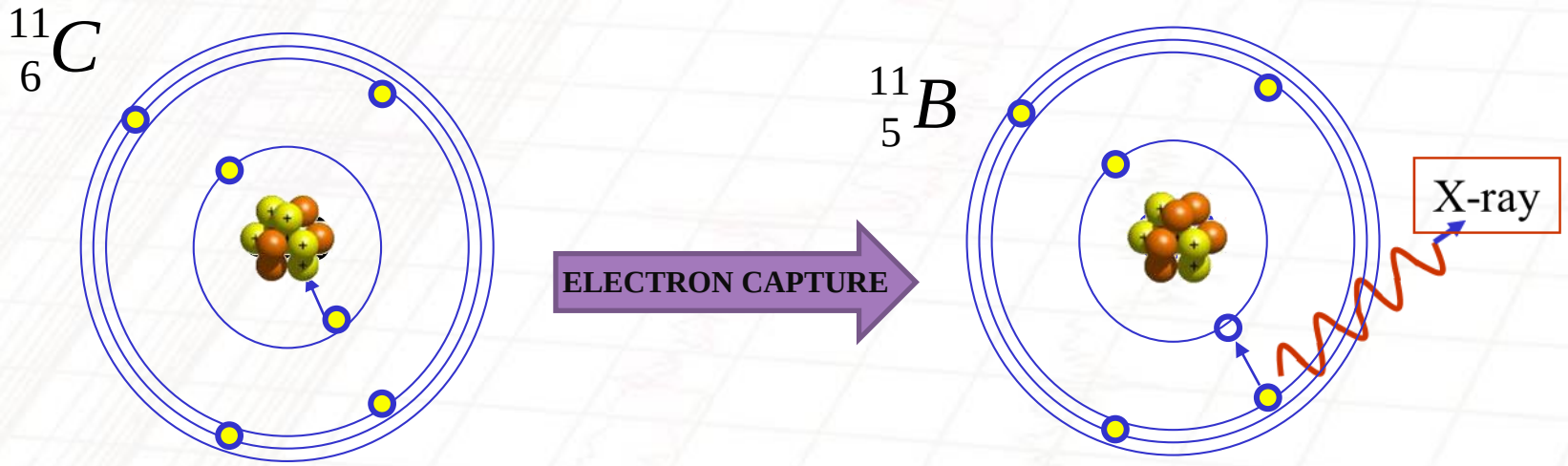
$$ {}^{1}_{1}\mathrm{p} \rightarrow {}^{1}_{0}\mathrm{n} + {}^{0}_{+1}\beta + \nu $$



$$ {}^{11}_{6}\mathrm{C} \rightarrow {}^{11}_{5}\mathrm{B} + {}^{0}_{+1}\beta + \nu $$

ELECTRON CAPTURE DECAY

A nucleus will undergo electron capture decay if it has **lower number of neutrons than it needs for stability** "**Lower (N/P) ratio**" "**Proton-rich**" "**too many protons**".

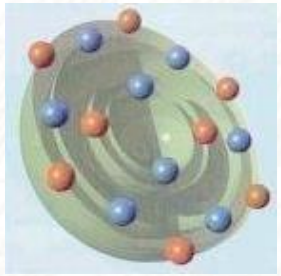


$$^{11}_{6}C \xrightarrow[20.3\ \text{min}]{\text{EC (0.21\%)}} {}^{11}_{5}B + \nu$$

GAMMA DECAY (γ)

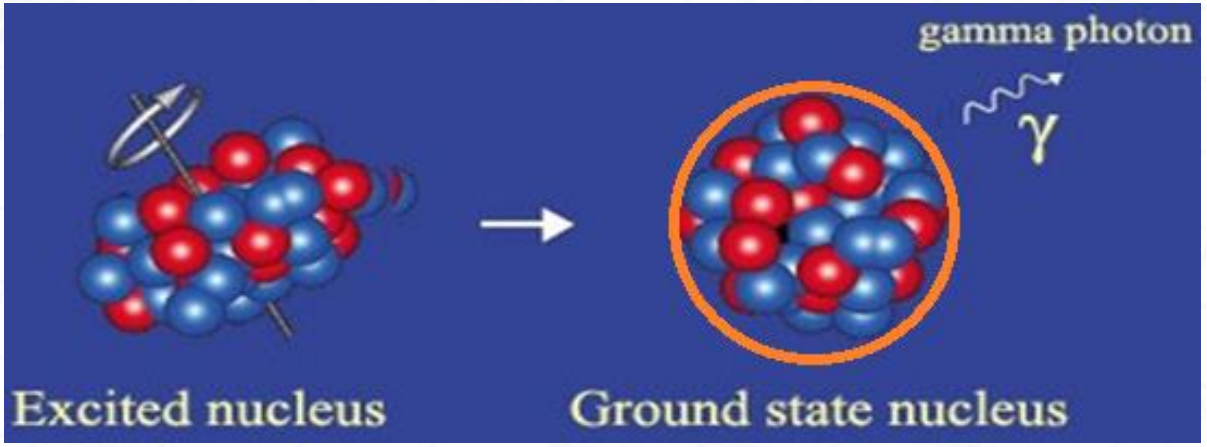
- After an α or β decay, the daughter nucleus may be in:

Ground State (Stable)



OR

Excited State

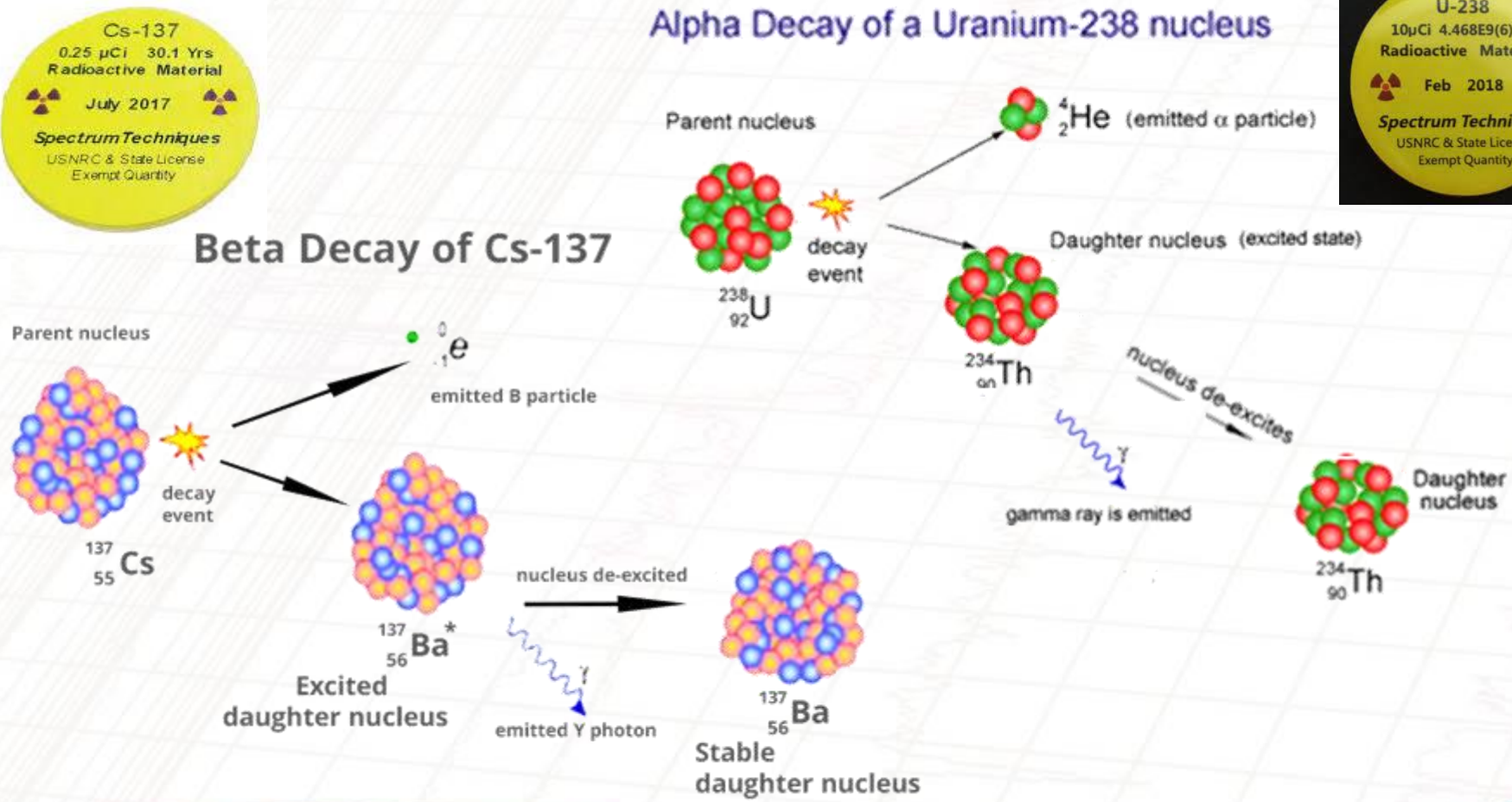


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GAMMA (Photon) DECAY

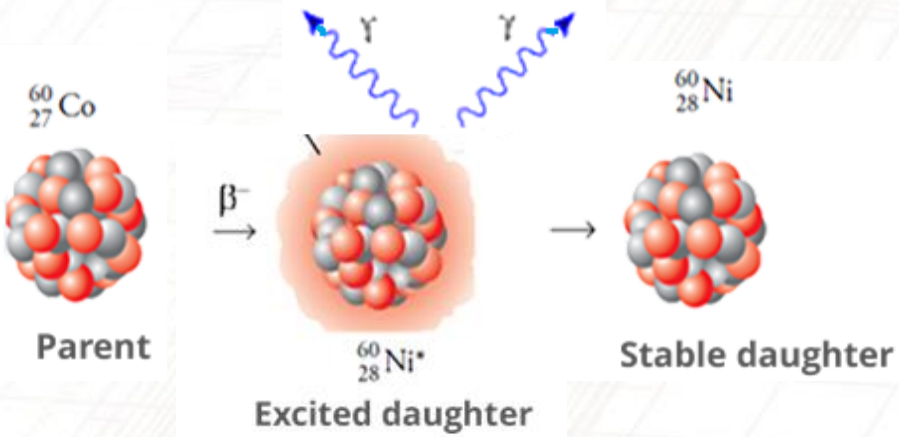
Example:

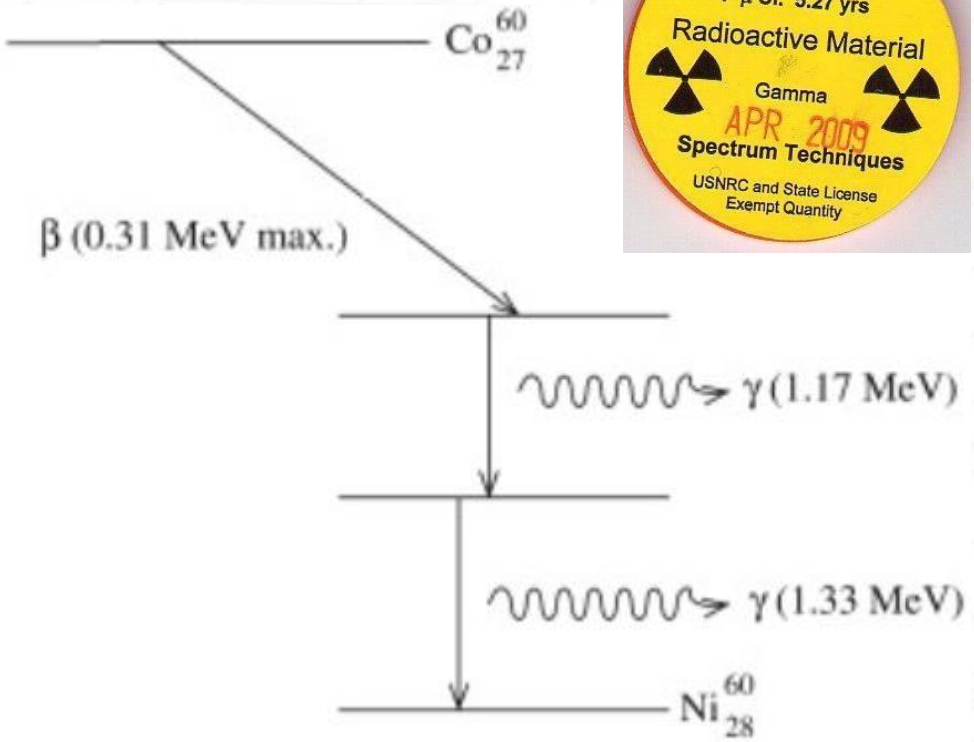


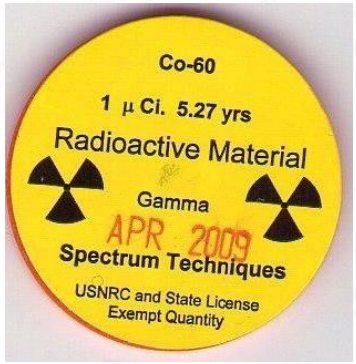


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GAMMA DECAY (γ)



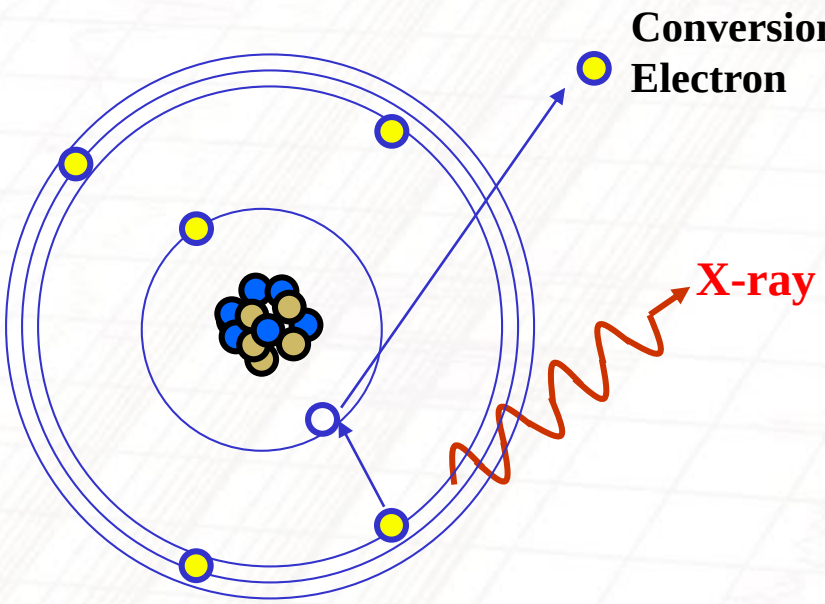




$$^{60}_{27}Co \rightarrow {}^{60}_{28}Ni^{*} + {}^{0}_{-1}\beta + \bar{\nu}$$

Decay of 1 atom of Co-60 emits: 1 β & 2 γ

INTERNAL CONVERSION DECAY



Alternative to gamma emission

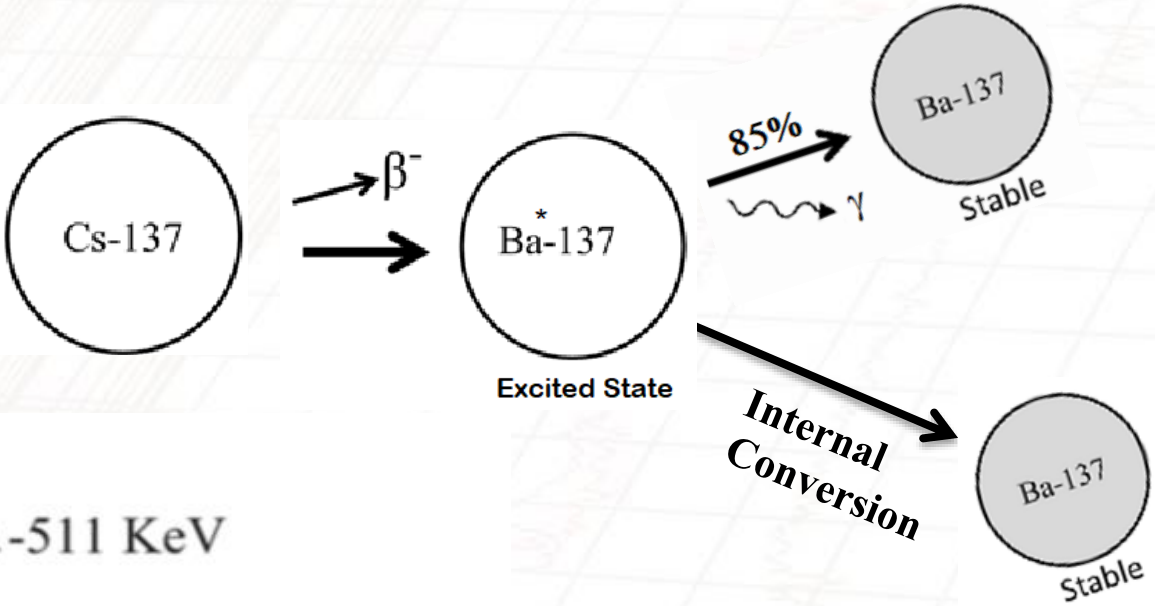
Exited Energy of the nucleus is transferred to one of the orbital electron and immediately ejected.

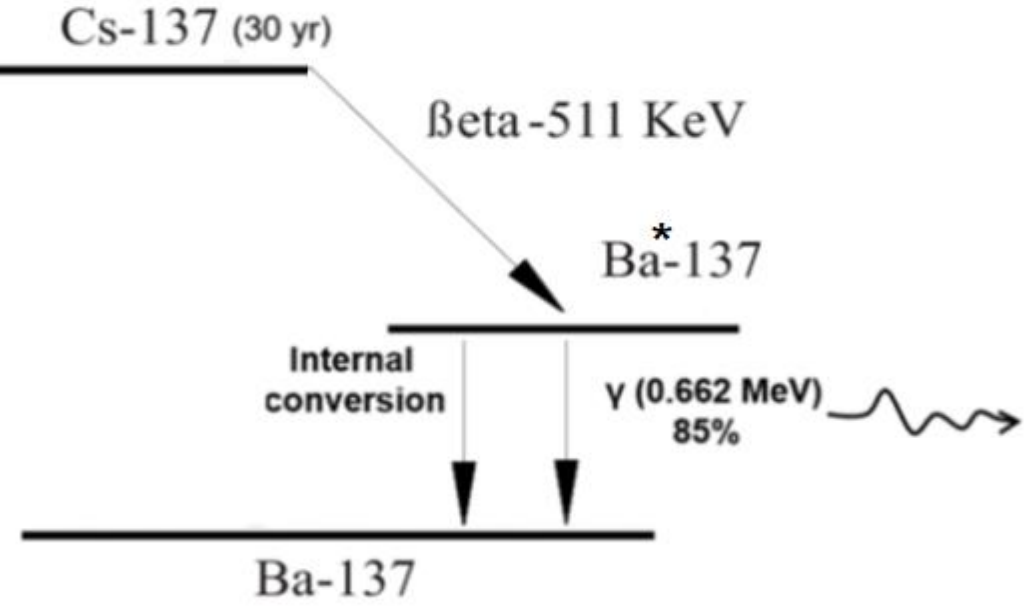
Characteristic x-rays are emitted as outer orbital electron fill the vacancy left by the ejected electron

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GAMMA & INTERNAL CONVERSION DECAY



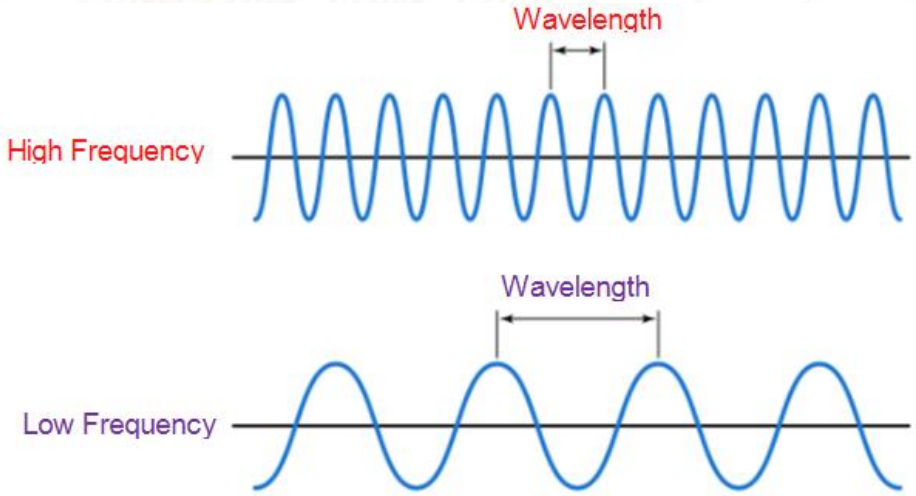




Decay of 1 atom of Cs-137 emits: 1 β & 0.85 γ

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Gamma Radiation



Example:

The wavelength of a gamma ray is $5x10^{-11}$ meter. Calculate the frequency of the ray (speed of light is $3x10^{8}$ m/s).

Solution:

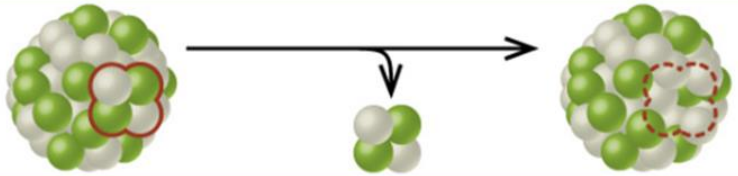
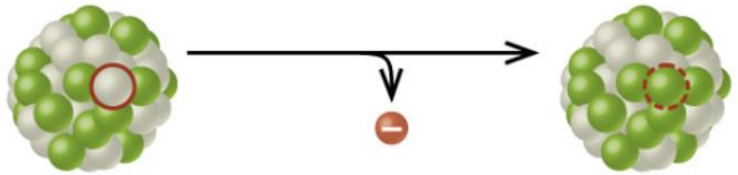
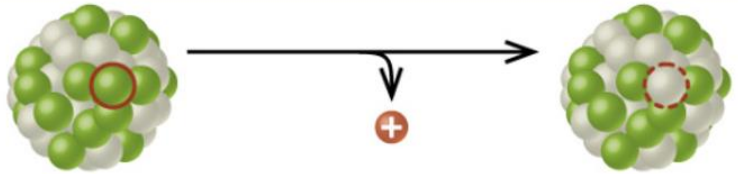
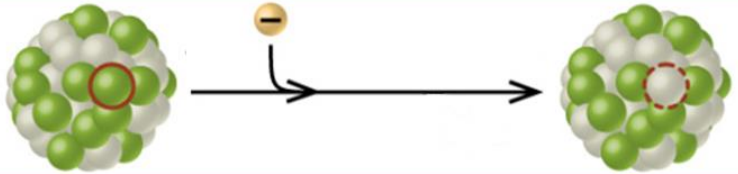
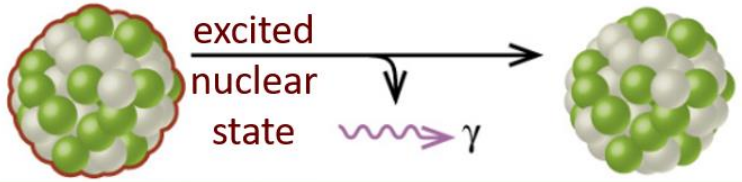
$$Speed\ of\ light = Wave\ length\ x\ Frequency$$

$$3x10^{8}\ \left(\frac{m}{s}\right) = 5x10^{-11}\ x\ Frequency$$

$$Frequency = \frac{3x10^{8}}{5x10^{-11}} = 6x10^{18}\ \ \frac{1}{Sec}\ (Hz)$$

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Summary

Type	Nuclear Equation	Representation
α decay	$^{A}_{Z}X \rightarrow {}^{4}_{2}He + {}^{A-4}_{Z-2}Y$	
β decay	$^{A}_{Z}X \rightarrow {}^{0}_{-1}\beta + {}^{A}_{Z+1}Y$	
β⁺ decay	$^{A}_{Z}X \rightarrow {}^{0}_{+1}\beta + {}^{A}_{Z-1}Y$	
electron capture	$^{A}_{Z}X + {}^{0}_{-1}e \rightarrow {}^{A}_{Z-1}Y$	
γ decay	$^{A}_{Z}X \rightarrow {}^{0}_{0}\gamma + {}^{A}_{Z}Y$	

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Common Sources and Applications

Promethium-147	Beta	Density & Thickness (Soft materials such as paper, plastic,……)
Thallium-204	Beta	
Krypton-85	Beta	
Strontium/Yttrium-90	Beta	
Phosphor-32	Beta	
Cobalt-60	Gamma	Density & Thickness (Hard materials such as steel, concrete, ……) Level Gauging & inspection
Caesium-137	Gamma	
Iridium-192	Gamma	
Americium-241	Gamma	
Americium-241/beryllium	neutron	Humidity (moisture)
Cf-252	neutron	
Iron-55	x-ray	Analysis (Chemical composition)
Cadmium-109	x-ray	
Florine-18	Positron	Imaging (medical)

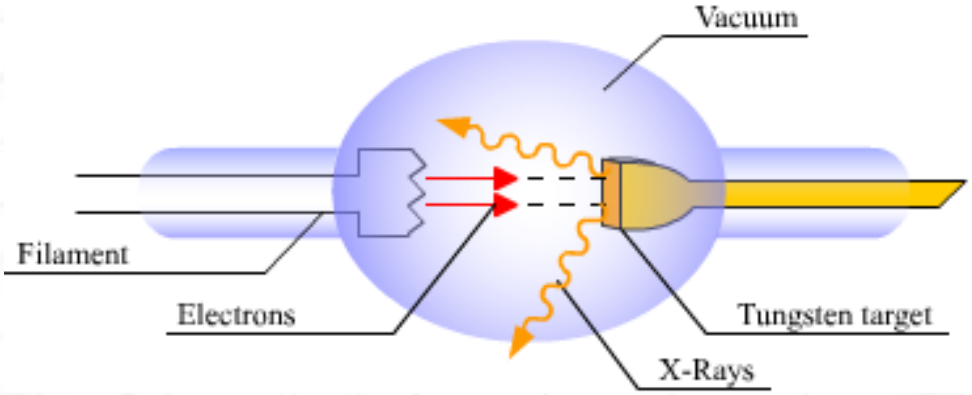
X-rays (Bremsstrahlung or "braking" radiation)

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Steps to produce x-ray:

1. Filament **current** applied through tungsten filament at cathode.
2. Heats up filament to produce enough energy to overcome binding energy of electrons. Electrons released from filament.
3. Tube voltage is applied across the x-ray tube. Electrons, therefore, are accelerated towards positively charged anode, which gives them a certain **energy**.
4. The electrons strike the anode and the energy released via interaction with the anode atoms produces **x-ray photons**.



Increasing the Tube Potential (kV)
Increased : Quantity of x-ray photons
Energy of x-ray beam

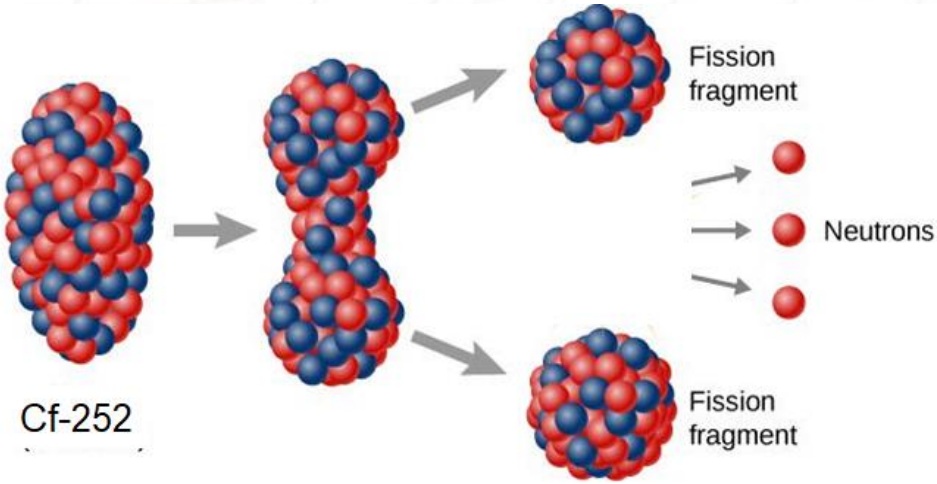
Increasing the Tube Current (mA)
Increased quantity of x-ray photons

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NEUTRON SOURCES

1.Californium-252 Source:

- A strong neutron source decays by both **Alpha** (97%) and **Self Fission** (3%)
- Cf-252 has a half-life of about 2.6 years



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NEUTRON SOURCES

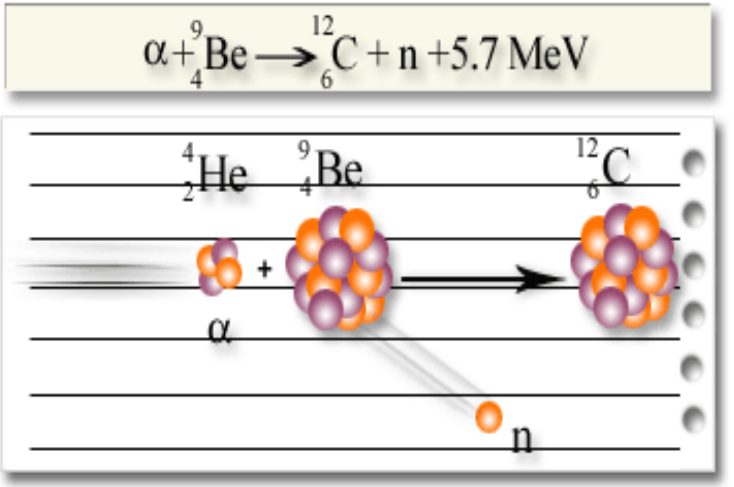
2. (α,n) Source:

- Mixed α source (***^{226}Ra, ^{241}Am, ..)*** with light elements (^{9}Be, Li, ..)
  It produce fast neutrons



Am-241 (α source) **Be-9** **Am-Be** **Am-Be Neutron**

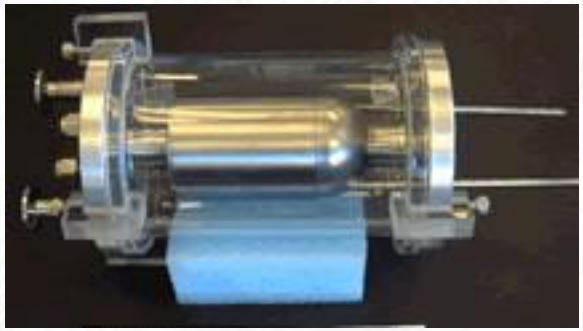
Source	Neutron Energy (MeV)	$T_{1/2}$
$^{226}Ra + {}^{9}Be$	5	1600 Year
$^{241}Am + {}^{9}Be$	5	433 Year
$^{239}Pu + {}^{9}Be$	4	24000 Year

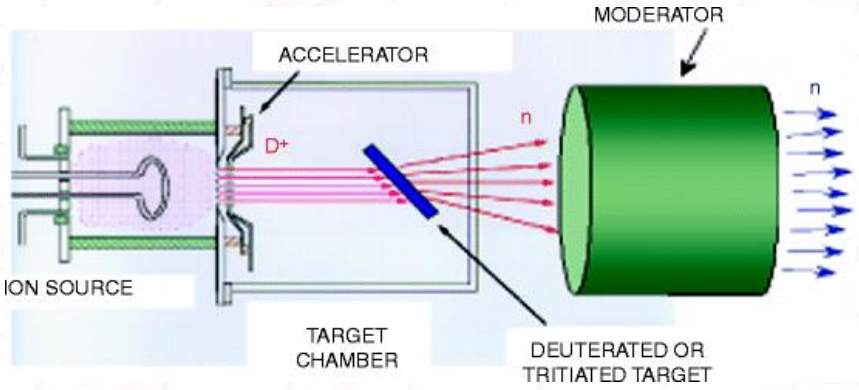


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NEUTRON SOURCES

3. Neutron Generator





D-D Reaction emits neutron with **2 MeV**

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} $$

D-T reaction emits neutron with **14 Mev**

$$ {}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} $$

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ACTIVITY

Radioactivity: It is a spontaneous decay of nuclei to more stable configurations via emission of "radiation".

Activity (A): **It is the number of atoms decay per second .**

Unit

Becquerel (Bq) - It is the activity of a radioactive source which undergoes 1 decay per second.

1 Bq = 1 atom decays per second

Curie (Ci) - It is the activity of a radioactive source which undergoes 37 billion decays per second (3.7×10^{10} dps)

$1\ Ci = 3.7 \times 10^{10}\ Bq$

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CONVERSION BETWEEN Ci & Bq

Prefixes

Name	*Prefix*	*Factor*
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

Example:

$$5\ \text{mCi} = ??\ \text{MBq}$$

$$5 \times 10^{-3} \times 3.7\times10^{10}\ \text{Bq}$$

$$= 5 \times 10^{-3} \times 3.7\times 10^{10}\ \frac{\text{M}}{\text{M}}\ \text{Bq}$$

$$= 5 \times 10^{-3} \times 3.7\times 10^{10}\ \frac{\text{M}}{10^{6}}\ \text{Bq}$$

$$= 185\ \text{MBq}$$

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CONVERSION BETWEEN Bq & Ci

Prefixes

Name	*Prefix*	*Factor*
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

Example:

$$100\ \text{MBq} = ??\ \text{mCi}$$

$$100 \times 10^{6} \times \frac{1}{3.7\times10^{10}}\ \text{Ci}$$

$$= 100 \times 10^{6} \times \frac{1}{3.7\times10^{10}}\ \frac{\text{m}}{\text{m}}\ \text{Ci}$$

$$= 100 \times 10^{6} \times \frac{1}{3.7\times10^{10}}\ \frac{\text{m}}{10^{-3}}\ \text{Ci}$$

$$= 2.702\ \text{mCi}$$

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ACTIVITY

Example: A radioactive source of 1,200,000 decays in 2 minuets. What is the activity of this sample in Bq and microcurie?

A = 1,200,000 /(2x60) = 10,000 dps (decay per second) = 10,000 Bq

$$\mathbf{A} = \frac{10{,}000}{(3.7\times10^{10})} = 2.7\times10^{-7}\ \text{Ci}$$

$$\mathbf{A} = 2.7\times10^{-7}\ \frac{\mu}{\mu}\ \text{Ci} = 2.7\times10^{-7}\ \frac{\mu}{10^{-6}}\ \mathbf{Ci} = 0.27\ \mu\text{Ci}$$

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SPECIFIC ACTIVITY (SA)

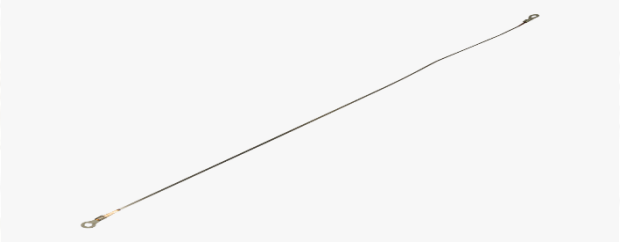
$(SA)_{mass}$ = Activity/Mass [Bq/kg]



$(SA)_{surface}$ = Activity/Area [Bq/m^2]



$(SA)_{Length}$ = Activity/Lenth [Bq/m]



$(SA)_{volume}$ = Activity/Volume [Bq/m^3]



Short "$t_{1/2}$" ⟶ High Specific Activity

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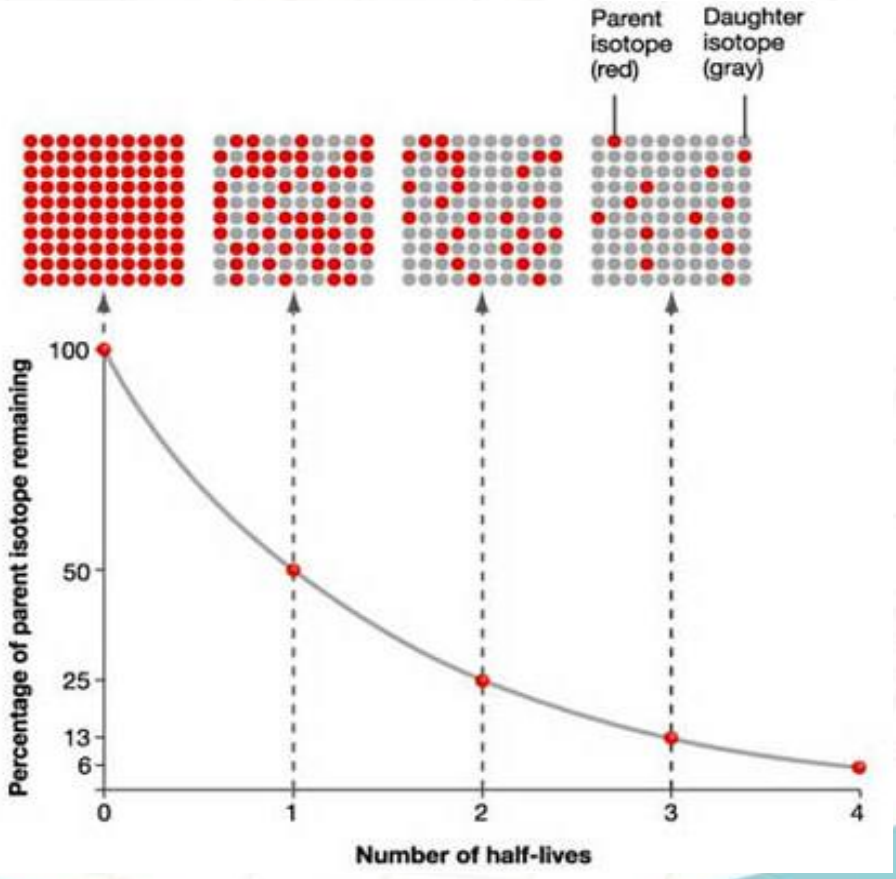
HALF LIFE

Half-life ($t_{1/2}$):

It is the required time for one-half of the initial number of radioactive atoms to decay. OR,

It is the required time for the activity of a radioactive source to decrease by half.

Example: If the activity of Co-60 is 12 mCi ($t_{1/2}$= 5.25 years), then the activity will decrease after 5.25 years to 6 mCi. After another 5.25 years, the activity will be 3 mCi, ……



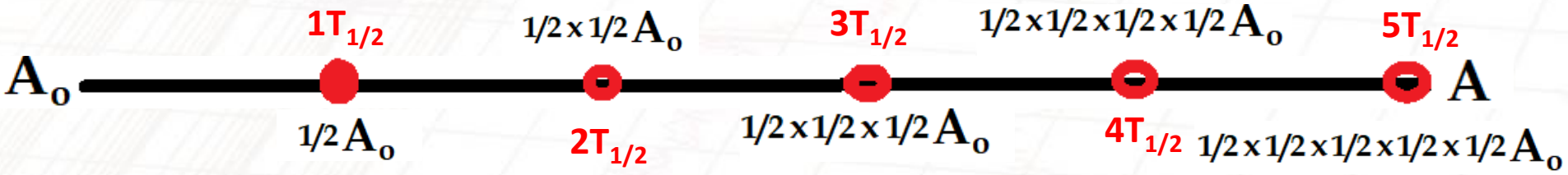
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HALF LIFE

Example: After five half-lives what will be the fraction of initial activity remaining?

(A) $(1/2)^{10}$ (B) $(1/2)^5$

(C) $(1/2)^4$ (D) ½



Solution: It is obvious that after five half-lives, the fraction of the initial activity remaining is given by $A/A_0 = (1/2)^5$

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HALF LIFE

Example: Half-life of an element is 30 days. How much part will remain after 90 days?

(A) 1/4th part (B) 1/16th part

(C) 1/8th part (D) 1/3rd part

Solution:

The half-life is given to be 30 days. Then we know the relation

90 days ÷ 30 days = 3 life-times

$$A/A_0 = (½)^3 = (1/8)$$

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ACTIVITY

- **Activity (A)** is directly proportional to number of radioactive atoms (N) of the radioactive source.

$$A = \lambda N$$

Where:

λ (Decay Constant): **It is the probability that a nucleus will decay in 1 Second.**

Units of λ are typically: $\left(\frac{1}{Sec}\right)$, (sec^{-1}), (per second)

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DECAY CONSTANT & HALF LIFE

$$\lambda = \frac{0.693}{t_{1/2}} \qquad t_{1/2} = \frac{0.693}{\lambda}$$

Example: What is meant by $\lambda = 0.00014\ Sec^{-1}$?

$$\lambda = 0.00014\ sec^{-1} = \frac{14}{100{,}000}\ \frac{1}{sec}$$

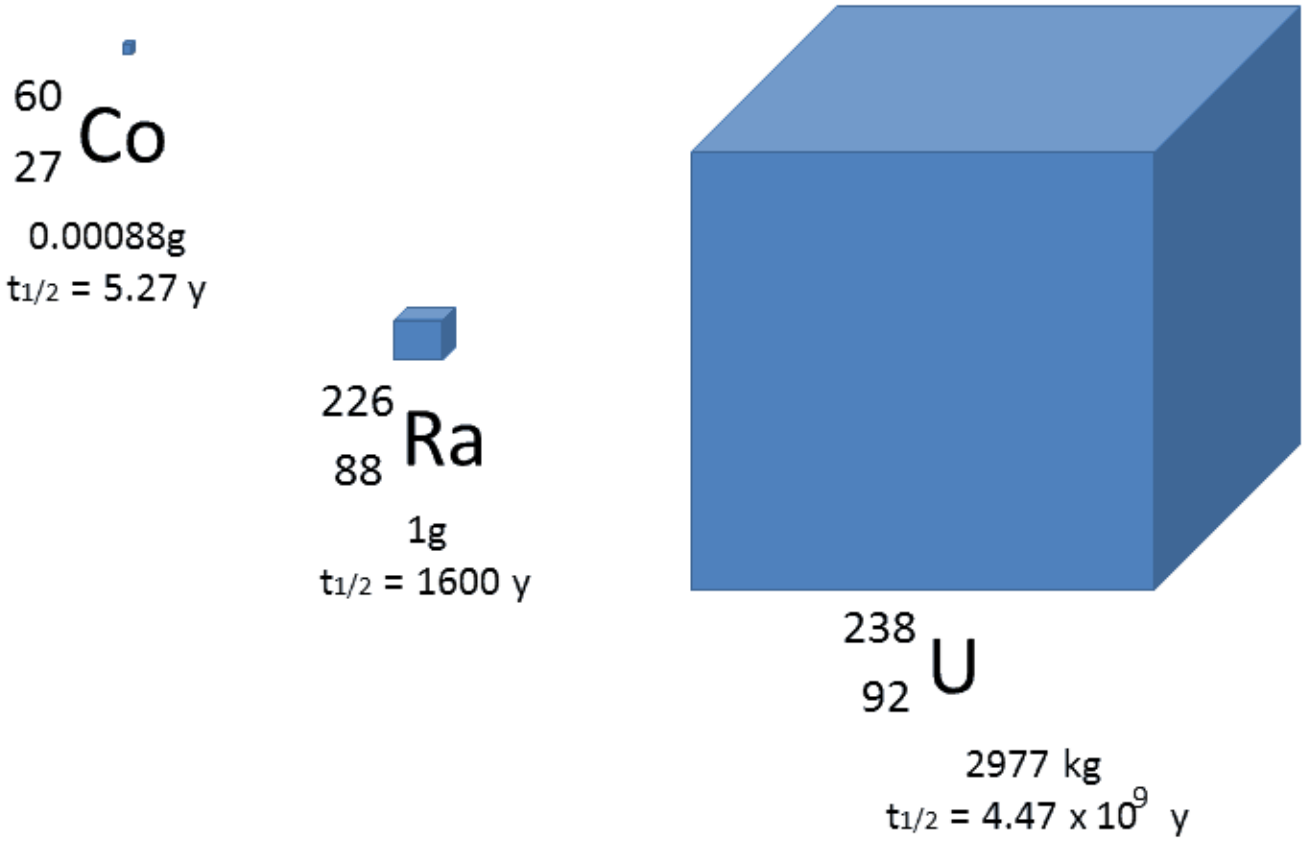
If the radioactive source has 100,000 radioactive atoms, 14 atoms will decay every 1 second.

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ACTIVITY CALCULATIONS

The amount of material necessary for 1 Ci

$1\ Ci = 3.7 \times 10^{10}\ Bq$



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ACTIVITY CALCULATIONS

Example:

A source of Cf-252 contains 10^{16} atoms, what is the activity of the source in MBq and mCi ($t_{1/2}$ = 2.65 years)?

Answer:

$$A = \lambda N$$

$$\lambda = 0.693 / (t_{1/2}) = 0.693/(2.65x365x24x60x60) = 8.3x10^{-9}\ Sec^{-1}$$

$$A = (8.3x10^{-9})x\ 10^{16} = 8.3\ x10^{7}\ Bq = 83\ MBq$$

$$A = 8.3\ x10^{7}\ Bq / (3.7x10^{10}) = 2.24x10^{-3}\ Ci = 2.24\ mCi$$

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ACTIVITY CALCULATIONS

Example:

3.64×10^{20} atoms of ^{248}Cm has activity 0.636 mCi. What is the half-life of ^{248}Cm?

Answer:

$$A = \lambda N$$

A = $(0.636 \times 10^{-3}) \times (3.7 \times 10^{10}) = 2.35 \times 10^{7}$ Bq

N = 3.64×10^{20} atoms

$\lambda = A/N = (2.35 \times 10^{7}) / (3.64 \times 10^{20}) = 6.46 \times 10^{-14}$ sec^{-1}

$t_{1/2} = 0.693/\lambda = 0.693/(6.46 \times 10^{-14})$ $sec^{-1} = 1.07 \times 10^{13}$ sec $= 3.4 \times 10^{5}$ years

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ACTIVITY CALCULATIONS

Example: Determine the activity of mCi and MBq units for a sample containing 10^{17} atoms of ^{137}Cs which has a half-life of 30 years?

Solution:

$$A=\lambda N$$

$\lambda = 0.693/(30\times3600\times24\times365)= 7.325\times10^{-10}\ sec^{-1}$

$N = 10^{17}$ atoms

$A = 7.325\times10^{-10}\ sec^{-1}\times 10^{17}$ atoms $= 73249619$ Bq

$= 73.25$ MBq

$= 1.98\times10^{-3}$ Ci $= 1.98$ mCi

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MEAN LIFE TIME (τ)

- The actual life of any particular atom can have any value between zero and infinity
- The average or mean life of a large number of identical radioactive atoms (τ) :

$$\text{Mean life}\,(\tau) = \frac{\text{Summation of all life of atoms}}{\text{Number of atoms}}$$

$$\tau = \frac{1}{\lambda} \qquad \tau = 1.44\, t_{1/2}$$

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MEAN LIFE TIME

Example:

A radionuclide has a half life of 20 days. What is the mean life?

Answer:

$$\text{Mean Life } (\tau) = 1.44\ t_{1/2}$$

$$= 1.44 \times 20 \text{ days}$$

$$= 28.8 \text{ days}$$

ACTIVITY

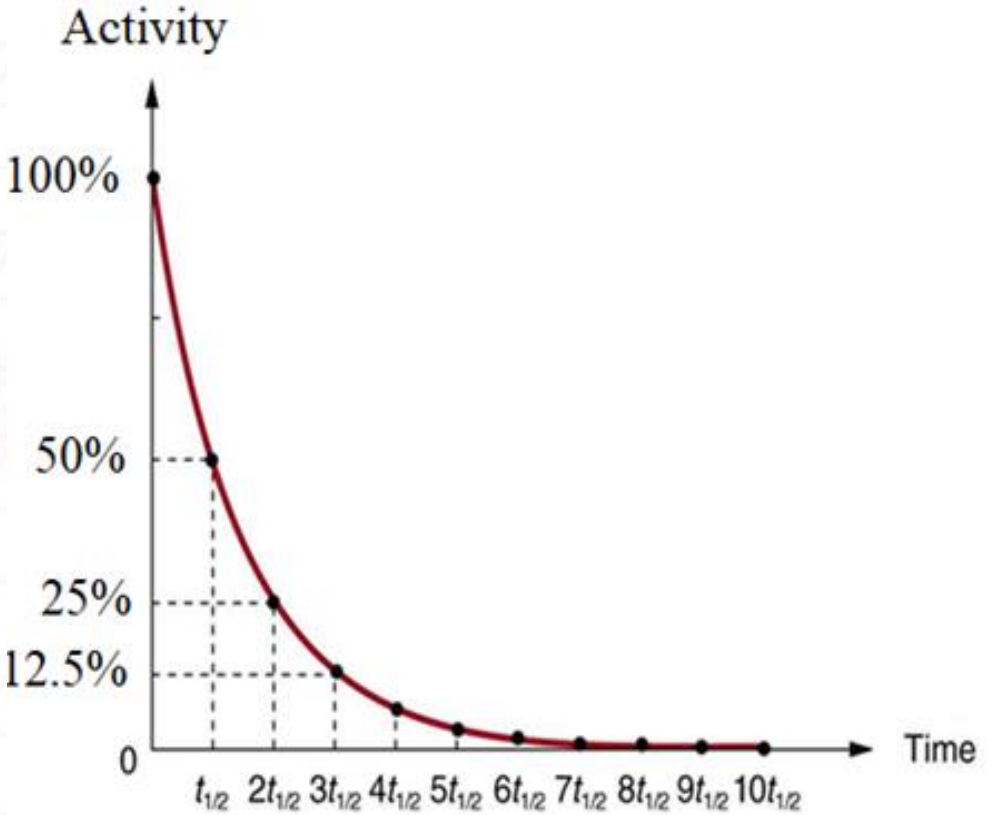
In any sample of radioactive material, the amount of radioactive material, hence the activity, constantly decreases with time because of radioactive decay.

$$A(t) = A_0 (0.5)^{\frac{t}{t_{1/2}}} = A_0 (e)^{-\lambda t}$$

Where:

A_0 : Initial activity

A(t) : Activity after time (t)



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ACTIVITY CALCULATIONS

Example:

We have 500 Ci of Sr-90 today. How many Ci of Sr-90 is left in 100 years? ($T_{1/2}$ of Sr-90 is 28.78 years).

Solution:

$$A(t) = A_0 (0.5)^{t/t_{\frac{1}{2}}}$$

$$\frac{t}{t_{\frac{1}{2}}} = \frac{100}{28.78} = 3.47$$

$$A(t) = 500 (0.5)^{3.47} = 45.1 \text{ Ci}$$

$$A(t) = A(0) \times e^{-\lambda t}$$

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{28.78 \text{ years}} = 0.0241 \text{ yr}^{-1}$$

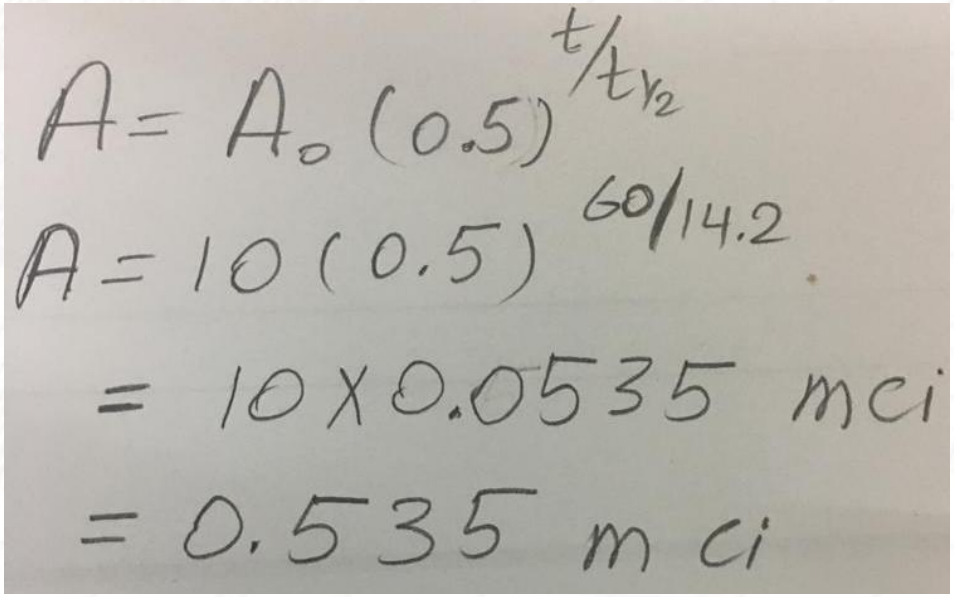
$$A = 500 \text{ Ci} \times e^{0.0241 \text{ yr}^{-1} \times 100 \text{ years}} = 45 \text{ Ci}$$

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ACTIVITY CALCULATIONS

Example:

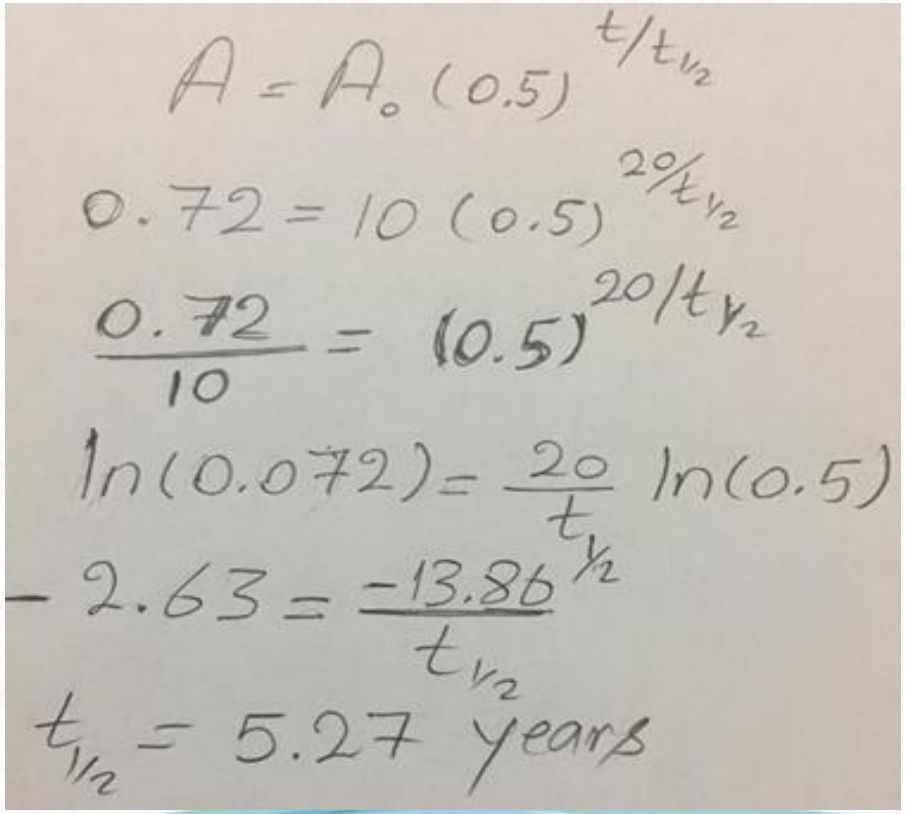
Given 10 mCi of ^{32}P, which has a half-life of 14.2 days, find the quantity remaining after 60 days.



ACTIVITY CALCULATIONS

Example:

20 years ago we received 10 Ci of Co-60. Today, 0.72 Ci remain. What is the $T_{1/2}$ of Co-60?



$$A(t) = A(0) \times e^{-\lambda t}$$

$$\ln\left(\frac{A(t)}{A(0)}\right) = -\lambda t$$

$$\lambda = -\frac{\ln\left(\frac{A(t)}{A(0)}\right)}{t} = -\frac{\ln\left(\frac{0.72\,Ci}{10\,Ci}\right)}{20\,years} = 0.1315\,yr^{-1}$$

$$T_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{0.1315\,yr^{-1}} = 5.27\ years$$

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ACTIVITY CALCULATIONS

Example: A source of 100 Ci needed to perform a specific task, if you know that the time of shipment of the source to reach the work site takes 50 days what is the activity of source to be manufactured (Half-Life 74 Days)?

$$A = A_0\left(\frac{1}{2}\right)^n, \quad n = \frac{t}{t_{1/2}}$$

$A = 100$ Ci, $t_{1/2} = 74$ days

$t = 50$ days, $n = \frac{50}{74} = 0.676$

$$A_0 = A/\left(\frac{1}{2}\right)^n \rightarrow A_0 = \frac{100}{\left(\frac{1}{2}\right)^{0.676}}$$

$A_0 = 159.7$ Ci

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ACTIVITY CALCULATIONS

Example: What will be the ratio of remaining activity of an Ir-192 source of 100 Ci after 3 years (Half-Life 74 Days)?

Ratio of remainig Activity is

$$\frac{A}{A_0} \Rightarrow \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

$$\frac{t}{t_{1/2}} = \frac{3*365}{74} = 14.8$$

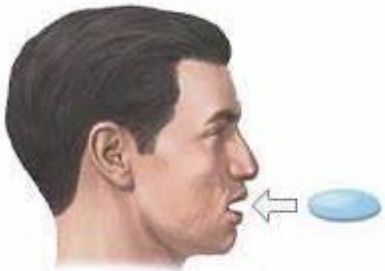
(14.8) means that there are 14.8 time of $t_{1/2}$ in 3 years

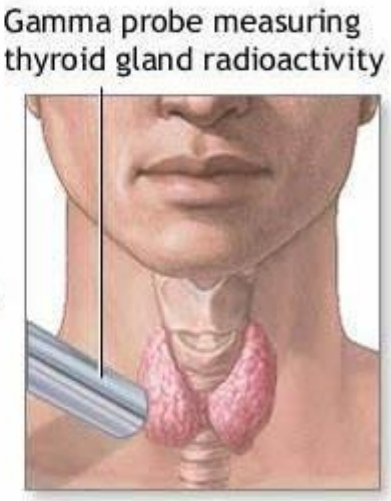
$$\frac{A}{A_0} = \left(\frac{1}{2}\right)^{14.8} = 0.000035 = 3.5 \times 10^{-5}$$

EFFECTIVE HALF-LIFE (t_{eff})









- The activity of the radioactive material in the body decreases by two methods:
    1. Physically (radioactive decay)
    2. Biologically (sweat, urine, feces,,..)

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EFFECTIVE HALF-LIFE (t_{eff})

- **Physical Half-Life** is the time required for a given amount of radioactive material to be reduced to half the initial amount by radioactive decay.
- **Biological Half-Life** is the time required for the human body to eliminate) half of the radioactive material taken into it (for example via urination)

Effective Half-Life is the time required to reduce the radioactivity level to exactly one half its original value biologically and physically.

	Half-life		
	Radiological	**Biological**	**Effective**
H-3	12 years	12 days	12 days
C-14	5560 years	10 days	10 days
P-32	14 days	257 days	14 days
S-35	87 days	90 days	44 days
Co-60	5 years	10 days	9.5 days
Sr-90	28 years	50 years	18 years
I-131	8 days	138 days	7.6 days
Po-210	138 days	60 days	42 days
Ra-226	1620 years	45 years	44 years

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EFFECTIVE HALF-LIFE (T_{eff})

Example: The physical half-life of iodine-131 is 8 days, the biological half-life of the thyroid is 138 days. Calculate the effective half-life of I-131?

$$t_{eff} = \frac{t_{bio} \times t_{ph}}{t_{bio} + t_{ph}} = \frac{138 \text{ x } 8}{138 + 8}$$

$$t_{eff} = 7.56 \text{ days}$$

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Technology Experts Company

Radiation Protection Center

Radiation Protection Training Course

Prepared by:

Eng. Ashraf Mohammed Enany

Master of Science in Nuclear Engineering

Head of Radiation Protection Center

Mobile : +966 568360241

Email : ashraf@ t-experts.com

Website : www.t-experts.com

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Interactions of ionizing Radiation with Matter

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Interactions of Radiation with Matter



What do we mean by "Radiation" ?

Radiation is **ENERGY** transmitted in the form of Waves (γ, x-rays) or Particles (α, β, neutrons).

How does Radiation "see" Matter ?

Matter is a huge amount of charged particles (negative orbital electrons and positive nuclei)

What do we mean by "Interactions" ?

Interactions: The ways that Radiation **loses** its energy inside the matter.

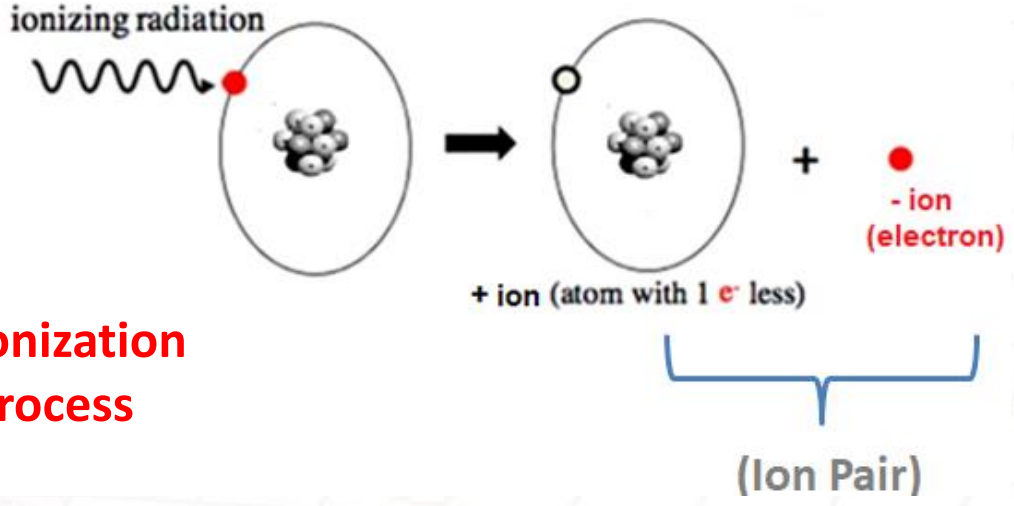
OR,

How a material **absorbs** the energy of incident radiation.

Ionizing Radiation

α , β , γ , x-rays and neutrons are classified as IONIZING RADIATIONS.

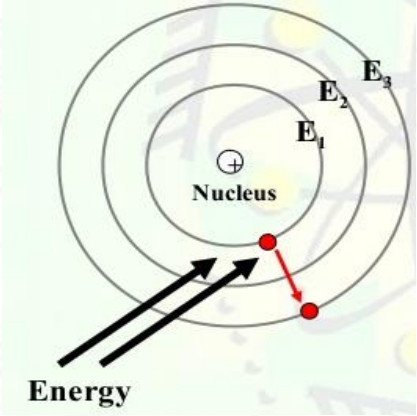
They have enough energy to remove an electron(s) from an atom or molecule, causing it to become ionized.



Ionization Process

Average Energy to create one ion pair in air, water and human body is 34 eV.

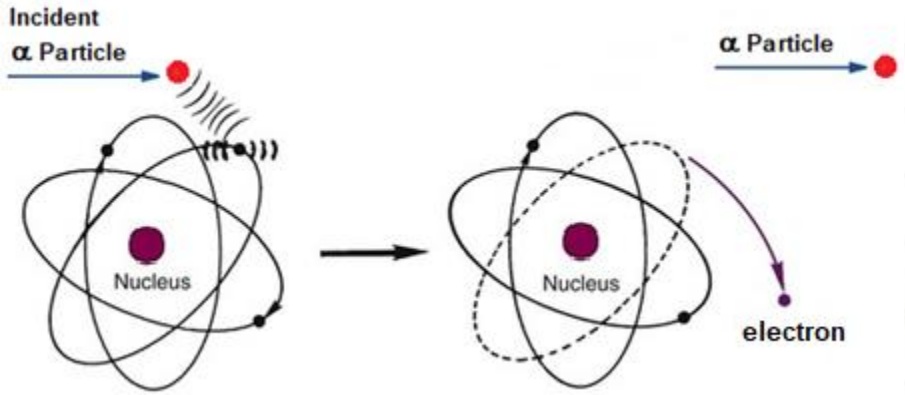
Excitation Process



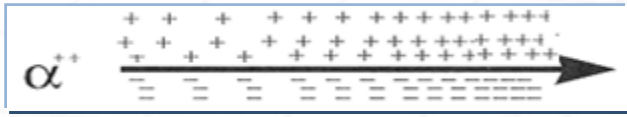
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Alpha Particles

- Heavy nuclear particle (a helium nucleus), doubly positively charged, relatively slow.
- **α particles** loss their energy **mainly** by **ionization** (as they interact with orbital electrons in the material, knocking them out of their atoms) and by **excitation** (by pulling inner orbital electrons to outer orbits).



- **α particles** travel within the material in a **straight line** due to their very large mass compared to the mass of electrons that interact with them during their penetration of matter.



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Alpha Particle Reaction

– The distance traveled by alpha particle before stopping is very short ➡ It loses all of its energy in a very short distance ➡ it creates many ion pairs in very short distance . ➡ **High LET (Linear Energy Transfer)**

5 MeV alpha particle penetration depths.

Absorber	Alpha Range	Comments
air (STP)	3.7 cm	
paper	53 μm	one sheet = 89 μm
water (soft tissue)	45 μm	will not penetrate skin

– **α does <u>not</u> constitute an external biological hazard.**

Linear Energy Transfer (LET): The amount of energy that an ionizing particle transfers to the material traversed per unit distance.
Unit: (Mev/cm, keV/μm, ..)

Example: LET = 150KeV/μm
It means that α-particle transfers 150 KeV of its energy to the medium in each μm by ionization and excitation processes

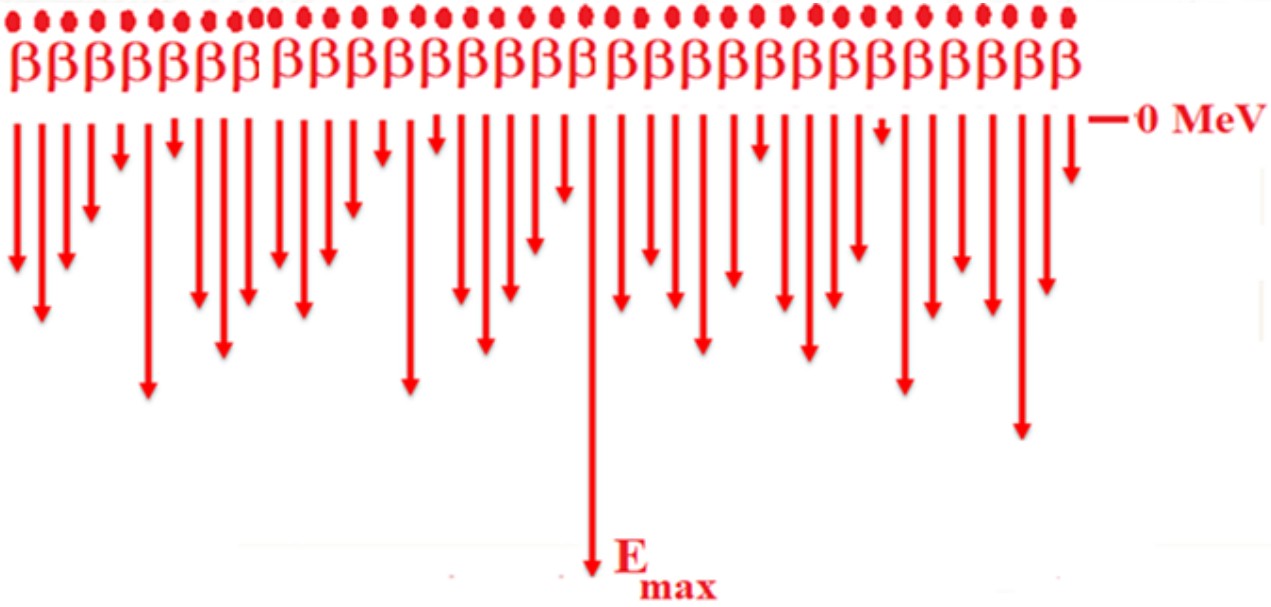
Shielding of Alpha Sources

- Thin piece of cardboard and plastic can be used to place alpha sources inside without danger from external exposure, where the thickness of a fraction of a millimeter of this material is enough to absorb the alpha particles.

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Beta (- or +) particle

- Physically, it is an electron but may be positively or negatively charged
- Emitted with a wide range of energies up to some maximum value with a continuous spectrum of energies for a given radionuclide.



Energy distribution of beta particles

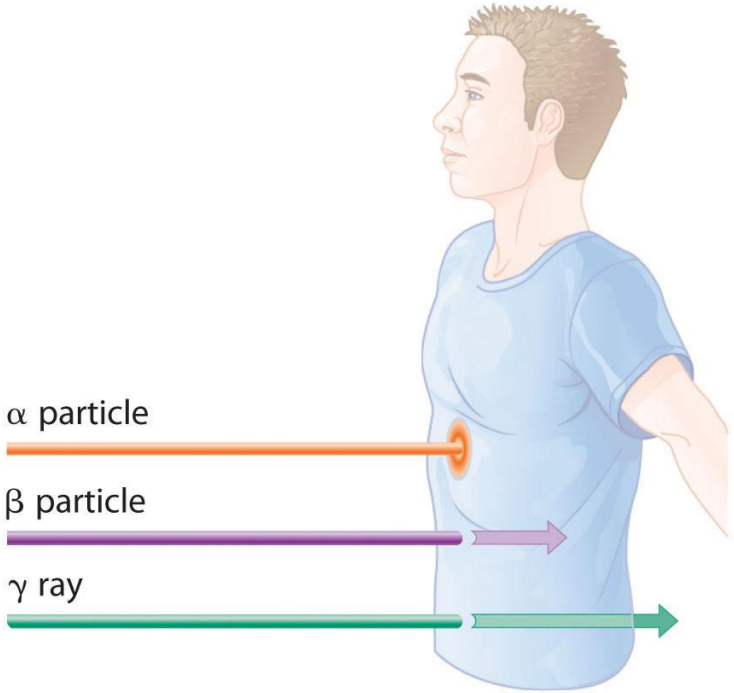
Beta (- or +) Particle

– Its range in matter depends on the **Energy** and the **Material type**.

More penetrating than alpha particles. For example:

(0.5) MeV of Beta particle has a range of 2 meters in air, 3 mm in tissue

β represents both external & internal hazards where β can penetrate till 1 cm in living tissue.



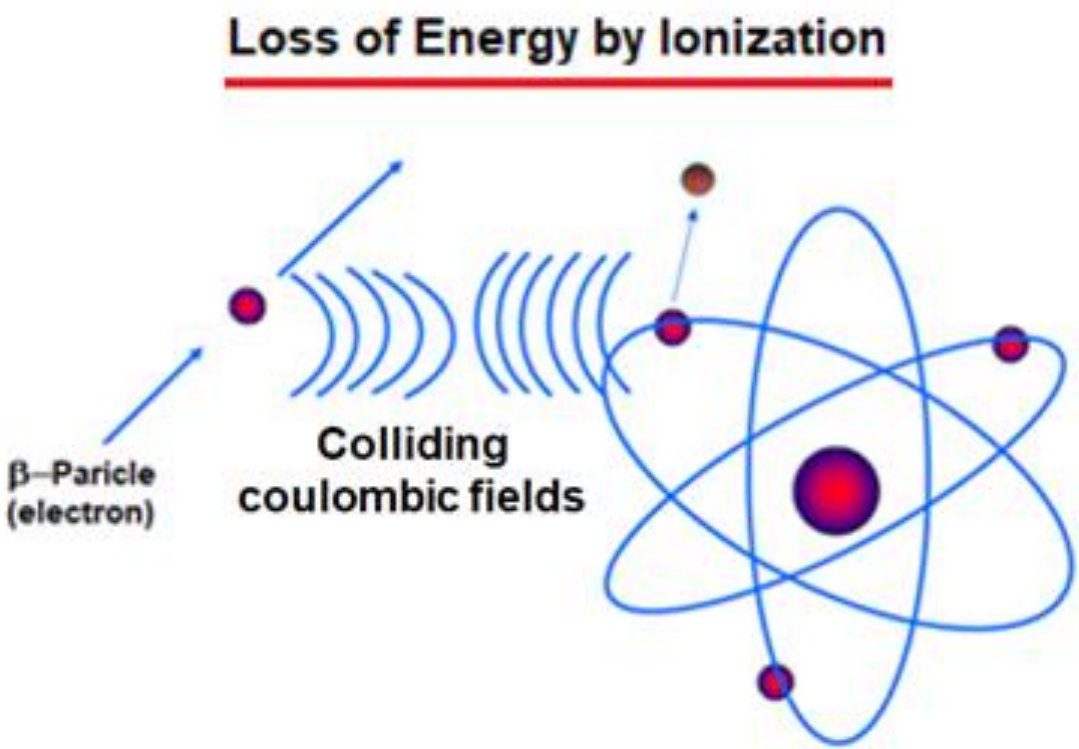
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Beta (- or +) particle

- Betas interact (loss of its energy) with the material by:

1. Beta with low energy (up to 0.5 MeV)

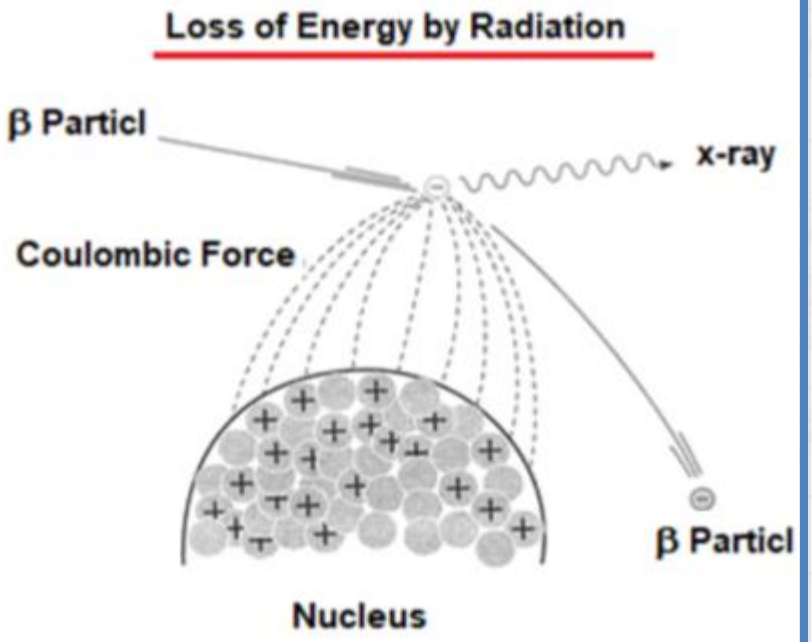
The main interaction is by **ionization and excitation** with the orbital electrons of the atoms.

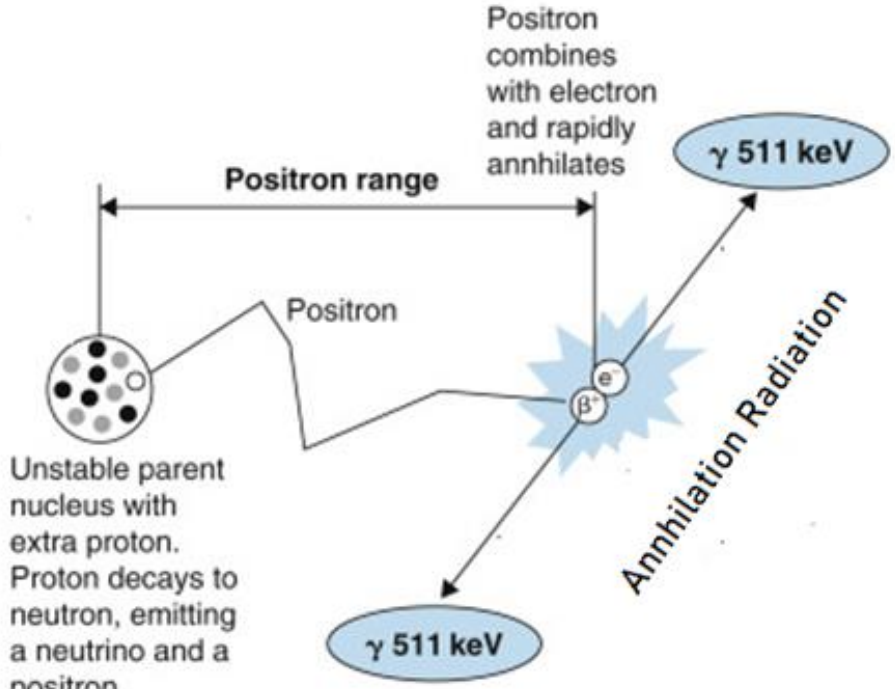


Beta (- or +) particle

2. Beta with high energy (> 1 MeV) ,

The main interaction is by emitting **X-rays (Bremsstrahlung) (braking radiation)** as a result of changing its direction when it becomes close to the **nucleus**.





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Shielding of Beta Sources

- Because of β is charged particle, β has a definite range (required distance to be **stopped**).
  - **Low Z materials** such as plastic (Perspex) or aluminum are common to avoid / reduce the production of Bremsstrahlung .

Radionuclide	Max energy (keV)	Range (cm) in air	water	aluminium
H-3	18.6	4.6	0.0005	0.00022
C-14	156	22.4	0.029	0.011
P-32	1700	610	0.79	0.29





- For **high β energy**, two layers is used as a shield. Inner layer material is low Z and outer layer with high Z material.

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Shielding of Beta Sources

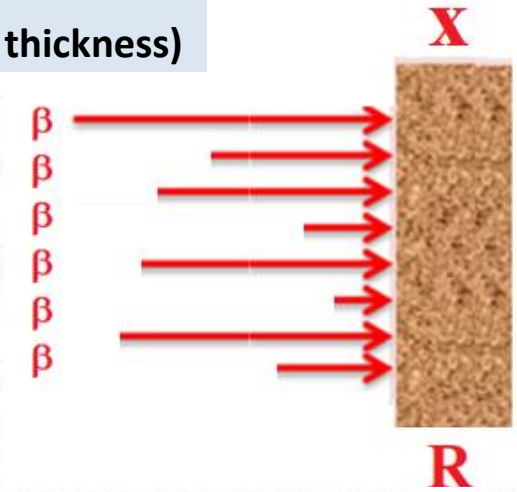
- The required thickness (**X**) of shielding material for β sources depends only on the **source (energy of β)** , and **material (Z)**.

R (maximum travelling distance in the shield) = **X** (minimum required thickness)



Linear Thickness

Mass Thickness





$$X_m = X\rho \quad (g/cm^2)$$

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Range of Beta Particle

The empirical range (g/cm^2) formulas for beta particles are:

$$R = 0.412(E_{\beta\ max})^{1.265-0.0954\ ln(E_{max})}$$ **For $E_{max} < 2.5$ MeV**

$$R = 0.530(E_{max}) - 0.106$$ **For $E_{max} > 2.5$ MeV**

- As we see, the range depends only on the maximum energy of beta particles and the type of material and does not depend on activity of the source.
- Example of pure β Sources: Sr-90 , P-32 , C-14

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Range of Beta Particle

Example: Find the minimum mass thickness required to make a shield for the Sr- 90 source. Then calculate the linear thickness of this shield when using an aluminum shield (aluminum density 2.7 g/cm^3 and maximum energy of Sr-90 = 2.284 MeV)

Solution: $R = 0.412(E_{max})^{1.265-0.0954\ \ln E_{max}}$

$$R = 0.412 \times (2.284)^{(1.265 - 0.0954 \ln 2.284)} = 1.09 \ g/cm^2$$

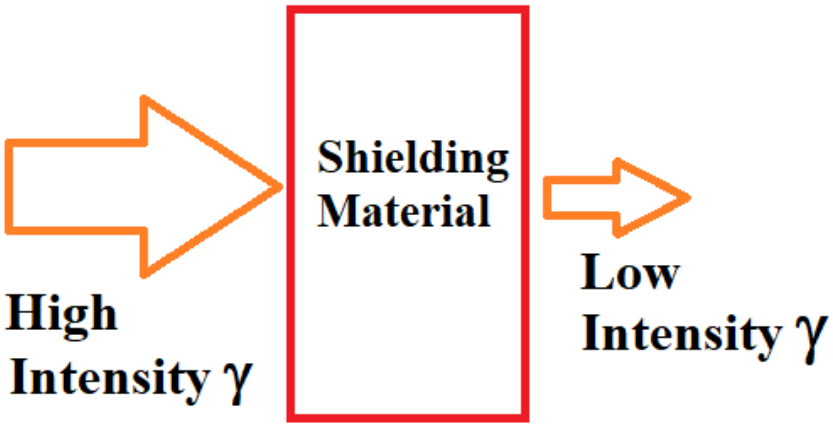
To find the minimum linear thickness, use the relationship:

$$X = X_m / \rho = 1.09 / 2.7 = 0.41 \ cm$$

- The thickness of shielding material for β (completely stopped β) **does NOT** depend on the Activity of the β source.

Gamma Rays

- High energy electromagnetic waves (photons) travel in straight line at the speed of light.
- Have no mass or charge.
- Able to travel significant distances. Gamma can travel tens or hundreds of meters in air.
- Gamma rays do not have a definite range, but instead the intensity of a gamma ray beam is attenuated by a combination of scattering and absorption processes so that it falls off steadily with distance.

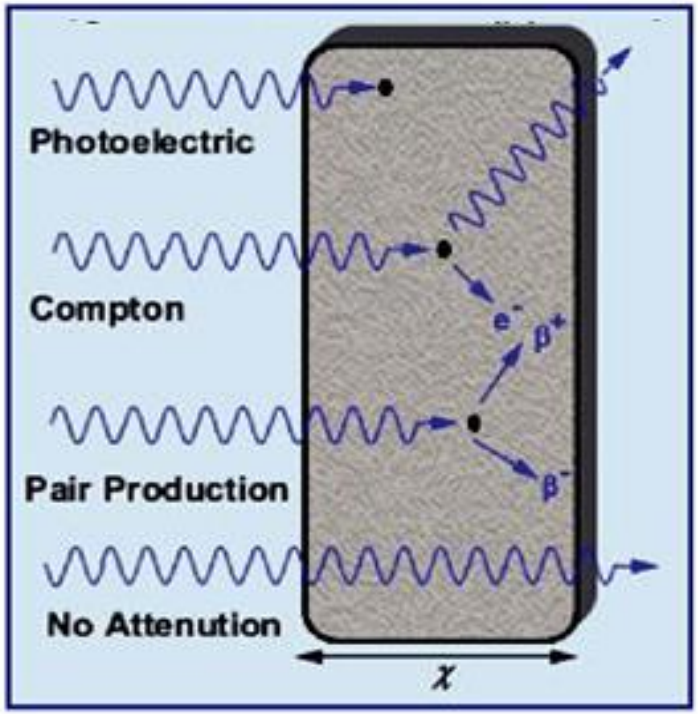


Attenuation
Reduction of the intensity (number) of incident radiation

Gamma Rays

- The methods of transmission of energy from gamma radiation (as well as x-ray) to the material differ significantly from those transmitted from the charged particles to the material.
- Gamma rays (x-rays) interact with the material and thus lose its energy through one of three main processes:

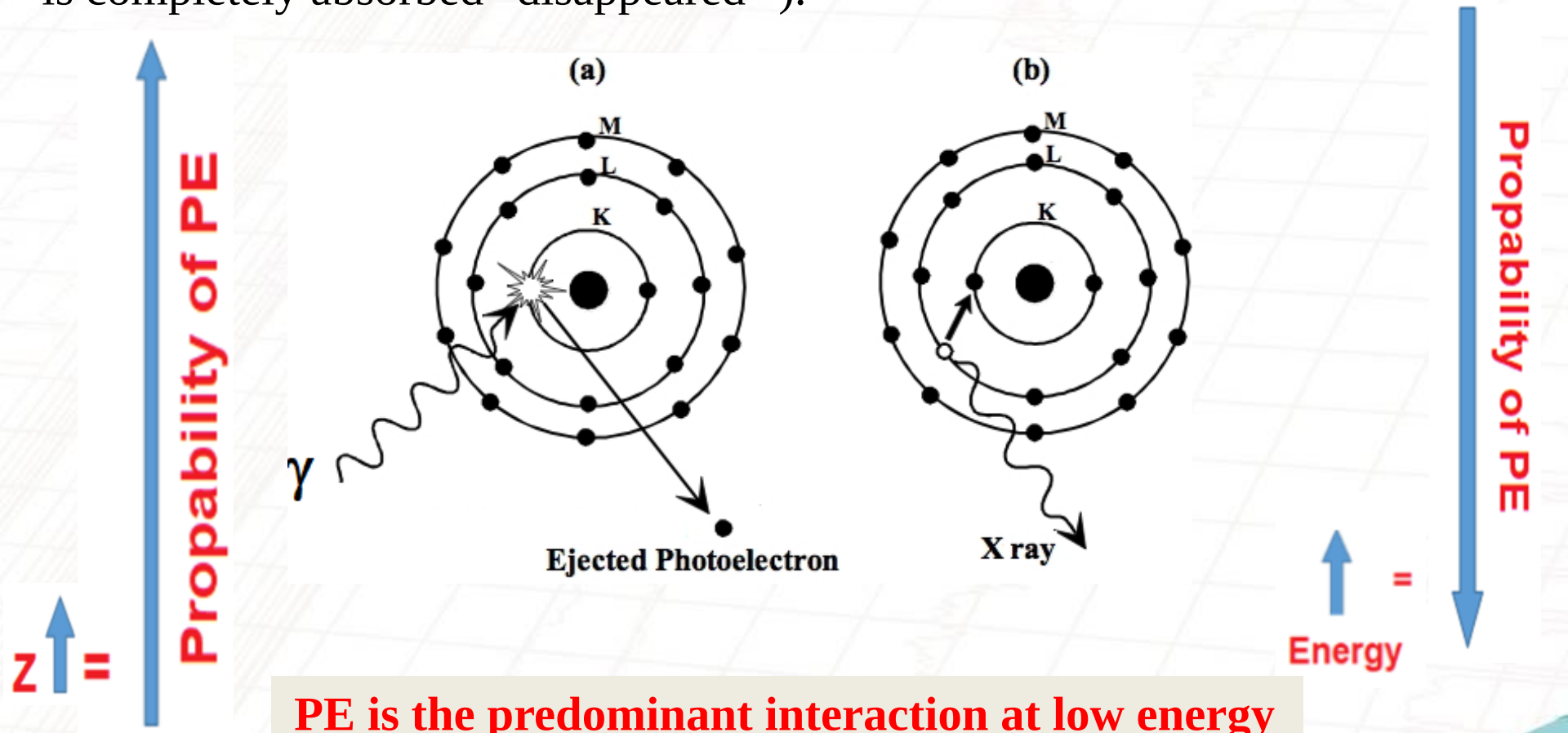
1. **Photoelectric effect.**
2. **Compton effect (scattering).**
3. **Pair production.**



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1. Photoelectric Effect

Gamma ray photon transfers **ALL** of its energy to an inner electron (gamma is completely absorbed "disappeared").

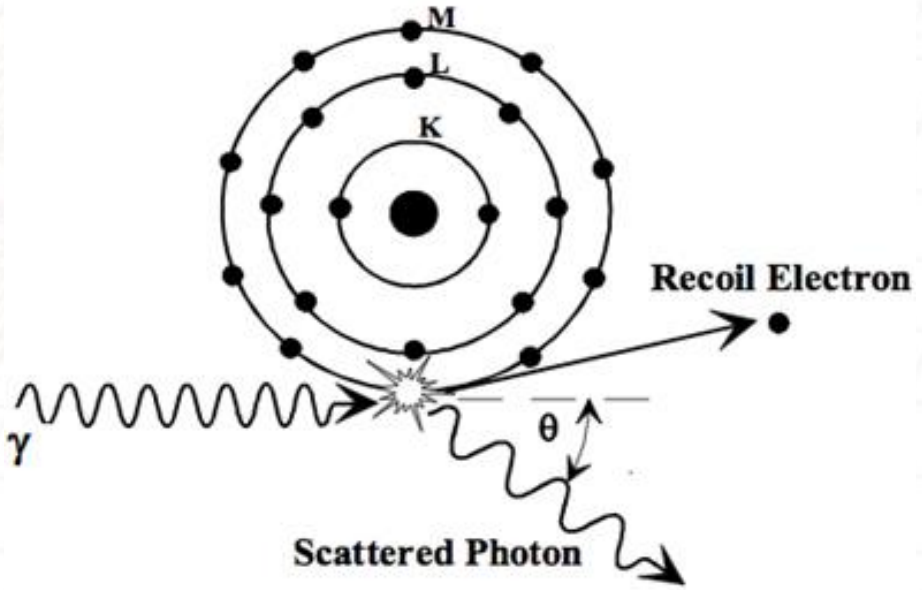


PE is the predominant interaction at low energy

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2. Compton Scattering

- The gamma photon transfers **PART** of its energy to one of the **free** or **weak** electrons in the atom.
- The electron is ejected (recoil electron) and the photon is scattered with a lower energy



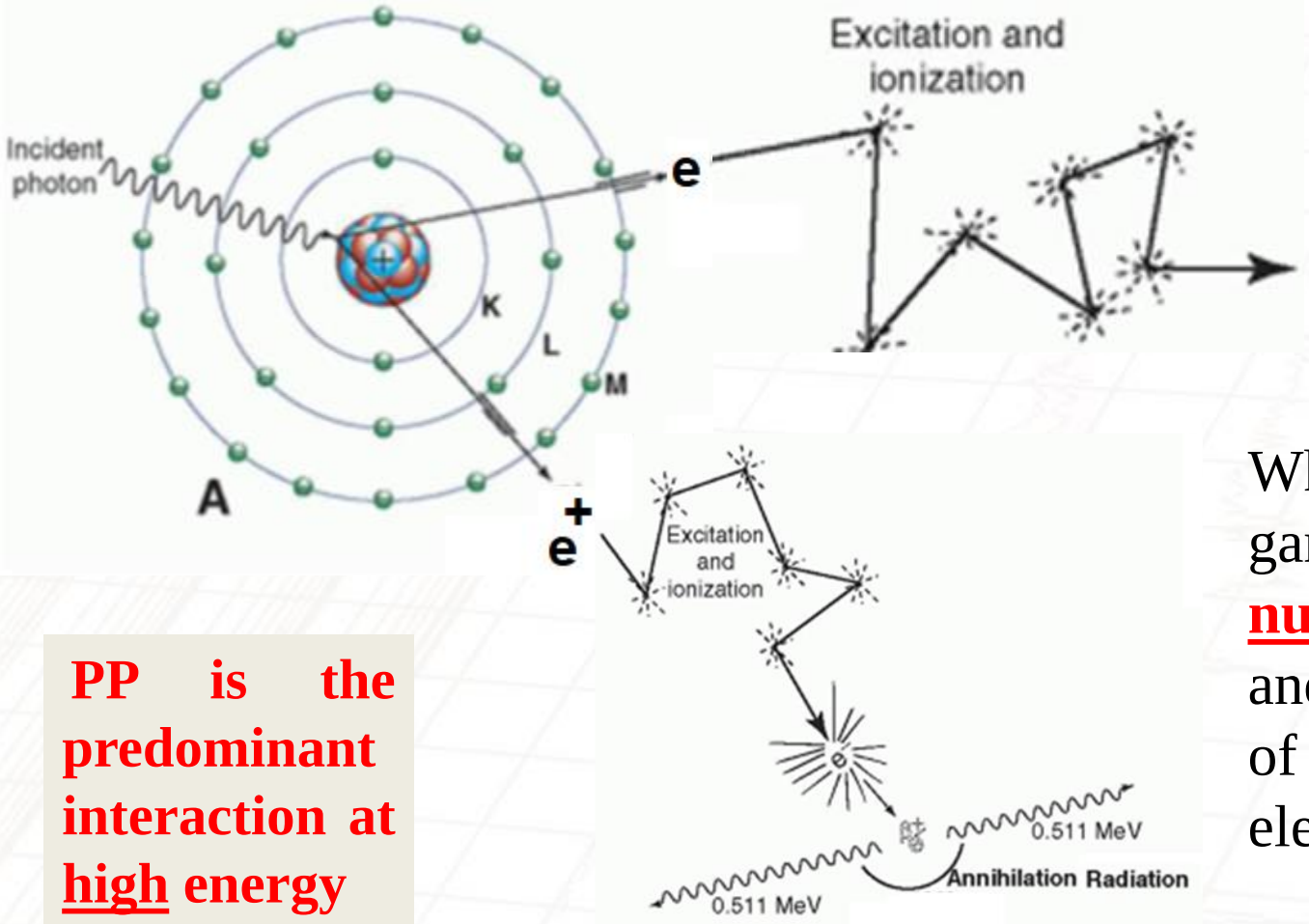


CS is the predominant interaction at intermediate energy

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3. Pair production

– **The photon of gamma must have energy of <u>at least 1.022 MeV</u>.**



PP is the predominant interaction at <u>high</u> energy

When a photon of gamma passes near the **<u>nucleus</u>**, it disappears and converts into a pair of particles - an electron and a positron.

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Attenuation of Gamma

- Attenuation of γ depends on:

1. Energy (Source).

Low Energy E< 300 keV	Intermediate Energy 0.5 - 5 MeV	High Energy E > 5 MeV
Photoelectric $(\propto 1/E^3)$	Compton Scattering	Pair Production

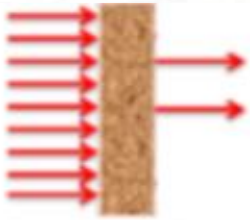
2. Material (Z).

Photoelectric	Compton Scattering	Pair Production
$\propto Z^4$	$\propto Z$	$\propto Z^2$

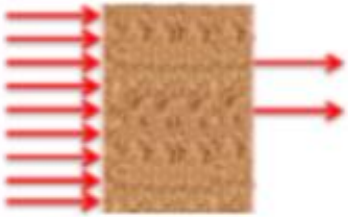
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Thickness of γ shielding

Materials (Z)



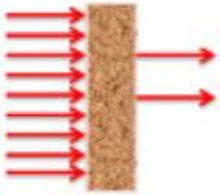
Lead
High Z



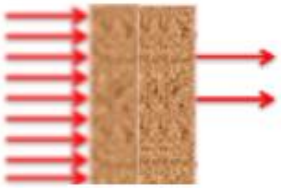
Concrete
Low Z
More Thickness

Energy (Source)

Low
Energy



intermediate
Energy

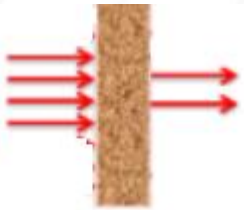


High
Energy



Activity of the source

Low
Activity



High
Activity



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Attenuation of Gamma

- **$X_{1/2}$ (half value thickness)** is the thickness of the shielding material at which the intensity of radiation entering it is reduced by **one half.**

Energy	Approximate Half Value Layers in cm				
	Uranium	Tungsten	Lead	Iron	Concrete
Ir-192	0.28	0.33	0.48	1.27	4.5
Cs-137			0.65	1.6	4.8
Co-60	0.69	0.79	1.2	2.1	6.2
Ra-226			1.66	2.2	6.9

- **Tenth value thickness ($X_{1/10}$)** . It is the thickness of the material at which the intensity of radiation entering it is reduced by one tenth.

$$X_{1/10} = 3.32\, X_{1/2}$$

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Attenuation of Gamma

- **μ** (linear attenuation coefficient), is the probability of an interaction per unit distance traveled (unit cm^{-1}).
- μ depends on photon energy and on the type of material (Z).

$$\mu = \frac{\ln 2}{X_{1/2}} = \frac{0.693}{X_{1/2}}$$

– μ_m (mass attenuation coefficient) is equal to the linear attenuation coefficient divided by the density of the attenuation material.

$$\mu_m = \frac{\mu}{\rho} \ (cm^2 / g)$$

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Attenuation of Gamma

Example: If a material has a half-value layer of 3 mm, what is its linear attenuation coefficient ? What is the mass attenuation coefficient? (density of material is 8 g/cm^3)?

Solution:

$$\mu = \frac{0.693}{X_{1/2}} = \frac{0.693}{3}$$

$$= 0.231 \ \frac{1}{mm} = 2.31 \ \frac{1}{cm}$$

Mass attenuation coefficient

$$\mu_m = \frac{\mu}{\rho} \ (cm^2 / g)$$

$$= \frac{2.31}{8} = 0.289 \ \frac{cm^2}{g}$$

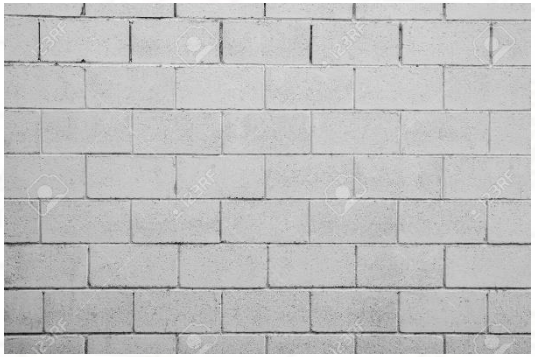
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Shielding of Gamma Sources

- Materials with **a high atomic number are used as shielding materials for gamma rays (and X-rays)** such as depleted uranium, lead and tungsten, mainly because the potential of the photoelectric effect increases dramatically as the atomic number of the shield material increases.

- The thickness of the shielding material of gamma sources depends on the **material type (Z)** , **type of source (Energy)** and **activity of the source**.







Neutrons

- Neutrons are uncharged particles:
  - No interactions with atomic electrons of material.
  - Interactions with the nucleus only.

– If the neutron does not react with the nucleus of the atom, the matter is like a vacuum. That is why neutrons are highly penetrating.

- Interactions of the Neutron depend strongly on its energy.
- Neutrons are classified according to the energy to:

1. Thermal 2. Fast

- Fact of neutrons " Neutrons are born fast".

Interactions of Neutron

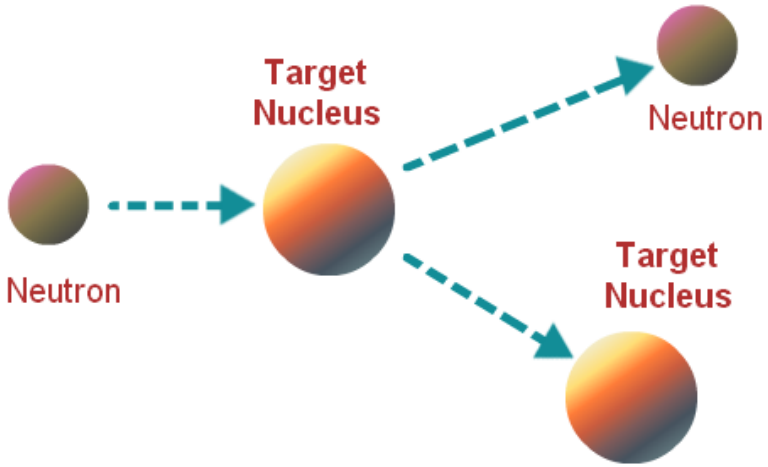
– Neutron interacts with the nucleus in several ways, but the most important interactions, from the radiation protection point of view, are:

1. **Elastic scattering**
2. **Radiative capture (absorption)**

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Elastic Scattering (collision)

- Elastic scattering is a billiard ball type of collision between a neutron and a nucleus.
- In this process, energy lost by the neutron is transferred to the recoiling nucleus.





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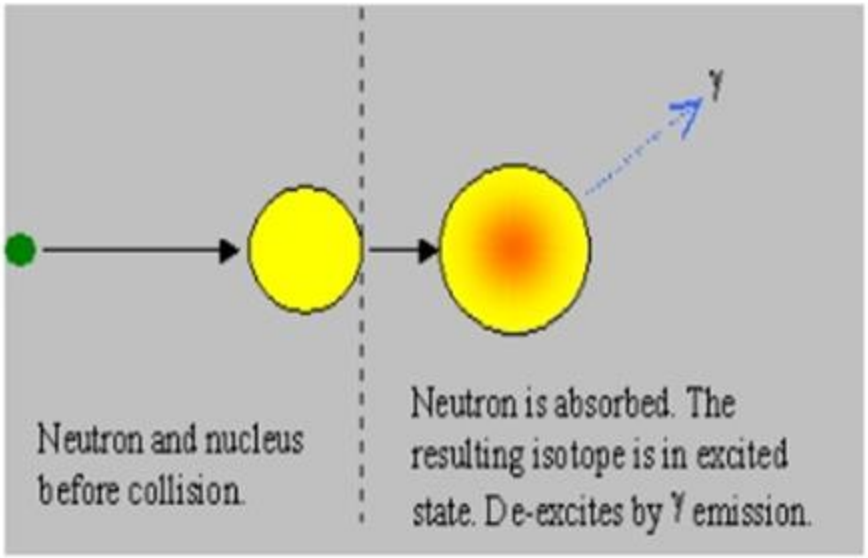
Elastic Scattering (collision)

- Elastic scattering is the most important process for slowing down neutrons.
- The hydrogen nuclei, (which are protons) are considered the **best moderator** for neutrons, and the materials which contain high concentration of hydrogen, such as wax, water, Polyethylene, and plastic are extensively used for effective slowing down of the fast neutrons.
- **Example**: In a single collision with a hydrogen nucleus (protons), the fast neutron looses, on average, 63 % of its energy.

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Radiative Capture

- The neutron is absorbed by the target nucleus to form the next higher isotope (of mass A+1), in an excited state of energy. The new isotope de-excites by emitting gamma rays The neutron is thus lost in this reaction.



$$ {}^{1}_{0}\mathrm{n} + {}^{114}_{47}\mathrm{Cd} \rightarrow {}^{115}_{47}\mathrm{Cd}^{*} \rightarrow {}^{115}_{47}\mathrm{Cd} + \gamma $$

Moderation of Neutron

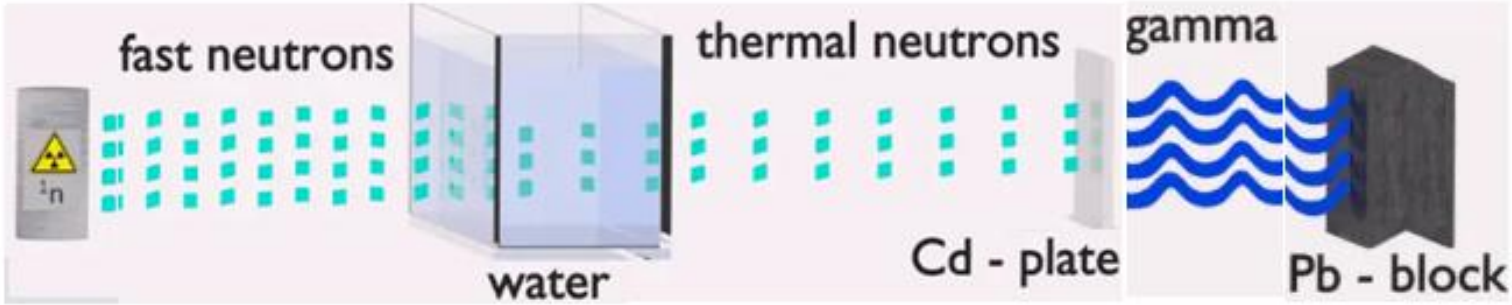
Example:

- Neutron loses an average of 63% of its energy in each collision with the hydrogen nucleus.
- Neutron loses an average of 0.5% of its energy in each collision with the lead nucleus.
- When using normal water as moderator, it is enough to use a layer of water of **20-25 cm** to convert the fast neutron into a thermal neutron.

Shielding of Neutron Sources

- The shielding of neutron sources consists of 3 different materials arranged as follows:

1. substance rich in hydrogen (polythene - paraffin - wax - water) with a thickness of approximately 25 cm to moderate fast neutron to become thermal neutron (elastic Scattering).
2. A thin layer of Cadmium or Boron (approximately 1 mm) to absorb thermal neutron (Radiative Capture)
3. A layer of Lead to absorb the gamma radiation generated by the absorption of thermal neutrons in Cadmium or Boron.



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Neutron Interactions within the Human Body

- Fast neutrons lose their energy within the human body through an elastic collision reaction with Hydrogen nuclei, which is the dominant component of the body.
- Thermal neutron loses its energy within the human body through the radiative capture interaction in Hydrogen nuclei.

- For E<10 keV the dose is mainly due to γ-rays resulting from thermal-neutron capture in hydrogen, $^{1}H(n,\gamma)^{2}H$
- For E>10 keV the dose is dominated by the contribution of recoil protons resulting from elastic scattering of hydrogen nuclei

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Radiation Protection Center

Radiation Protection Training Course

Prepared by:

Eng. Ashraf Mohammed Enany

Master of Science in Nuclear Engineering

Head of Radiation Protection Center

Mobile : +966 568360241

Email : ashraf@ t-experts.com

Website : www.t-experts.com

Dosimetric Quantities
&
Biological Effects

Exposure

- **Exposure (X) :** is the amount of **ionization** caused in dry **air** by **x-ray or gamma** radiation.



- **Exposure unit: Roentgen (R)**

Roentgen (R): The exposure which yields a charge of 2.58×10^{-4} Coulomb in 1Kg of dry air.

$$1R = 2.58 \times 10^{-4}\ C/kg_{air}$$

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Absorbed Dose

- **Absorbed Dose (D):** The amount of deposited **ENERGY** per **unit mass** of matter.



- **Units: Gray (Gy) or rad**

1 Gray: The absorption of one joule of radiation energy per kilogram of matter (**1 joul/Kg**) (SI unit)

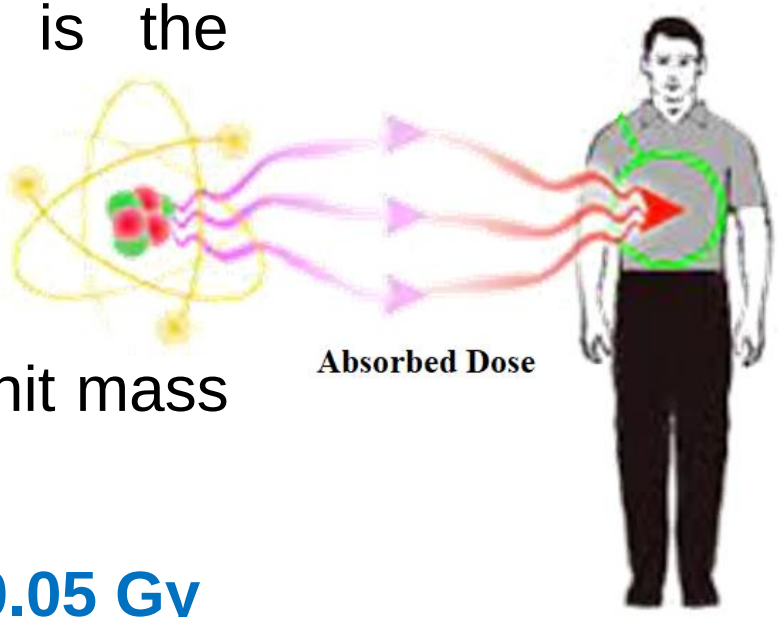
1 rad = 100 erg /gram = 0.01 J/kg (Old unit)

1 Gy = 100 rad

Absorbed Dose

Example:

A worker of weight 70 kg is exposed to radiation of gamma which deposited 3.5 Joule in his whole body. What is the absorbed dose that he received?



Answer:

"**Dose** is the deposited energy per unit mass of material"

Dose = 3.5J / 70kg = 0.05 J/kg = 0.05 Gy

= 5 rad

Absorbed Dose and Exposure

1 R = 0.86 rad (air)

1 R = 0.96 rad (tissue)

Approximately,

1 R ≅ 1 rad (tissue)



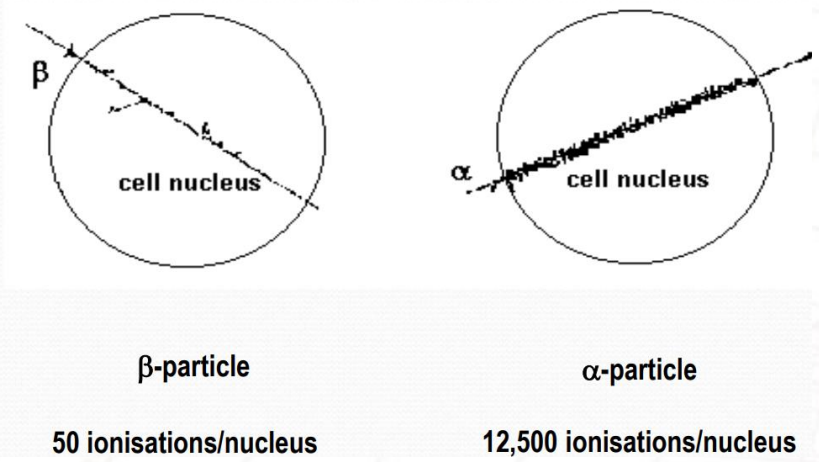
Biological Effects

For the same absorbed dose value

Different types of ionizing radiations cause different amount of biological effects.

Example:

Biological damage from a dose of 1 rem of α is **20** times more dangerous than the same dose of β radiation



Some radiations are biologically more harmful than others

Radiation Weighting Factor - W_R

So, it is necessary to adopt the concept of the absorbed dose in order to take into account the dependence of amount of biological damage on the Type of Radiation .



Radiation Weighting Factor W_R

(W_R is a unitless factor).

Radiation Weighting Factors (W_R)

Type and energy range	W_R
Photons, all energies	1
Electrons, all energies	1
Neutrons, energy < 10 keV	5
10 keV to 100 keV	10
100 keV to 2 MeV	20
2 MeV to 20 MeV	10
>20 MeV	5
Protons, other than recoil protons, energy > 2 MeV	5
Alpha particles, fission fragments, heavy nuclei	20

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Equivalent Dose H_T
(Organ Dose)

H_T: A quantity derived from the absorbed dose (D) of radiation type (R) for a certain tissue (T).

$$H_T = W_R \times D_{T,R}$$

Units:

Sievert (Sv): Weighted Joule per kilogram (SI unit).

rem : 0.01 Weighted Joule/kilogram (Traditional unit)

1Sv = 100 rem

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Equivalent Dose Rate

Example:

A worker exposed to radiation doses of : γ–rays (4 mGy) to stomach , slow neutrons (1 mGy) to stomach and fast neutrons (1MeV) to skin (0.5 mGy). What equivalent dose would be received?

- γ rays: D = 4 mGy ⟹ **H = (1×4) = 4 mSv Stomach**
- Slow neutrons: D = 1 mGy ⟹ **H = (5 ×1) = 5 mSv Stomach**
- Fast neutrons : D = 0.5 mGy ⟹ **H = (20 ×0.5) = 10 mSv Skin**

So, the worker received (9 mSv) in Stomach and (10 mSv) in Skin

DO NOT SAY THAT THE WORKER RECEIVED 19 mSv

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The question is:

Will any dose received by a person cause cancer and/or genetic effects (stochastic effects)?

The Answer is:

It is not certain , but it is a probable.

Can we determine this probability?

Yes

How?

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Tissue Weighting Factors - W_T

First, We must know that:

The probability of causing stochastic effects depends on the organ (tissue) exposed, where the organs have different sensitivities to the induction of these effects.

This is taken into account by applying the "Tissue Weighting Factor $\mathbf{W_T}$.

Tissue Weighting Factors (W_T)

tissue or organ	tissue weighting factor w_T
breast	0.12
bone marrow	0.12
colon	0.12
lung	0.12
remainder	0.12
stomach	0.12
gonads	0.08
bladder	0.04
liver	0.04
oesophagus	0.04
thyroid	0.04
bone surfaces	0.01
brain	0.01
salivary glands	0.01
skin	0.01
TOTAL	1.00

Organ or tissue weighting factors (ICRP 2007)

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Effective Dose
(Whole Body Dose)

Second, We need to calculate the Effective Dose (E):

$$E = \sum w_T H_T$$

H_T : The equivalent dose in tissue or organ T

W_T : The weighting factor for tissue T.

The effective dose is the basic for determining the probability of causing the stochastic effects.

Effective Dose

Example:

A worker receives the following equivalent doses: 100 mSv in the lungs, 20 mSv in the thyroid and 800 mSv in the skin. What is the effective dose he received? What is the probability of cancer induction (Risk factor = 0.05 Sv^{-1})?

Answer:

Effective Dose = (100*0.12) + (20*0.04) + (800*0.01) = 20.8 mSv
= 0.0208 **Sv**

Probability of cancer induction = 0.0208 x 0.05 = 0.00104

Effective Dose

Example:

What is the probability of stochastic effects from 0.8 rad/yr absorbed dose of alpha radiation in lung for 30 years (Risk factor = 0.05 1/Sv)?

Answer:

Total absorbed dose in 30 years	= 0.8 x 30 = 24	rad
Total Equivalent dose from alpha in Lung	= 24x 20 = 480	rem
Total Effective dose	= 480 x 0.12 = 57.6	rem
	= 0.576 **Sv**	

Probability = 0.576 x 0.05 = 0.0288

Types of Exposure

1. Acute Dose

- Large radiation dose in a short period of time
- Large doses may result in observable health effects
  - Early: Nausea & vomiting
  - Hair loss, fatigue, & medical complications
  - Burns and wounds heal slowly
- Examples: Accidental exposure to sealed sources



2. Chronic Dose

- Small radiation dose received over a long period of time
- Body more easily repairs damage from chronic doses
- Does not usually result in observable effects
- Examples: Background Radiation and Internal Deposition



Inhalation

Biological effects

Two types of biological effects:

1. Stochastic Effects (Random Effects)

- Stochastic effects are associated with long-term, low-level (chronic) exposure to radiation., where cells mutated in a such a way as to cause
  - Cancer
  - Hereditary disease
- There is NO threshold for the effects to be observed.
- Risk proportional to dose
- Minimise risk by keeping doses as low as reasonably achievable (ALARA)

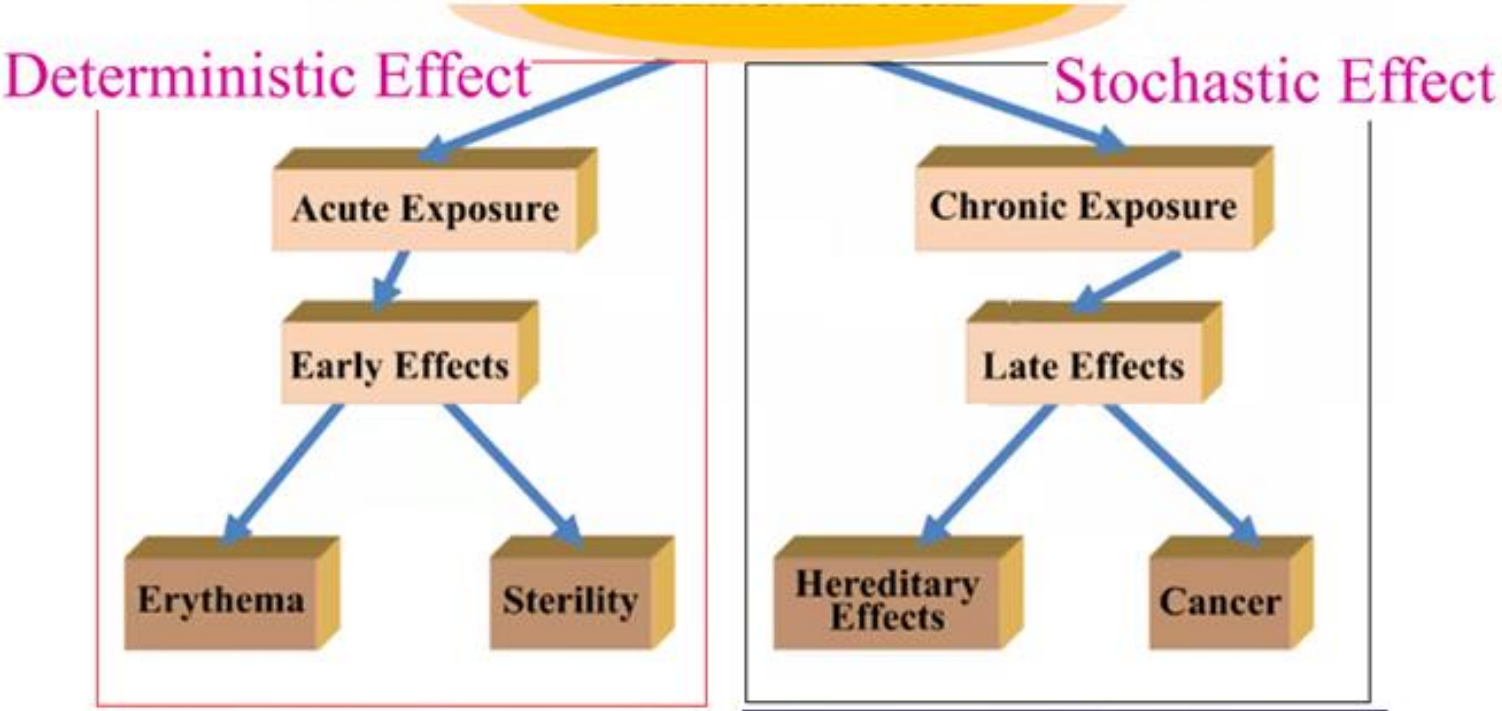
Biological effects

2. Deterministic Effects
(Non-stochastic Effects)

- Severity of the result is related to the dose (usually high dose).
- Generally related to a large dose over a short time (acute exposure).
- Adverse effect happens soon after exposure and can be directly linked to exposure.
- There is a threshold level - observed effects follow typical distribution around a dose.
- Example:
  - erythema
  - cataract
  - sterility

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Summary of Radiation Effects



Deterministic Effect	Stochastic Effect
Has a threshold dose	Has no threshold
Deterministic in nature	Probabilistic in nature
Severity increases with dose	Probability increases with dose
Occurs only at high doses	Occurs even at low doses
Can be completely avoided	Cannot be completely avoided

Collective Dose

Collective Dose (CD):

- **This is used to measure the total impact of a radiation practice or source on all the exposed persons, for example diagnostic radiology**
- **CD is measured in man-sievert (man-Sv)**
- **There are many typed of CD, for example: Collective Effective Dose, Collective Equivalent Dose, ……..**

Collective Equivalent Dose

Example:

250 Nuclear workers → Equivalent dose in lunge per worker, 6 mSv.

What is the Collective equivalent dose in lung?

Collective equivalent dose in lung = $6 \times 250 = 1500$ man-mSv

$= 1.5$ man-Sv

Collective Dose

Example:

What is the collective effective dose of the world's population for one year?

Answer:

Average Dose from all sources per year = 3.6 mSv

World population 6,000 million

Collective Effective Dose = $3.6 \times 10^{-3} \times 6 \times 10^{9} = 21.6 \times 10^{6}$ man-Sv

Summary

Dose Quantities Hierarchy

Absorbed Dose: Energy per unit mass of tissue

↓

Equivalent Dose: Includes Radiation weighting

↓

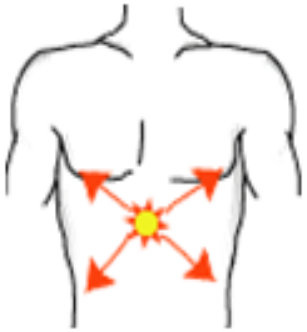
Effective Dose: Includes Tissue weighting

↓

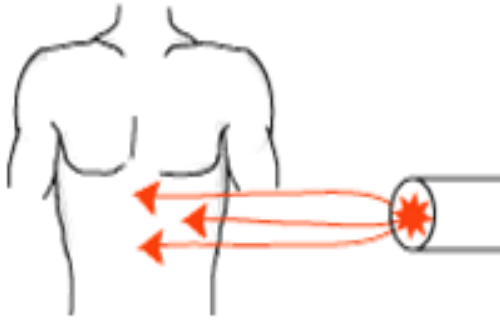
Collective Dose: Effective dose to a group or population.

Internal Dose
(Intakes of Radioactive Material)

- **All radiation types present a hazard**
- **When radioactive material is inhaled or ingested, the result is an intake into the body**
- **Intakes of radioactive material are usually expressed in units of Bq.**



Internal Radiation



External Radiation

Ways of Entry into the Body:
1. Inhalation
2. Ingestion
3. Skin

Internal Dose
(Intakes of Radioactive Material)

- **Two Types of Internal Doses:**
  - **Committed Equivalent Dose, CDE (specific to a particular organ)**
  - **Committed Effective Dose, CED (Whole body)**

KERMA

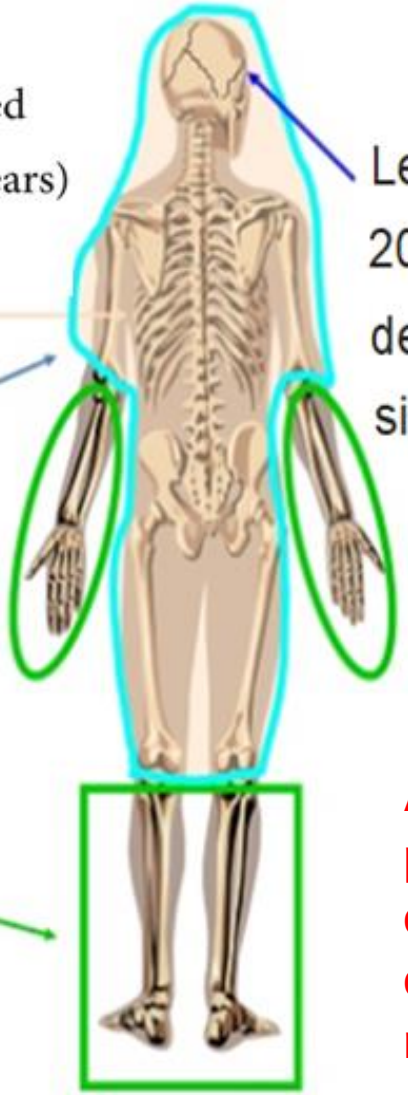
KERMA "**kinetic energy released per unit mass**", defined as:

"The sum of the initial kinetic energies of all the charged particles liberated by indirectly ionizing radiation (such as photons and neutrons) in a sample of matter, divided by the mass of the sample".

Unit: Gy (J/kg)

Occupational Dose Limits

An effective dose of 20 mSv per year averaged over five consecutive years (100 mSv in 5 years) and of 50 mSv in any single year;



Lens of the Eye
20 mSv per year, averaged over defined periods of 5 years, with no single year exceeding 50 mSv

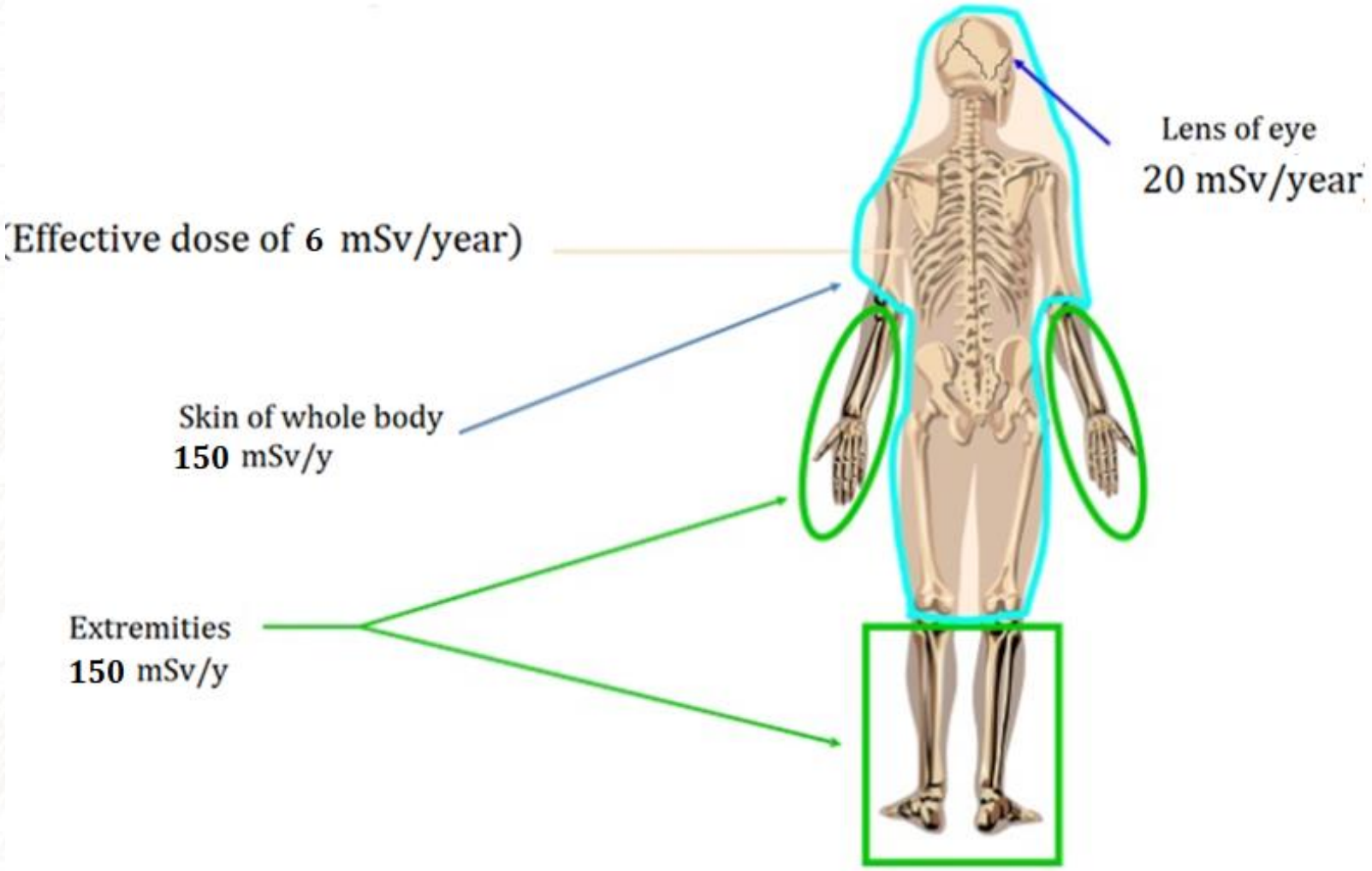
Skin of whole body
500 mSv/y

Extremities
500 mSv/y

After a worker declares a pregnancy, the dose to the embryo/fetus should not exceed 1 mSv during the remainder of the pregnancy

Trainees & Students Dose Limits

(16-18 years old)



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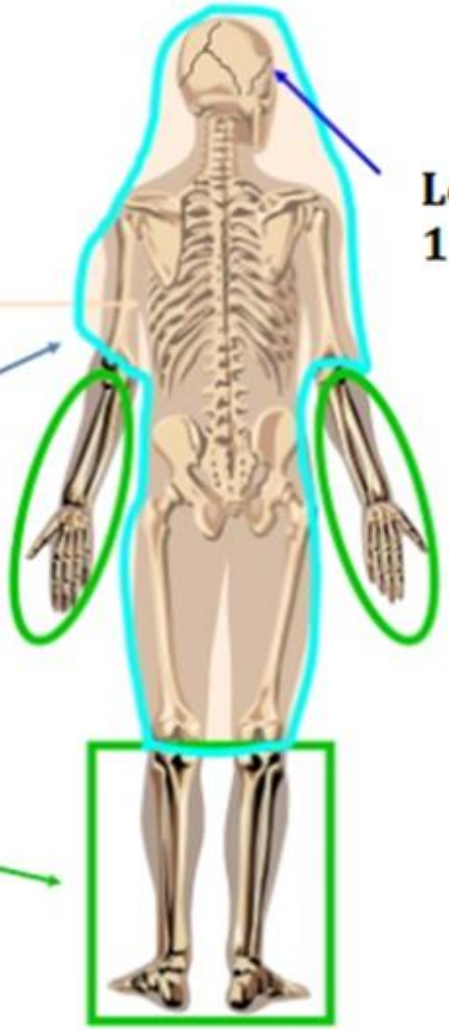
Public Dose Limits

Whole Body 1 mSv/y

In special circumstances, a higher value could be allowed in a single year, provided that the average over 5 years does not exceed 1 mSv per year

Skin
50 mSv/y

Extremities
50 mSv/y



Lens
15mSv/y

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Dose Measurements











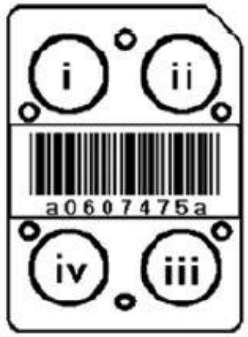




Typical TLD Card Configuration

Position/Element 1:
Low Energy Photon Discrimination and Lens dose (Hp(3))
- ~300 mg/cm² filtration

Position/Element 2:
Deep dose (Hp(10))
- 1000 mg/cm² filtration



Position/Element 4:
Mid-Energy Photon Discrimination and Neutron dose
- Li-6 isotope
- Various filtration

Position/Element 3:
Shallow dose (Hp(0.07))
- ~17 mg/cm² filtration

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Acute Exposure Effects

AVG DOSE	DAMAGE
> 7 Sv	Death
3 - 5 Sv	LD $_{50\text{-}30}$ (Possible death)
1 - 2 Sv	Radiation Sickness (Nausea, vomiting, hair loss, fatigue,..)
0.25 – 0.50 Sv	Slight Blood Changes
< 0.25 Sv	No detectable symptoms or blood changes

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Lethal Dose

- Lethal Dose ($LD_{50/30}$) is used to refer to a dose that without treatment will be lethal to 50% of the population within (respectively) 30 days.

- $LD_{50/30}$ = 3-5 Sv

Technology Experts Company

Radiation Protection Center

Radiation Protection Training Course

Prepared by:

Eng. Ashraf Mohammed Enany

Master of Science in Nuclear Engineering

Head of Radiation Protection Center

Mobile : +966 568360241

Email : ashraf@ t-experts.com

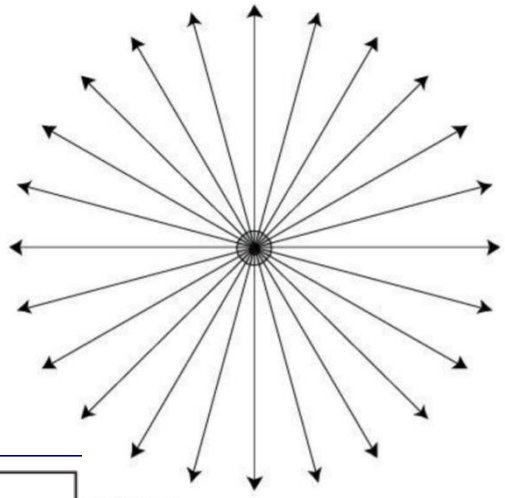
Website : www.t-experts.com

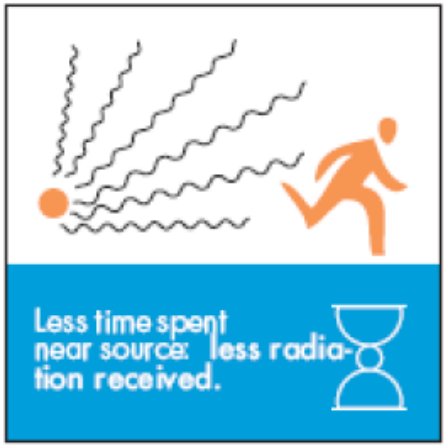
Radiation Protection Bases

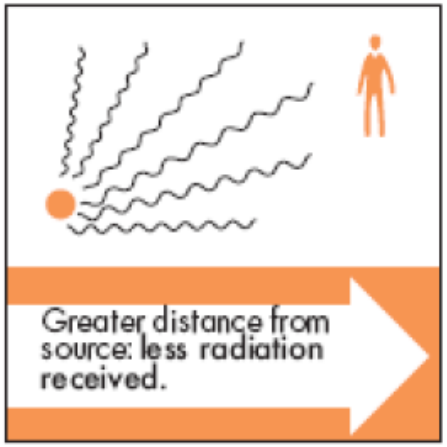
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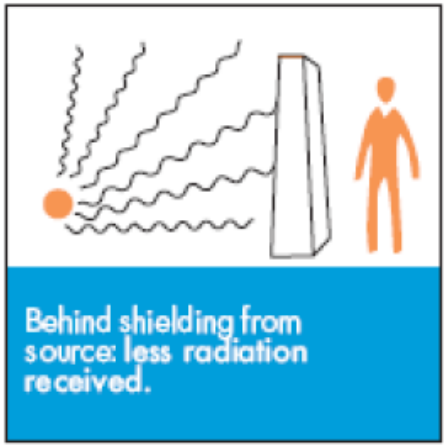
The methods of reducing external radiation Dose

- Decrease time of exposure
- Increase distance from the source
- Use shielding











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Variation of Dose with Time

Dose received is a linear function of the exposure time

$$D = \dot{D} \times t$$

Where

D is the radiation dose

$\dot{D}$ is the radiation dose rate

t is the exposure time

Dose Limits

Occupational:

- Annual Dose limit = 20 mSv/y
- Weekly Dose limit = 20/50 = 0.4 mSv/w = 400 μSv/w
- Daily Dose limit = 0.4/5 = 0.08 mSv/d = 80 μSv/day
- Hourly Dose limit = 10 μSv/h

Example

Example:

Estimate the time required for a radiation worker working in a 2.5 mSv/h. so that the worker should not exceed the daily limit?

$$D = \dot{D} \times t$$

Answer:

- We know that the derived limit = 10 μSv/h.
- 2.5 mSv/h = 2500 μSv/h is greater than the derived limit by 250 times (2500/10)
- So the time must be reduced by the same ratio, so
- He can works (2000/250 = 8 h/y)

Variation of Dose with Distance

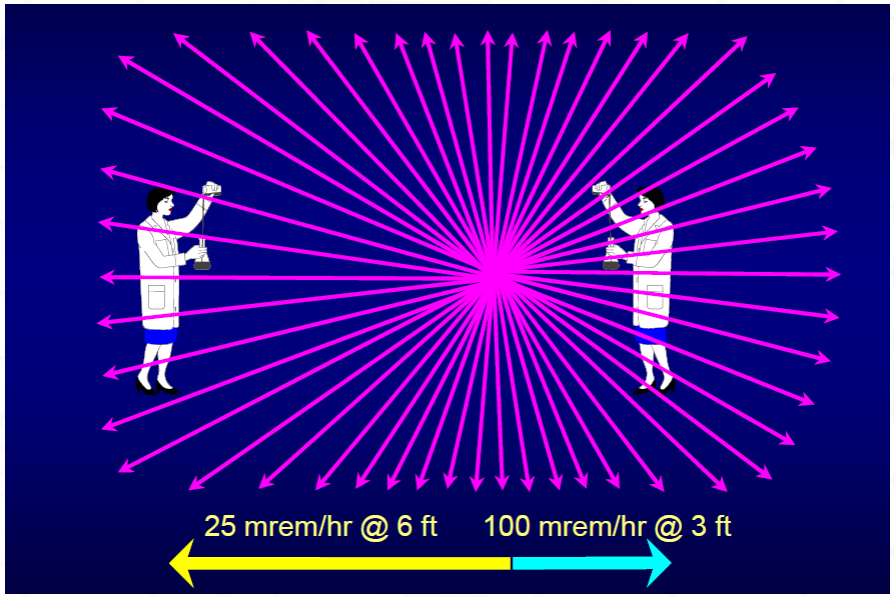
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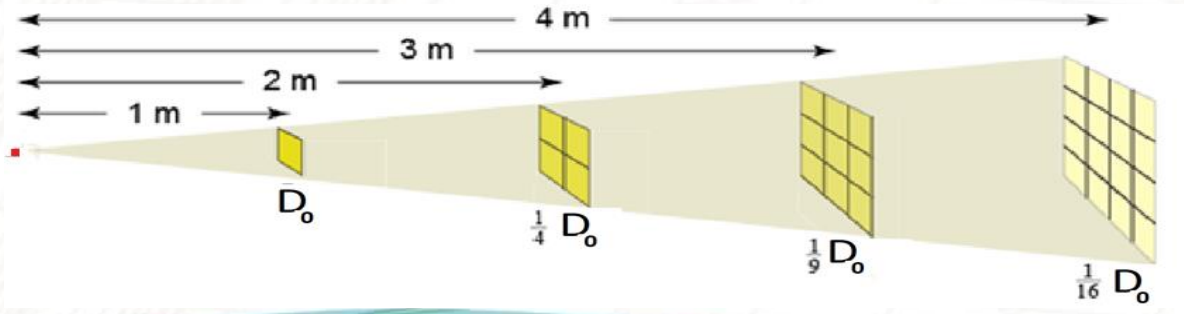
Radiation dose inversely proportion with the square of the distance.

$$\dot{D}_1 d_1^2 = \dot{D}_2 d_2^2$$

Where

- $\dot{D}_1$ **dose rate at distance 1**
- $\dot{D}_2$ **dose rate at distance 2**
- d_1 **is distance at 1**
- d_2 **is distance at 2**





Variation of Dose with Distance

Example: The dose rate at 2 meters away from a gamma source measures 100 mSv/h. What is the dose rate at 4 meters?

Solution:

$$\dot{D}_1 d_1^2 = \dot{D}_2 d_2^2$$

$$\dot{D}_2 = \dot{D}_1 \times (d_1^2/d_2^2) = 100 \times (2^2/4^2) = 25 \text{ mSv/h}$$

Variation of Dose with Distance

Example

An operator is conducting a corrective action with a source located at 25 cm from him for 10 minutes and his dose will be 180 mSv, If he can do the same action using a long handle so that the distance to the source will be 150 cm but the time will be 15 min, then the incurred dose is?

Solution:

$$\dot{D}_1 d_1^2 = \dot{D}_2 d_2^2$$

$$\frac{180}{10} x\ 25^2 = \dot{D}\ x\ 150^2$$

$$\dot{D} = 0.5\ \frac{mSv}{min}$$

$$D = 0.5\ x\ 15 = 7.5\ mSv$$

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Estimation of Dose Rate from Gamma Sources

- The effective dose rate from a gamma source is calculated by:

$$\dot{D} = \frac{\Gamma \times A}{d^2}$$

- where
  - $\dot{D}$ is the effective dose rate in µSv/h
  - Γ is the gamma factor (dose rate constant) in µSv-m^2/(MBq-h)
  - **A** is the activity of the source in MBq
  - **d** is the distance in meters from the source

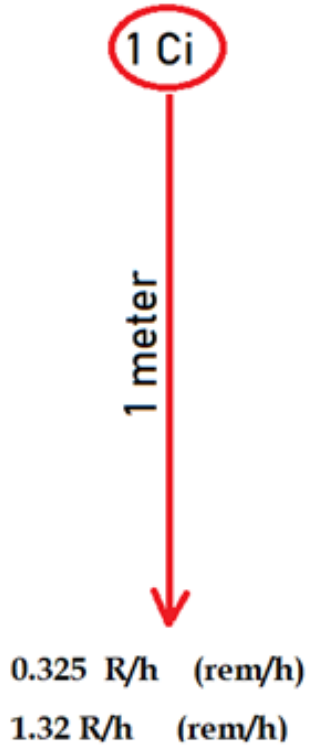
Specific Gamma Dose Rate Constant

Γ is the Effective Dose Rate in µSv/h at 1 meter from 1 MBq of the radionuclides in µSv.m^2/(MBq.h) (SI)

Γ is s the Exposure Rate in R/h at 1 meter from 1 Ci of the radionuclides in R.m^2/(Ci.h) (old unit)



The radionuclide	The gamma specific constant Γ	
	(µSv.m^2/h.MBq)	(R.m^2/h.Curie)
Cesium Cs-137	0.087	0.325
Cobalt Co-60	0.356	1.32
Gold Au-198	0.0622	0.23
Iodine I-131	0.0595	0.22
Iridium Ir-192	0.13	0.48
Radium Ra-226	0.223	0.825
Sodium Na-24	0.497	1.84



Effective dose of gamma sources

Example: **Calculate the effective dose rate at a distance of 5 m from an Ir-192 gamma source which has an activity of 400 MBq.**

Solution:

$$\dot{D} = \frac{\Gamma A}{d^2} = \frac{0.13 \times 400}{5^2} = 2.08\ \mu Sv/h$$

$$= 0.208\ \text{mrem/h}$$

Example: **The effective dose rate from an exposed Cs-137 source of 3 Ci activity at a distance 25 cm from the source is:**

Solution:

$$\dot{D} = \frac{\Gamma A}{d^2} = \frac{0.325 \times 3}{0.25^2} = 15.6\ \text{R/h}$$

$$= 15.6\ \text{rem/h}$$

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Estimation of Dose Rate from Beta Sources

- The effective dose rate from a beta source is calculated by:

$$\dot{D} = \frac{5 \times A \times E_{ave}}{d^2}$$

- Where
  - $\dot{D}$ is the effective dose rate in µSv/h
  - A is the activity of the source in MBq
  - E_{ave} is the average energy = $E_{max}/3$ in MeV
  - d is the distance in meters from the source

Example

Calculate the effective dose rate at a distance of 20 cm from a thin unshielded P-32 beta source which has n activity of 20 mCi (E_{max} = 1.71 MeV).

$$\dot{D} = \frac{5 \times A \times E_{ave}}{d^2}$$

Answer:

- **Activity in MBq = $20x10^{-3}$ x $3.7x10^{10}$ x 10^{-6} = 740 MBq**
- **$E_{avg} = E_{max}/3 = 1.71 / 3 = 0.57$ MeV**
- **$D^{\bullet} = 5 \times 740 \times 0.57 / (0.2)^2 = 52700$ µSv/h**
- **= 5270 mrem/h**

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Estimation of Dose Rate from Neutron Sources

- The effective dose rate from an isotropic neutron source can be calculated as follows:

$$\dot{D} = \frac{0.08 \times C \times N}{d^2}$$

- where
  - $\dot{D}$ is the effective dose rate in µSv/h
  - **N** is the neutron fluence in neutrons/s
  - **C** is conversion factor in (µSv/h)/(n/m^2.sec)
  - **d** is the distance in meters from the source

Effective Dose Rate Conversion Factor

Neutron Energy	Conversion factor
1 KeV	3.74×10^{-6}
10 KeV	3.56×10^{-6}
100 KeV	2.17×10^{-5}
500 KeV	9.25×10^{-5}
1 MeV	1.32×10^{-4}
5 MeV	1.56×10^{-4}
10 MeV	1.47×10^{-4}

Example

Calculate the effective dose rate at 0.5 m from an Am-Be neutron source that emits 3×10^7 n/s, assuming the average energy of the neutron is 1 MeV.

$$\dot{D} = \frac{0.08 \times C \times N}{d^2}$$

Answer:

- From table C for 1 MeV = 1.32×10^{-4} $(\mu Sv/h)/(n/m^2 s)$
- $\dot{D} = (0.08 \times 1.32 \times 10^{-4} \times 3 \times 10^7) / (0.5)^2$

  $= 1.27 \times 10^3$ µSv/h.

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Intensity of Neutron Sources

1 μg of Cf-252 emits 2.3×10^6 neutron/sec

1 Ci of Am-Be emits 2.2×10^6 neutron/sec

Flux of Neutron Sources (ϕ)

$$\phi = \frac{\text{Intensity}}{4\pi d^2}$$

d is the distance between the point and the source

Estimation of Dose Rate from Alpha Sources

- Alpha particles do not present an external radiation hazard.
- External dose calculations are generally not required.

Annual Limit of Intake (ALI)

- The intake of a given radionuclides (**Bq**) in a year by reference man which result in a committed dose equal to the relevant annual dose limit of 20 mSv.

Isotope	ALI (Bq)
Tritium	1×10^9
Technetium-99	3.1×10^7
Iodine-131	1.0×10^6
Caesium-135	1.0×10^7
Plutonium-239	8.0×10^4

Derived Air Concentration (DAC)

DAC is the maximum concentration (Bq/m^3) of a single radionuclide in the air at the working place of an occupational reference man, breathing of which during the whole working hours through the year (2000 hours/year), would result in a committed effective dose equal to the relevant occupational dose limit (i.e 20 mSv).

radionuclide	DAC [Bq/m^3]
^{3}H	460 000
^{60}Co	290
^{90}Sr	56
^{131}I	1100
^{137}Cs	1700
^{239}Pu	0.18

Example

A worker was exposed internally to 0.4 ALI and inhaled 1000 DAC.h so the external dose should not exceed in that year:

$$ALI \rightarrow 20mSv$$
$$0.4ALI \rightarrow \ ?$$

$$E_1 = \frac{20 \times 0.4}{1} = 8(mSv)$$

$$E_{\text{Internal}} = E_1 + E_2 = 10 + 8 = 18(mSv)$$

$$20 - E_{\text{Internal}} = 20 - 18 = 2(mSv)$$

$$2000DAC.h \rightarrow 20mSv$$
$$1000DAC.h \rightarrow \ ?$$

$$E_2 = \frac{1000 \times 20}{2000} = 10(mSv)$$

Example

A worker intake $8x10^5$ Bq of radioactive isotope, the annual limit of its intake (ALI) is $1.6x10^6$ and inhaled 400 DAC.h, so its total dose in this year is:

$$1.6 \times 10^6 \rightarrow 20mSv$$
$$8 \times 10^5 \rightarrow \ ?$$

$$E_1 = \frac{8 \times 10^5 \times 20}{1.6 \times 10^6} = 10(mSv)$$

$$2000DAC.h \rightarrow 20mSv$$
$$400DAC.h \rightarrow \ ?$$

$$E_2 = \frac{400 \times 20}{2000} = 4(mSv)$$

$$E = E_1 + E_2 = 10 + 4 = 14(mSv)$$

Shielding Calculations

I is the radiation intensity after traversing a thickness (x)

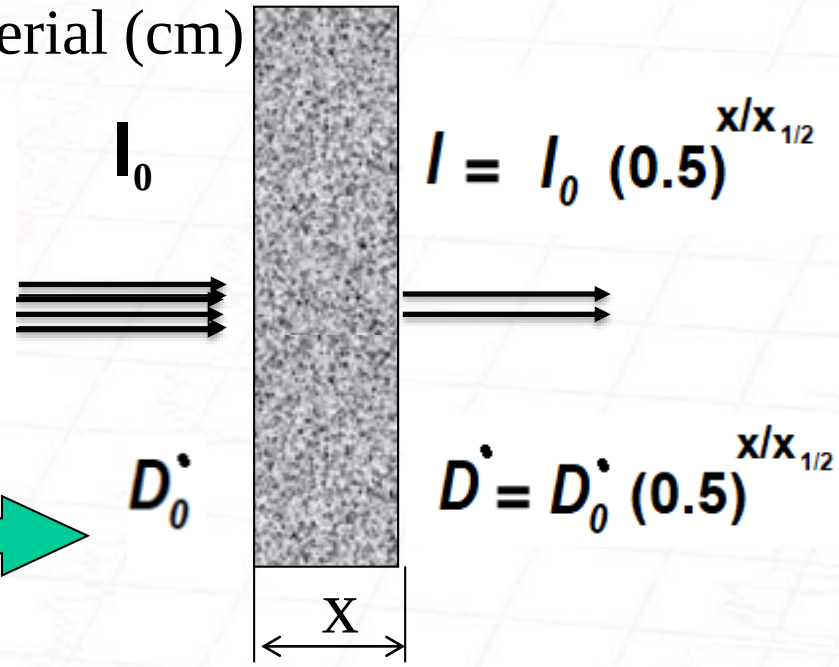
I_o is the original radiation intensity

$x_{1/2}$ is the half value layer (cm)

x is the thickness of attenuating material (cm)

$\dot{D}$ is the dose rate after traversing a thickness x

$\dot{D}_o$ is the original radiation dose rate



Good Geometry

(small thickness and narrow beam of γ)

Buildup factor =1

Shielding Calculations

Example:

The dose rate from an Ir-192 exposed source with activity 60 Ci at a distance 50 cm is _____ , when this source will be shielded by a material with a HVL equal 5 mm and with thickness of 10 cm then the dose rate at the same distance will be :

Solution (without Shield):

$$\dot{D_o} = \frac{\Gamma \times A}{d^2}$$

$$\dot{D_o} = \frac{0.48x\ 60}{0.50^2} = 115.2\frac{R}{h} = 115.2\frac{rem}{h} = 1.152\frac{Sv}{h}$$

With shield:

$$\dot{D} = \dot{D_0}\ (0.5)^{x/x_{1/2}}$$

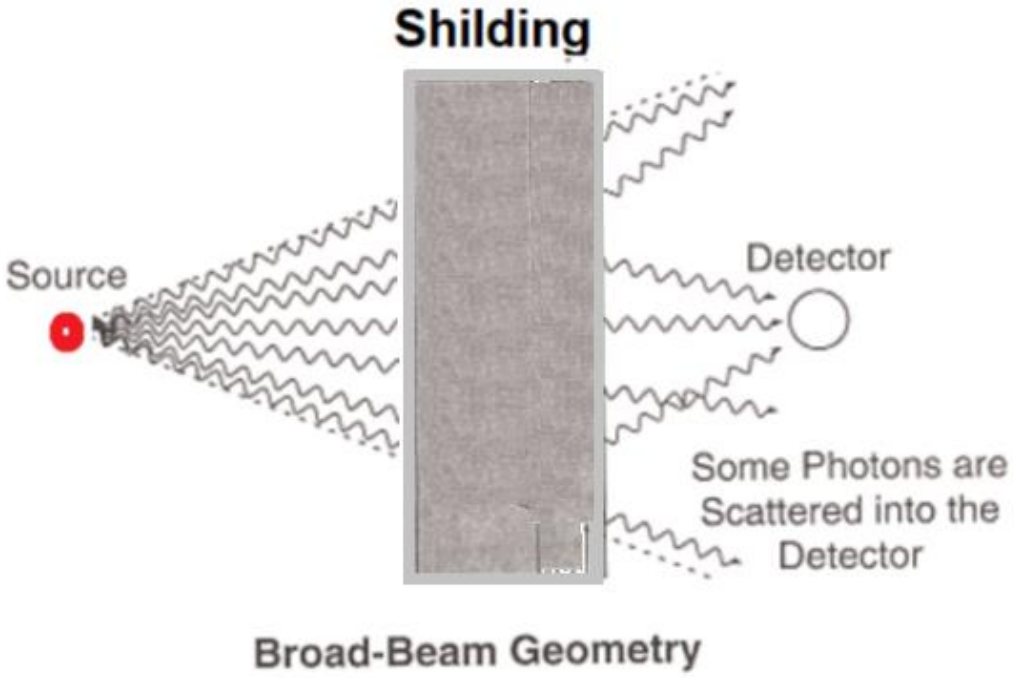
$$\dot{D} = 1.152\ x\ (0.5)^{\frac{10}{0.5}} = 0.00011\frac{Sv}{h} = 1.1\ \mu Sv/h$$

Build-up Factor (B)

$$B = \frac{\text{Primary photons + Scattered photons}}{\text{Primary photons}}$$

$$\dot{D} = B\dot{D}_0 (0.5)^{x/x_{1/2}}$$

For Bad Geometry (large thickness and wide beam of γ)



Broad-Beam Geometry

Buildup factor > 1

Z ↑ B ↑

X ↑ B ↑

Build-up Factor (B)

Example

Build-up factor for a shielding material at a certain point behind a shield equals 5, what is the percentage of the gamma radiation reaches this point from the radiation source directly (primary)?

Solution

$$B = 5 = \frac{I_t}{I_p} = \frac{5}{1} = \frac{I_p + I_s}{I_p} = \frac{1+4}{1}$$

the percentage of the gamma radiation reaches this point from the radiation source directly = 1/5 = 20%

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Technology Experts Company

Radiation Protection Center

Radiation Protection Training Course

Prepared by:

Eng. Ashraf Mohammed Enany

Master of Science in Nuclear Engineering

Head of Radiation Protection Center

Mobile : +966 568360241

Email : ashraf@ t-experts.com

Website : www.t-experts.com

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Radiation Detection
&
Measurement

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Types of Detectors

- ***Gas-filled Detector*** **It consists of a volume of gas between two electrodes (Ionization process).**

- ***Scintillation Detector*** **It consists of scintillation materials that become excited when interact with ionizing radiation.**

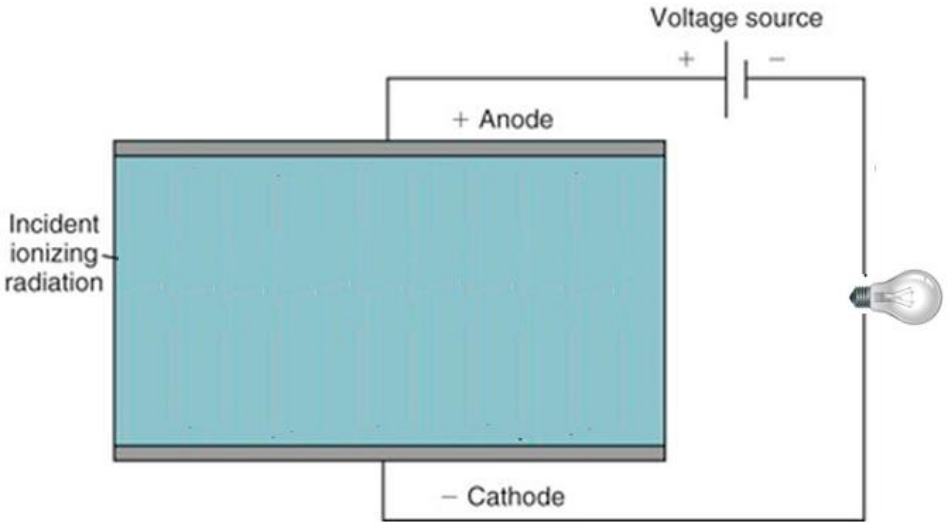
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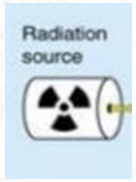
Types of Gas-filled Detectors

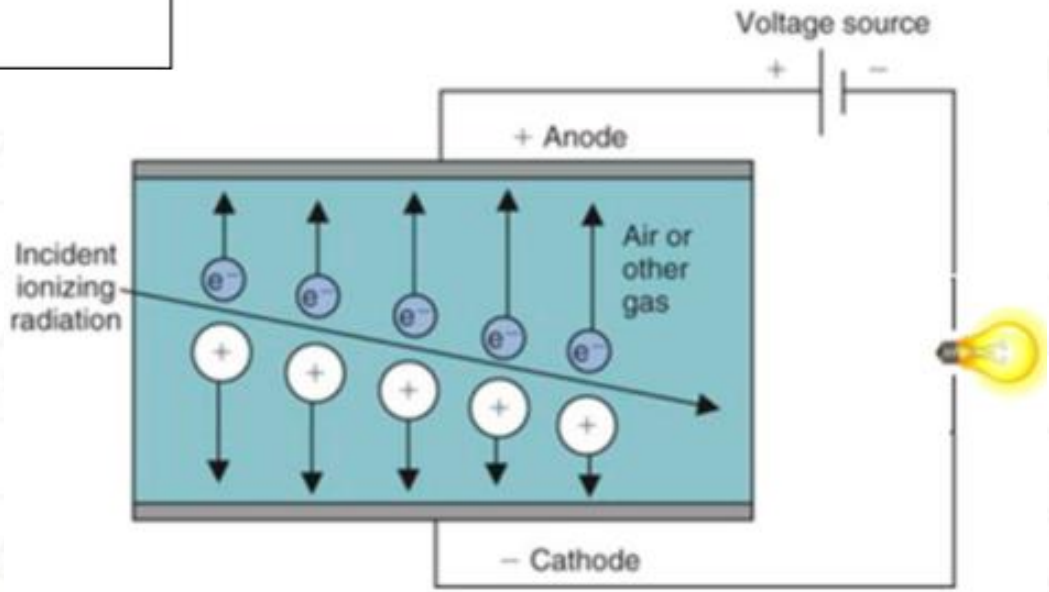
- Three types of gas-filled detectors in common use:
  - Ionization chambers
  - Proportional counters
  - Geiger-Mueller (GM) counters
- Type determined primarily by the voltage applied between the two electrodes

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Gas Detector

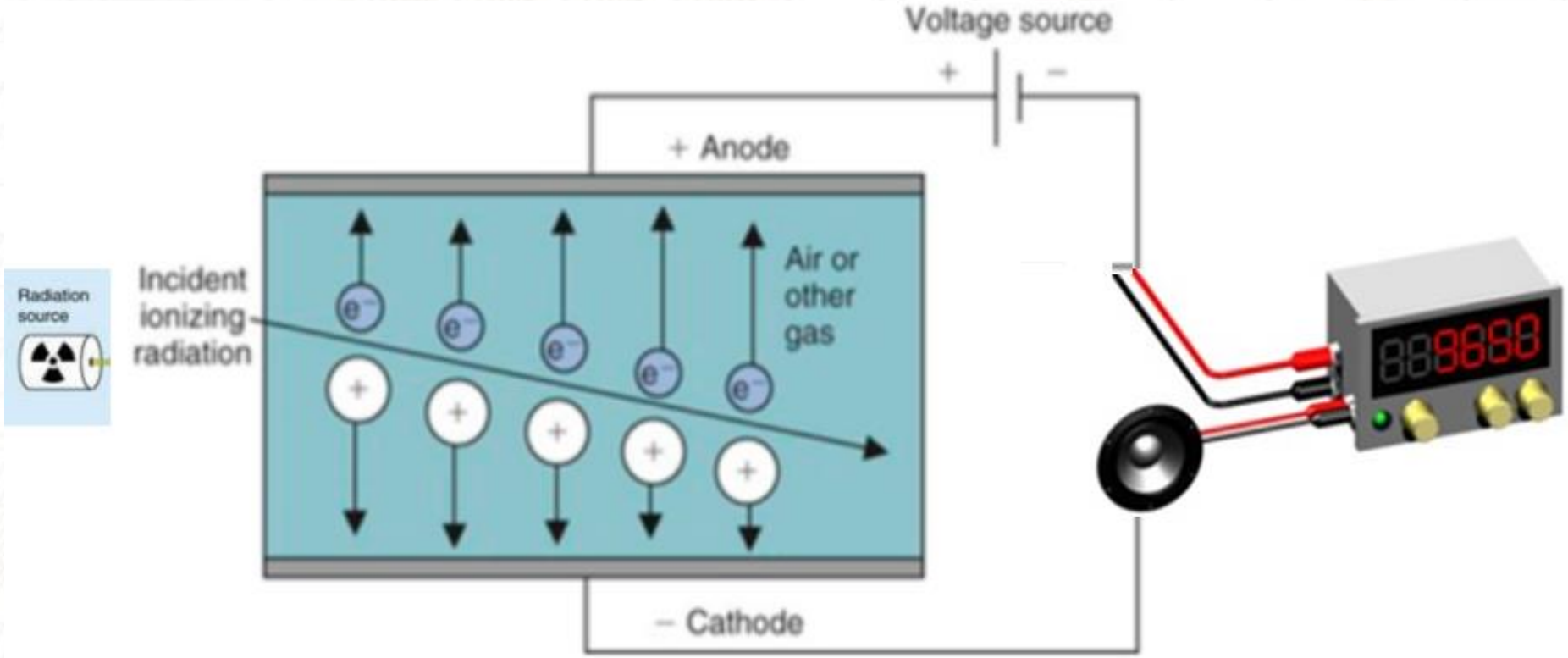






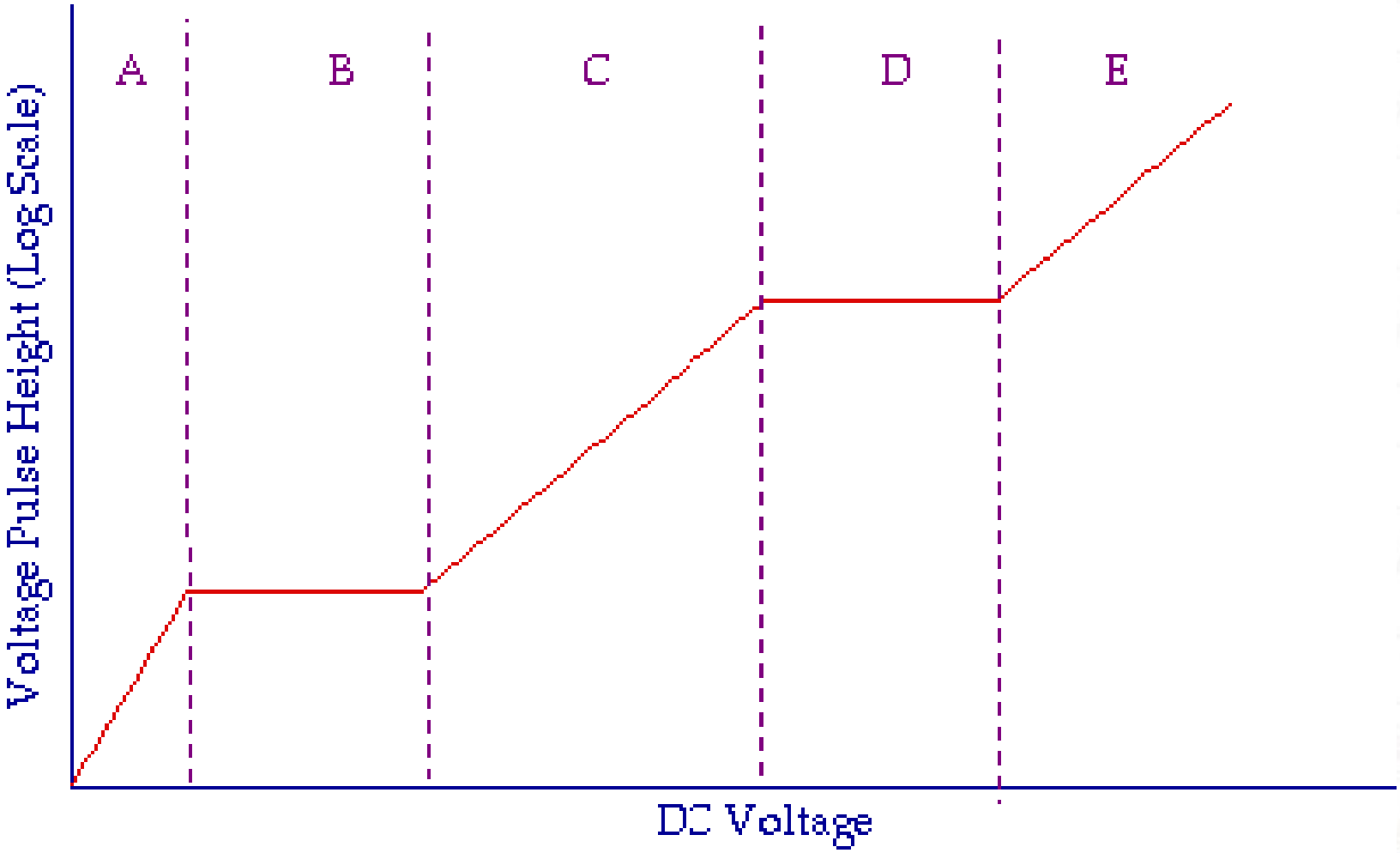
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Gas Detector



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DC Voltage Dependence



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Energy Value of Ionization

In gases, the energy needed of ionization is

30-35 eV

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Ionization Chamber

- Multiplication Factor = 1
- Used in high radiation field➔ Have short dead time
- Good energy resolution
- Used for calibration
- Direct exposure measurment
- Relatively flat response curve (better for multi radiation sources)



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Proportional Counters

- Multiplication Factor is greater than 1
- Typical amplification of factors of ~10^4
- Proportional counter filled with (BF_3) OR (He-3) gas is used to detect neutrons
- It can distinguish between Alpha and Beta radiation.
- Typical gas: P10 (90% Ar and 10% methane)

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GM Counters

- Multiplication Factor is greater than 1
- It can NOT distinguish between radiation type. (Insensitive to the types of radiation)
- It has long dead time
- Can not be used in high radiation field because it becomes saturated.



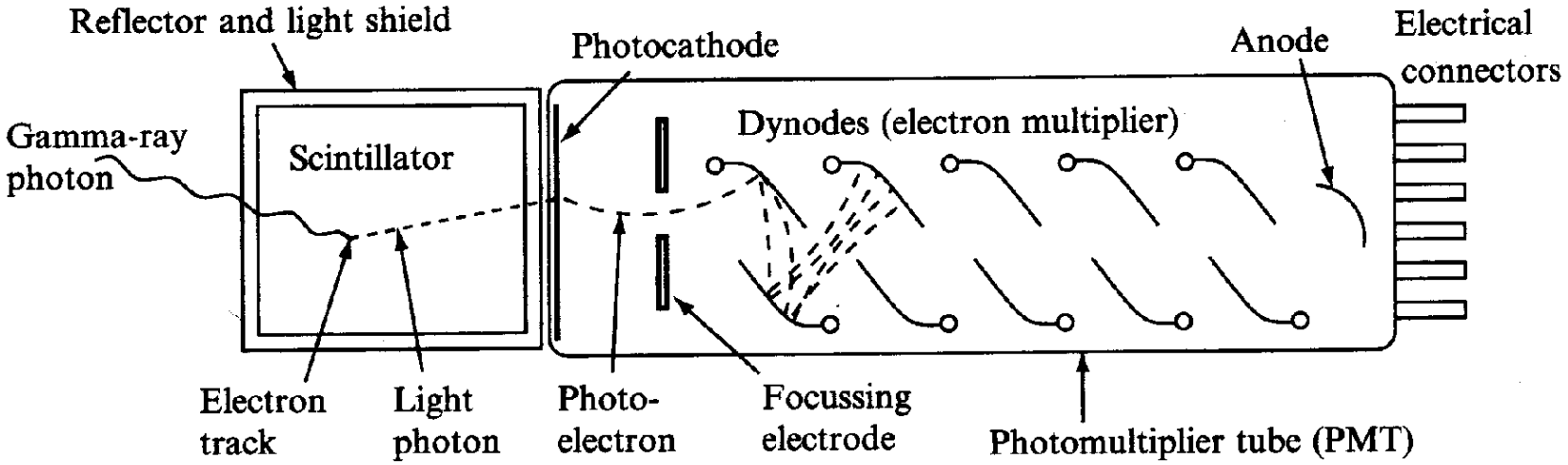




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Scintillation Detectors



Two general types: Organic and Inorganic

- The detection efficiency of Scintillation detectors is close to 30% for gamma rays.
- Good energy resolution.

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Scintillation Detectors

Material	Radiation
Liquid	Beta
ZnS(Ag)	Alpha
NaI(Tl)	Gamma
ZnS with wax	Fast neutrons
Li^6 I(Ti)	Thermal neutrons

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Dead Time

- Main problem with detectors in pulse mode is that two interactions must be separated by a finite amount of time if they are to produce distinct signals
- This interval is called the ***dead time*** of the system
- If a second interaction occurs in this interval, its signal will be lost.
- GM Detector has the longest dead time.

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Dead Time

- R_{true} : The "true" counting rate.

$$R_{true} = \frac{R_o}{1 - R_o\tau}$$

R_o : The observed (measured) counting rate.

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Dead Time

Example:

If the measured count rate using a device with 250 microsecond is 30000 cpm, then the corrected count rate is _____ cpm:

Solution:

$$R_{true} = \frac{R_o}{1 - R_o\tau}$$

$$R_{true} = \frac{30000}{1 - \frac{30000x250x10^{-6}}{60}} = 34286\ cpm$$

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Dead Time

Example:

During a one-minute sample count, a detector with a resolving time of 100 microseconds yields 175,000 net counts. What is the true count rate of the sample?

1. 17,502 cpm
2. 100,000 cpm
3. 166,666 cpm
4. 247,058 cpm

Answer:

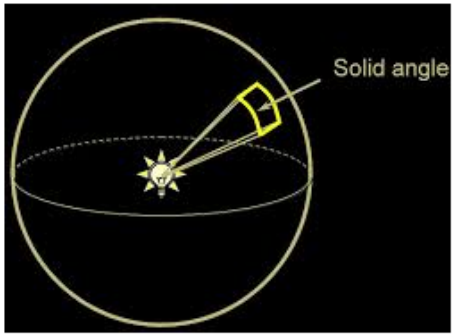
$$R_{true} = 175{,}000 / [1 - 175{,}000\ (100\ \mu sec * 1E\text{-}06\ sec/\ \mu sec * 1/60\ min/sec)]$$

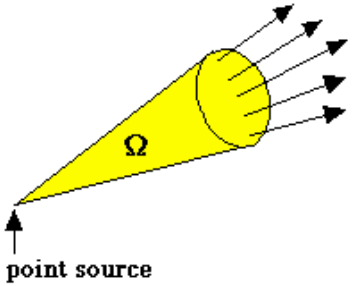
$$= 247{,}058\ cpm$$

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Detection efficiency

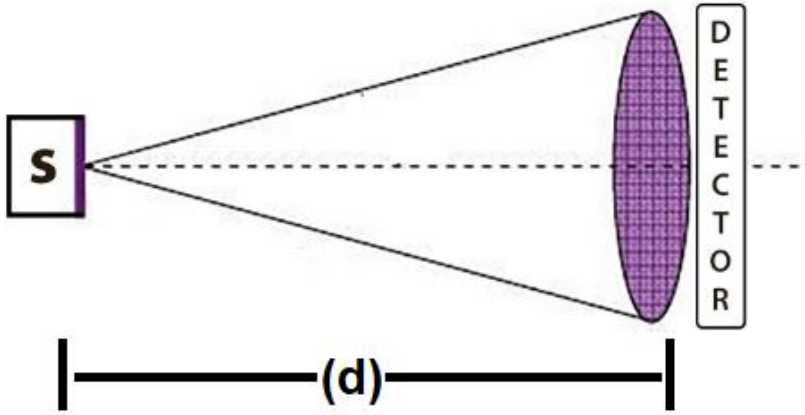
- The *efficiency* of a detector is a measure of its ability to detect radiation
- Efficiency of a detection system is defined as the probability that a particle or photon emitted by a source will be detected





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Detection efficiency



(d) is the distance between the source and detector

$$\text{Efficiency} = \varepsilon = \frac{\text{Detector Count Rate}}{\text{Impacting Rate on Detector}} = \frac{C}{f A \Omega}$$

$$\textit{Solid Angle} = \Omega = \frac{\textit{Surface Area of detector}}{4\pi d^2}$$

- (A) is the activity of the source (Bq)
- (f) is the number of emitted photons per decay

Example :

A cesium-137source of activity 1 μCi at a distance of 30 cm from a circular detector. If the radius of the detector is 1.25 cm and its efficiency is 0.04. What is the counting rate of the detector (Cs-137 emits 85 photons with per 100 decay) ?

Answer:

$$C = A \Omega f \varepsilon$$

$$\Omega = \frac{r^2}{4\,d^2} = \frac{(1.25)^2}{4(30)^2} = 4.34 \times 10^{-4}$$

$$A = 1 \times 3.7 \times 10^4 = 3.7 \times 10^4 \text{ Bq}$$

$$C = \varepsilon \Omega f A = 0.04 \times 0.85 \times 4.34 \times 10^{-4} \times 3.7 \times 10^4 = 0.55 \text{ Counts/Sec}$$

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Counting Statistics

$$Error = Standard\ Deviation\ (\sigma)$$

$$\sigma = \pm \sqrt{\frac{Count\ Rate}{Counting\ Time}}$$

Example: **A survey meter recorded 3600 counts through 10 seconds, what is the count rate with the second level of confidence?**

Solution:

$$\sigma = \pm \sqrt{\frac{3600/10}{10}} = \pm 6\ cps$$

$$360 \pm 12\ cps \quad @\ 2^{nd}\ C.L$$

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Counting Statistics

$$\sigma = \pm \sqrt{\frac{Count\ Rate}{Counting\ Time}}$$

Example:

A survey meter recorded 100 counts through a certain period the device displayed it as 15 µSv/h, what is the dose rate with 99% level of confidence

Solution:

$$\text{Error in count rate} = \sigma = \pm\sqrt{100} = \pm 10\ cps$$

100 cps ⟶ **15 µSv/h**

10 cps ⟶ **???**

$$\text{Error in dose rate} = \sigma = \pm\left(\frac{10x15}{100}\right) = \pm 1.5\ \frac{\mu Sv}{h}$$

$$\text{Dose rate} = 15 \pm 4.5\ \frac{\mu Sv}{h} \qquad @\ 3rd\ C.L$$

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Error Calculations

Example:

The gross count of a sample was 2,000 counts for a 10 minute. The background was 200 counts for a 5 minute count.

What is the net count rate and associated uncertainty?

Answer:

$$R_{net} = R_g - R_b$$

$$\sigma_{net} = \sqrt{\sigma_g^2 + \sigma_b^2}$$

$$R_{net} = \left(\frac{2{,}000\ counts}{10\ m}\right) - \left(\frac{200\ counts}{5\ m}\right) = 160\ cpm$$

$$\sigma_{net} = \sqrt{\frac{200\ cpm}{10\ m} + \frac{40\ cpm}{5\ m}} = 5.3\ cpm$$

$$160\ cpm \pm 5.3\ cpm$$